

LONG TERM PERFORMANCE OF SOIL AND BACKFILL SYSTEMS

**WORKING GROUP
B1.41**

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LONG TERM PERFORMANCE OF SOIL AND BACKFILL SYSTEMS

WG B1.41

Members

W. ZENGER, Convenor	US
J.M. ARGUELLES	ES
M. BURCEANU	BE
M. CHATZIPANOS	GR
Y. DOUIMA	FR
M. GENOVESI	IT
J. MILLAR	FI
F. OLIVEIRA	BR
C. SONDEREN	NL

J. HEAD, Secretary	UK
H. BRAKELMANN	DE
J. CAIRNS	AU
N. COWAP	IE
Y. FAN	CN
A. MAKOVOZ	US
C. MOREAU	FR
D. PARMAR	CA

Invited Expert

S. RAPOPORT	UK
-------------	----

Retired Member

E. THUNBERG	SE
-------------	----

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EXECUTIVE SUMMARY

The work of WG B1.41 was set up to study the often-neglected subject of the technical performance of the backfill for cable systems, which is used to reinstate the trench around a buried cable. This material is a critical engineering component of the cable system, which in conjunction with the surrounding material is the means of conduction of the heat, produced by the operation of the cable, to the ground surface where it is dissipated to the atmosphere. Therefore, the thermal properties of these materials, assumed in the design of the cable system, are as important to the correct operation of the system as is the correct thermal design in the first place. This design will generally be derived using detailed calculations often involving finite element analysis by computer, whereas the backfill may be installed by manual labourers with no knowledge of the performance requirements of a backfill, other than perhaps the fairly obvious one of supporting whatever is above (such as a road surface).

In addition to the above considerations, the load on both unmodified and up-rated cable systems may be increasing progressively. In all cases higher loads result in higher operating temperatures for the backfill even if the rated maximum operating temperatures remain the same. Many of the existing circuits have been in service for 40 or more years when engineered backfills were in their infancy. Limited knowledge is available of past backfill design and how it will change over time. Recent work showed that the condition of a backfill can alter, for example changes such as a reduced degree of compaction and stratification of backfill components have been reported. Of particular interest is how high load conditions, change in physical properties and environmental changes will impact aged backfill and soil conditions.

There are several well-known incidents in which the performance and /or condition of the backfill has contributed to the failure of a cable system. One of the most notable was in Auckland in 1998, which suffered failure of all 4 of the 110kV cable circuits feeding the Central Business District. The initial cable breakdown was attributed to either gas loss (from the gas pressure cable) or thermo-mechanical problems; the subsequent 2 circuit failures were due to thermo-mechanical problems (gas pressure and fluid filled circuits) whilst the final (fluid filled) circuit failed due to thermal runaway. One of the main factors contributing to thermo-mechanical and thermal runaway problems was cited as the high thermal resistivity of the backfill material when it was dry, the original system design being based on thermal resistivity values that were not achieved in the actual installation.

The aim of the WG was to investigate the subject widely to determine the extent of known problems with backfills. Initially the available technical literature on the subject was reviewed and a survey of users of cable systems was carried out in order to investigate the extent of known problems. The experience of practices in the countries known to the working group members was also gathered.

In order to inform readers not familiar with the particular subject area the types of soil (soil being generally used as a term to describe the indigenous ground material through which the cables are laid) and backfill are described, followed by a comprehensive list of the terminology used in relation to soils, backfills and their thermal performance.

At a technical level, one of the most important aspects affecting the thermal behaviour of backfills is that of moisture migration. Chapter 5 is devoted to this phenomenon, which has a major effect on heat transfer from the cable to the environment via the ground surface. The key aspect to understand in the case of moisture migration is that there is a critical moisture content of soils and backfills such that if the material dries below the critical level then there is a rapid increase in thermal resistance of the material which in extreme cases can lead to thermal runaway and resulting cable failure.

Two subsequent chapters of the TB are devoted to describing:

- the usual installation processes for backfilling the cable trench
- the Standards and related test instrumentation commonly applicable to measurements of soil / backfill physical and in particular thermal properties.

Chapters 8, 9 and 10 describe the anticipated changes or mechanisms by which deterioration of the properties of an installed backfill might occur, methods for assessment of such changes, and the potential consequences of such changes on the cable system.

Chapter 11 examines how deterioration of the thermal performance of a cable system might be mitigated and / or avoided by maintenance procedures or rectification or changes to the installation.

Probably the most useful chapter follows, which forms a “Users Guide” in relation to the assessment of thermal backfill performance. Two flow charts are provided; the first guides the user through the desktop process of assessing the rating of a circuit with a view to deciding whether to progress to a field survey. Should a field survey be found to be necessary, then the second flowchart links this process, via specific clause references to the information provided in the earlier chapters of the TB. This guides the user in the process of a full field and laboratory assessment of the backfill / soil performance. The data deriving from the assessment can then be fed back into the rating assessment to allow decisions to be made regarding the need for modifications to the circuit (e.g. work to mitigate particular hotspots) or perhaps in extreme cases to replace the circuit or reinforce the connection in some way.

The conclusions are that while many users have not reported known problems with backfill, there are nevertheless a number of instances worldwide where backfill performance has been severely compromised for example due to movement of fines by uncontrolled water flows in the cable trench.

The performance of the backfill ranks in importance with the accuracy of finite element rating calculations and hence should be considered as an important aspect of a cable installation when determining quality control requirements for initial installation.

The TB provides information as to possible mechanisms of backfill deterioration and indicates possible actions to mitigate problems. It details appropriate maintenance strategies which if adopted will reduce the probability of undetected problems arising due to changes in backfill condition and the cable local environment.

The economic viability of corrective actions is also considered briefly.

References and appendices with some specific case studies are also provided.

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1. INTRODUCTION AND SCOPE

The backfill used to reinstate the trench around a buried cable is a critical engineering component of the cable system, which in conjunction with the surrounding material is the means of conduction of the heat, produced by the operation of the cable, to the ground surface where it is dissipated to the atmosphere. Therefore the thermal properties of these materials, assumed in the design of the cable system, are as important to the correct operation of the system as is the correct thermal design in the first place. This design will generally be derived using detailed calculations according to IEC [1][2] or other Standards, possibly using finite element analysis by computer [3], whereas the backfill may be installed by manual labourers with no knowledge of the performance requirements of a backfill, other than perhaps the fairly obvious one of supporting whatever is above (such as a road surface).

In addition to the above considerations the load on both unmodified and up-rated cable systems may be increasing progressively. In all cases higher loads result in higher operating temperatures for the backfill even if the rated maximum operating temperatures remain the same. Many of the existing circuits have been in service for 40 or more years when engineered backfills were in their infancy. Limited knowledge is available of past backfill design and how it will change over time. Recent work showed that the condition of a backfill can alter, for example, changes such as a reduced degree of compaction and stratification of backfill components have been reported. Of particular interest is how high load conditions, change in physical properties and environmental changes will impact aged backfill and soil conditions.

The scope of the work of WG B1.41 was to cover the following topics (see Appendix A for the official Terms of Reference (ToR) of the WG):

- To review the literature (experience, history) on the subject
- To establish the appropriate terminology and characterization parameters
- To review methods to measure the thermal, mechanical and chemical soil / backfill properties and stability
- To review methods to measure the aging and long-term stability of soil and backfill properties over system life
- To review technical methods how to mitigate deterioration of soil and backfill conditions including moisture depletion by vegetation or other utilities
- To evaluate the consequences, if no action is taken, such as a reduction of current rating (ampacity), including cost and overheating of the cable system
- To integrate the information in a practical users guide
- To apply to extruded, paper, and paper-laminate cable systems
- To apply to HV AC and DC land cable systems including direct buried, direct buried ducts or pipe, duct bank / manhole systems, and Horizontal Directional Drill (HDD) installations
- To apply to HV AC and DC submarine cable systems, beach landing area only, including ploughing, jetting, trenching and HDD installations
- To apply to MV AC cable systems of high importance

It should also be noted that from a survey described in a recently published Technical Brochure (TB) 640 [4] it was found that in the majority of cases there was reported to be little control of the actual quality of backfills installed on new installations. Thus, the information in this TB can also be applied not only in assessing the ageing or degradation of backfills, but also to assess the actual thermal performance which may not have changed since the original installation but nevertheless could be of inferior performance to that assumed in the thermal rating calculations of a circuit.

This TB comprises 14 chapters; in the next chapter a review of the literature is presented, details of the papers reviewed being presented in Appendix B. Chapter 3 discusses the results of a survey conducted by the WG into utility experience in relation to long term performance and problems experienced with cable

backfills. The detailed results of the survey are shown in Appendix C. In addition, this chapter includes a summary of local experience gathered from the WG members.

In order to assist those new to the subject of backfills Chapter 4 describes main types of backfills and gives the meaning of the terminology in general use. Chapter 5 introduces the concepts moisture migration and drying out, with an overview of the physics involved and some practical measurements. The following two chapters (6 & 7) describe methods used for installing backfills in commonly met applications and give test methods used for determining backfill properties and stability; if required, the details of the relevant test methods can be found in Appendix D.

Chapter 8 then moves on to the main subject of the Technical Brochure, describing possible changes which may occur to backfills during the life of a cable system. Chapter 9 then gives methods of determining potential degradation of performance which may have occurred. Chapters 10 & 11 describe the consequences to the cable system of performance changes of the backfill and methods of mitigating such changes and their effects.

The key chapter in the TB for users seeking to assess the condition of their cable system is Chapter 12 "Users Guide". This provides flowcharts guiding the user as to how to evaluate the need for an assessment of the backfill condition and then how to go about such an assessment, with specific references to the various preceding chapters of the TB.

Chapter 13 then gives the overall conclusions of the TB, followed, in the final chapter, by a list of the references cited.

2. REVIEW OF LITERATURE

A comprehensive literature survey ranging from academic to industry-oriented publications has yielded very little material directly related to the scope of WG B1.41, although there is a wealth of material that comes close. In some ways this is disappointing, but in other respects lends meaning to the core activities of this working group. At the time of writing, WG B1.35 had recently completed an extensive brochure on cable rating, and so our literature survey focuses on the cable environment and installation practices, in order to address how these aspects of HV and important MV underground cables may potentially be affected by aging or by possible external changes to the cable environment over time.

Brief summaries of the literary surveys from the various sources are given in this section, but more detailed reviews and tables of the papers reviewed can be found in Appendix B.

2.1 CIGRE

CIGRE papers, brochures and articles from 1960 to 2012 were checked in terms of their treatment of the thermal resistance of soils, backfilling properties, etc.. Some interesting articles were found, but the number of contributions touching this WG topic is low compared to other subjects, such as thermal monitoring systems or cable rating. The relevant papers are listed in Table B.1 in Appendix B.

2.2 JICABLE

The Jicable papers from 1984 to 2016 were filtered and checked with the following keywords: thermal, stability, controlled, backfill, moisture, long term, ageing, and soil. Interesting articles were found, but, as for CIGRE, the number of papers directly related to the WG topic is low compared to other issues, in particular the ageing of cable insulations.

Some papers, however, do report a change in ground parameters over time. Original design values of thermal resistivity are compared with actual present values at critical conditions [5]. Reference [6] covers a change in cable type to cope with dry seasons, city extension and overload conditions in Mexico City. Another climate change related paper [7] relates the effect of hotter and drier summers in Victoria, Australia.

Environmental impacts are explored in [8]. Using a critical heat flux rather than the more traditional critical temperature rise is investigated in the development of a real-time application of 2-zone moisture migration modelling in [9].

A more complete summary of these papers can be found in Appendix B.2, with a wider selection of papers given in Table B.2.

2.3 IEEE EXPLORE

The papers listed in Table B.3 include publications available from the IEEE explore website, e.g., AIEE publications, Electrical Engineering (which became IEEE spectrum), IET and IEE publications, and some conferences, such as CIREN. Once again, there is a conspicuous lack of literature relating directly to the core task of WG B1.41. There are, however, many of the now classic papers relating to cable rating (ampacity), moisture migration, installation configurations, etc. A summary is given in Appendix B.3 for the most relevant papers shown in Table B.3, listed by date, which were judged to be relevant to the working group. It should be noted that in 2006 the IEE and IIE (UK) combined to become the IET.

A few of the most significant papers are summarised below, and more full comments can be found in Appendix B.3.

An early paper [10] that includes extensive on-site testing related to thermal and mechanical problems on a 138 kV circuit in New Jersey, if revisited, might give interesting information regarding long-term behaviour.

Reference [11] contains a clear description of moisture movement in porous media and good overall data from the south of England, plus results from 16 field studies spanning seven years. The authors make the sobering point that “where soil surrounding a cable is dried to a stage at which the capillary film of moisture between soil particles is broken, recovery of moisture is very slow. In soils subjected to a temperature gradient with no saturation water available, recovery of moisture may be impossible”.

Reference [12] follows the famous 1962 cable failures in Belvedere, London, which caused a revision of the critical temperature for moisture migration for fully loaded cables.

Some long-term monitoring is contained in [13], which makes the point that if sands are installed in wet conditions or without adequate compaction, severe degradation may eventually occur after long-term moisture migration yields conditions that may not be detected in installation tests.

The evolution of backfills in the UK up to 1975 is covered in [14].

Reference [15] investigated the application of a special backfill to mitigate hot-spots in an existing connection, but otherwise noted that the temperature responses along the cable route did not change much in 8 years.

Reference [16] covers the failures, in 1998, in the 110 kV cables feeding the Auckland Central Business District. This was not due to environmental degradation but points to the need to have an accurate knowledge of the thermal environment of important cable connections, especially when loading patterns change over time.

Long term thermal resistivity (TR) measurements in Queensland, Australia revealed strong correlations of TR with rainfall [17].

Of particular interest to WG1.41, [18] points out that ground subsidence can be a real problem in cable environments.

Reference [19] is one of the few found that deal with HVDC cables, and suggests the compromising of underwater connections if the pore spaces in the protective rock berm become blocked with fine material, a condition that could be interpreted as long-term thermal degradation.

Appendix B.3 also includes many significant contributions to cable rating and moisture migration modelling that impinge indirectly on the concerns of WG B1.41. It should be understood that moisture migration is a phenomenon that can take years to develop and may not manifest until well into the life cycle of a cable system, when loads may have become higher, more stochastic, and strict N-1 operation principles may have given way to more probabilistic reliability philosophies that take away the operation margins – just when aging cables are least able to cope with additional thermal and thermo-mechanical burdens.

2.4 RESEARCH ORGANISATIONS AND MISCELLANEOUS REFERENCES

Appendix B.4 also lists papers from EPRI and ERA, as well as a report entitled, "Physical principles and calculation methods of moisture and heat transfer in cable trenches".

There have been numerous intensive research and testing projects related to cable ampacity and the thermal environment of cable systems around the world. This brochure cannot cover them all, but several have had outstanding impact in cable rating.

KEMA, now DNV GL, is mentioned in Section 3.3.12 and Appendix B.5.8, which covers some of the extensive work on the thermal properties of soil and backfills performed and published in the Netherlands over the last half century. Several Canadian utilities, including BC Hydro and IREQ, have had large research facilities in the past. Ontario Hydro forms a large part of the testing reported in Section 3.3.14. The Central Electricity Research Laboratories, CERL, in the UK also carried out considerable underground cable and installation research up until the 1990s.

2.5 GENERAL CONCLUSIONS

The survey of literature going back over 60 years has produced very few papers that directly address or describe evidence of changes of the thermal performance of backfill materials. In most cases, it is not only the thermal performance of the material that changes per se, but also the surrounding environment (including the ambient conditions) changes and /or there is some physical movement of backfill (or a component of it) which results in a change of the thermal resistance.

In some cases, there can be a reduction of thermal resistance, e.g. due to an improvement in backfill compaction with time, but more generally the thermal resistance deteriorates (increases), which can be due to a number of factors such as:

- washing away of backfill leaving air spaces

- washing away of fines in a backfill
- changes in the moisture content of the environment leading to increased chances of drying out
- changes in the environment (air and ground) ambient temperature

Generally, there is no comment on mechanical problems with backfills, i.e. cable movement or loss of physical support of a cable. However, in one case (in the Netherlands), severe ground and cable subsidence is cited as a major factor in reducing cable life.

No papers have been found related to chemical changes to backfill materials; this aspect is nevertheless addressed in Section 7.5 of this document.

3. SURVEY OF UTILITY EXPERIENCE AND LOCAL EXPERIENCE IN DIFFERENT COUNTRIES

3.1 GENERAL

One of the most important factors affecting cable rating is the thermal resistivity of the soil, which may change over time due to moisture migration and soil particle movement.

The thermal resistance of the cable environment is mainly related to the thermal resistivity of the trench backfill, surrounding native soil and the burial depth of the cables. A cable's depth and /or arrangement may change over time due to construction activities in the vicinity of the cable, ground subsidence between cables in the trench, cable movement due to thermal expansion and contraction etc..

The scope of this survey was to gather available information on the long term thermal performance of underground cable installations, asking to share the experience of visual or measured changes in soil/backfill parameters, flooding, soil migration, cable failures due to local overheating, or suspect water/sewer breaks in the vicinity of the cable trenches.

The working group circulated a simple 2 page questionnaire, shown in Appendix C, to utilities worldwide.

A total of 46 responses were received from 17 countries. Of those replies, 22 from 11 countries gave a positive response to question 1, indicating that they had experienced a problem with thermal backfill.

A summary table of the positive replies is given in Appendix C.

Figure 3.1 shows a map of the countries with replies to the questionnaire.



Figure 3.1 Map showing countries with replies to the survey (in green)

In addition a summary of experience provided from different countries is given in 3.3.

3.2 SURVEY

3.2.1 Discussion

Upon review of the 46 utility survey replies, several observations can be made.

First of all, where the answer has been a simple 'No', the utility may not be aware of any problems if the system is working satisfactorily and they have not conducted a site survey. The system may have been designed and installed properly, but also the cable loads may have been significantly lower than the design load. Cable systems are most of the time managed without periodic controls or checks, unless a problem arises. Distributed Temperature Sensing (DTS) systems have only been recently introduced and are usually installed on major systems and many are not used up to the present time.

Secondly, while thermal resistivity measurements are made during the design stage and/or at the time of installation, measurements are rarely made after putting a circuit into operation, except in case of a fault. A periodical control of the backfill quality and properties seems not widespread at all; only one reply mentioned periodic thermal resistivity tests. The same is applicable for the conditions around the cables: sometime there is an awareness of environmental changes around the cables but nothing is done, sometime the situation is unknown.

Direct buried cables are more subject to problems than those in pipes. Only two cases of problems were reported for cables in pipes and one case for cables installed in troughs. As for backfill type, sand is the most critical, followed by native soil. Issues with cement bound sand / weak mix are reported in two cases only. It is not clear from the survey whether this is due to a less widespread use of this type of backfill (it is common in some markets) or is due to this type of material having stable thermal properties in service.

Flooding had been a problem in 12 replies out of 22. However the backfill had only been affected in a couple of cases, in which either cement bound sand or sand had been used. In general, flooding was not a problem, at least in the short term.

When the soil or backfill was noticed to be completely dry or rock hard, high sustained cable loads seemed to be the cause for moisture migration. It could also be attributed to poor quality of backfill and installation.

Regarding higher soil temperatures, only one reply pointed to the possibility of global warming, while many others are addressing the drying out of soils for various reasons, such as:

- poor backfill
- higher loads
- installation not as designed
- replacement of backfill material surrounding the cable with materials of poor thermal properties
- installation of heating pipes

Sometimes the cause of higher soil temperature was not so clear; in one case a lack of rain rather than ambient temperature increase seems to have been responsible for thermal problems.

In general, rather than changing natural environmental conditions, human activities have been the main factor causing the working conditions of cable circuits to deteriorate.

3.2.2 Conclusions of survey

As a conclusion, it can be affirmed that, in general, at the design stage the properties and characteristics of cable backfills are correctly selected with the right safety margins.

However, during the installation phase, backfills are in some way "forgotten", and left to the experience of the construction company sometimes with poor or no control.

Increasing circuit loads and changing environmental conditions are eroding the safety margins. A change of assumptions during the design stage can lead to cable faults. A change in what was assumed at the design stage, for example, a shift in the time of peak loading from winter to summer due to widespread implementation of air-conditioning or construction of air-conditioned buildings, can lead to cable faults.

As a final consideration, it appears there are no common standards or guidelines on how and when to check backfill properties, or the type of tests to be performed. Very often measurements are made only after a

fault has occurred, to determine the cause. In general, each Utility has its own procedures, and long-term performance issues of backfills are not being addressed.

3.3 LOCAL EXPERIENCE IN DIFFERENT COUNTRIES

3.3.1 Australia

In Australia, a significant amount of experience has been gained with the deterioration of backfills, in general due to physical disturbance due to water or ground movement. Some technical details of a number of these incidents are given in Appendix B.5.1.

However, historically very little or no information was available on the soil thermal characteristics for the same reason as listed below for North America. Government research agencies (Australian Research Council, Geoscience Australia and Department of Agriculture and Water Resources) were a good source of information but seldom contacted by the utilities. On turn-key projects, it was relied upon the expertise and experience of the cable manufacturers who were (in many cases) responsible for the selection of local civil and mechanical contractors for the construction. The practices and standards for the selection of the cable trench backfills were based on their 'home base' experiences. In some cases, the selected parameters were questionable; especially for the ambient temperatures and soil mineralogical classifications. Utilities with in-house research and testing facilities (civil engineering discipline) had a better understanding of the thermal/mechanical requirements of the backfills and thus, they would provide specifications for the type and performance of the selected backfills. In most cases, these were based on the performance of the 'sand:cement' mixtures. The standard mixes were: 7:1, 14:1, 20:1. Sand and native soil have also previously been used. The choice of backfill varied on the utility, voltage and cable location.

Since 2001, fluidized backfill was accepted as a standard material for directly buried HV and EHV, XLPE cable installations. Local suppliers of ready-mix concrete have taken keen interest and advantage of this newly introduced material and its applications. Duct bank installations may be either concrete or fluidized backfill. Route soil sampling has also become a standard practice for most HV and EHV cable projects. DTS (distributed temperature sensing) systems for cable temperature monitoring have been used on a number of installations, however, they have not always been used effectively.

3.3.2 Belgium

In Medium Voltage Networks for directly buried cables, common sand is used for cable bedding and then the native soil is used to fill the cable trench. Visual soil changes (soil dry-out) were seen after failures around the MV cables and around several joints.

In HV networks, for the directly buried cables thermal backfill is used from 1990s and bentonite is used in cable ducts for directional drilling when necessary. The thermal resistivity at different depths is measured for each cable project in order to evaluate the necessity of using bentonite in ducts. Until now, no incident has been reported regarding the soil/backfill changes. More information about the soil investigation in HV network is given in Appendix B.5.2

3.3.3 Brazil

One location in Brazil suffered problems due to changes in the level of the water table. A report of extensive laboratory study work to understand the situation is given in Appendix B.5.3.

3.3.4 China

China is a country that has been undergoing massive infrastructure projects at the time of writing, and extensive underground cabling is one of them. A series of thermal resistivity tests were carried out at the test centre in Wuhan, China, in 2008. The results of the research provided by the Chinese member of the WG are reproduced in Appendix B Section B.5.5.

In addition, Figure 3.2, below, shows the results from a series of measurements showing the seasonal temperature variation at a typical cable burial depth.

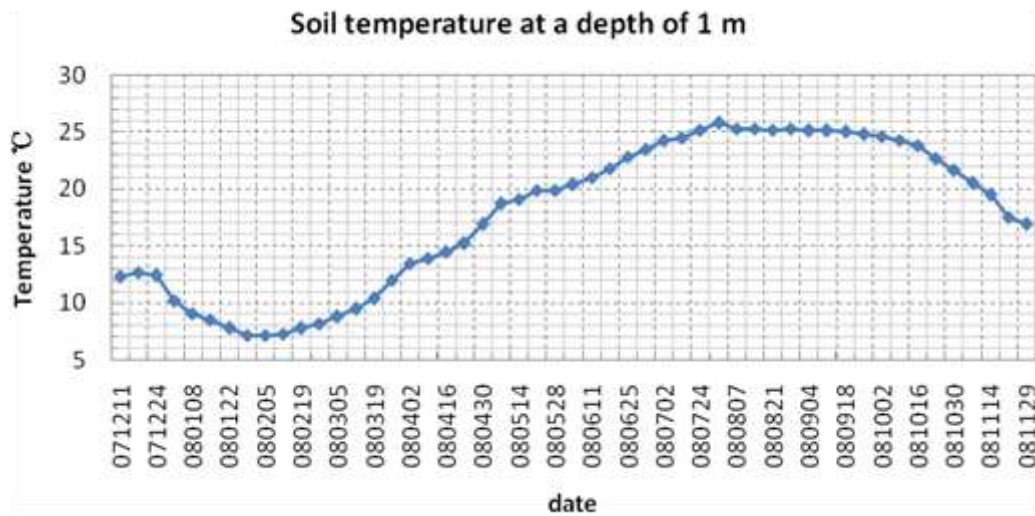


Figure 3.2 Seasonal soil temperature variation at a depth of 1m

3.3.5 Finland

The transmission network (>200kV) in Finland is almost entirely overhead due to the long distances and difficult terrain for trenching, except for submarine HVDC connections to Sweden and Estonia.

110 kV sub-transmission cables in the Helsinki region are usually installed in trefoil in backfilled concrete troughs (about 50x40 cm). Peak loads are mostly in the winter, although there are a few connections in Helsinki that are starting to carry higher loads during the summer, due to minimal generation from local CHP (Combined Heat and Power) in the summer months. No degradation of the thermal environment of cables has been recorded, although it has been noted that moisture migration in the graded sand and crushed rock backfills typically used can occur at temperature rises much lower than the assumed safe 35K above ambient (corresponding to the well-known 50 °C isotherm). Appendix B.5.6 gives more information.

3.3.6 France

France has seasonal and geographically defined ambient temperatures and thermal resistivities. The country is subdivided into three zones, each with summer and winter ambient temperatures to be used in cable rating. In addition, the "by default" standard thermal resistivities of native soils in France are: 1.20 K·m/W during the summer and 0.85 K·m/W during the winter. Appendix B.5.7 shows the map and the respective temperatures.

3.3.7 Germany

Rating calculations are guided by IEC-publications. A combination of the Cox/Coates-approach [20] for the two-layer-model and the Neher/McGrath-approach [21] for the consideration of cyclic loads provides the basis for DIN/VDE-rules and rating tables.

If no specific information about soil and backfill is available, the tables in DIN/VDE use the following standard parameters:

- thermal conductivity of moist soil/backfill: $\lambda_m = 1,0 \text{ W/(K}\cdot\text{m)}$ (TR = 1.0 K·m/W)
- thermal conductivity of dried-out soil: $\lambda_d = 0,4 \text{ W/(K}\cdot\text{m)}$ (TR = 2.5 K·m/W)

The critical temperature rise for onset of drying-out is dependent on the load characteristic. It is taken as:

- $\Delta\theta_{cr} = 15 \text{ K}$ for continuous load

and

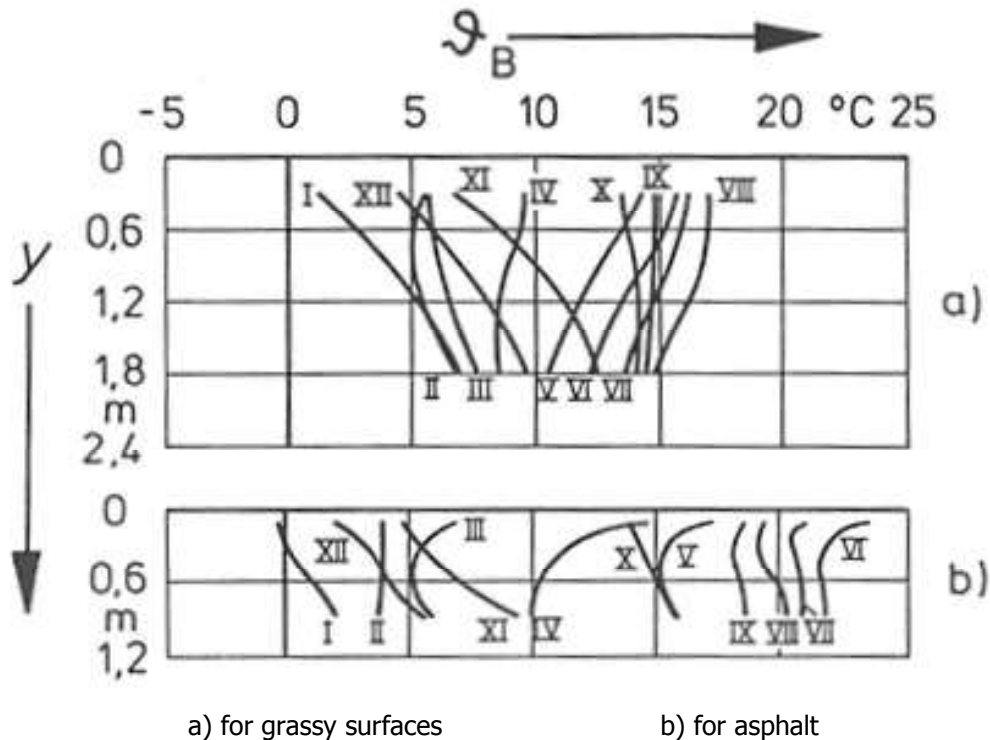
- $\Delta\theta_{cr} = 15 \text{ K} + (1.0-m) 100 \text{ K}/3$

for cyclic load with m daily load factor (e.g. $\Delta\theta_{cr} = 31.7$ K for $m = 0.50$).

soil ambient temperature:

- $\theta_a = 20^\circ\text{C}$ for laying depths up to approx. 1 m
- $\theta_a = 25^\circ\text{C}$ under paved surfaces (asphalt etc.)
- $\theta_a = 15^\circ\text{C}$ for laying depths up more than 1.5 m

Examples of measured monthly mean soil temperatures [22] are shown in Figure 3.3, as function of the laying depth y and of time.



Roman numerals refer to the month (I = January to XII = December).

Figure 3.3 Monthly mean values of soil temperature vs. depth under a) grassy surface b) asphalt surface

3.3.8 Greece

Generally compacted gravel (sand) is used as a backfill.

In special cases other options have also been used in order to increase current carrying capacity:

- Bentonite-filled pipes
- Weak-mix backfilling
- Horizontal directional drilling (HDD) with Bentonite-filled pipes

For the underground section of submarine cables the proposals of the contractors for appropriate installation (usually cement backfilling) are being followed.

A series of studies was undertaken by the National Technical University of Athens (Mechanical Engineering Dept., Heterogeneous Mixtures and Combustions Systems Lab) in order to measure the thermal resistivity of used materials for backfilling in Greece. The results of the study were used to select the appropriate backfilling materials for the installation of underground cables in Greece and current carrying capacity calculations.

Samples of bentonite, C12/15 and lightweight concrete were taken into investigation as well as three types of mortar sand of different material and grain size (exclusively types that are available in Greek quarries) in different conditions of moisture and pressure. Additionally, three types of weak mix were investigated as well as the gravel (Local Norm: PTP 3A) that is used in Greece. The measurements took place according to ISO-8894-1, DIN 51046 and ASTM D2326-70 by using the CT-Metre instrument and proprietary testing equipment.

According to the results:

C12/15 concrete is by far the best option among concretes with a measured thermal resistivity between 0.52 and 0.43 K·m/W.

Weak Mix (Concrete, bentonite, sand and water in 2:2:4:1 mixing ratio) was the best option among weak mixes with a measured thermal resistivity of 0.99 K·m/W.

Mortar sand was found to have a maximum thermal resistivity in dry conditions between 2.86 and 2.50 K·m/W (with a strong decrease in presence of moisture and application of pressure).

For the gravel, the measured thermal resistivity varies from 3.03 K·m/W (0% moisture, 0.01 bar pressure) to 0.89 K·m/W (6% moisture, 0.75 bar pressure). Thermal resistivity varies poorly with pressure when the gravel is completely dried.

3.3.9 India

Design and installation of cable systems in most turn-key projects were handled by cable suppliers. In the recent past, the local utility as well as local cable manufacturers have developed in-house capability to conduct route thermal surveys and for the sourcing / formulation of corrective thermal backfills. Standards for the type, quality and installation of these backfills have been developed and implemented. Government research institutes have taken interest in the same subject to acquire technology to help local projects.

3.3.10 Israel

The design and installation of cable systems in most turn-key projects were left in the hands of the cable suppliers. In the recent past, the local utility has developed in-house capability to conduct route thermal surveys as well as sourcing and formulation of corrective thermal backfills. Standards for the type, quality and installation of these backfills have been developed and implemented. A data-base of the thermal /geotechnical characteristics of the native soils and that of potential backfills the across the country was also undertaken.

3.3.11 Italy and Switzerland

National data for soil temperature over a long range of time:

there is not much data available regarding long-term changes in soil temperatures, but some is available for air temperatures.

A report by ISTAT (Italian National Institute of Statistics) [23] shows data relevant to air temperature and rainfall changes: in the 2000-2009 period the average air temperature increased in Italy by 0.8°C, while rainfall amounts declined by about 4% , compared with the 1971-2000 period.

A study published in 2015 by ISPRA (Italian National Institute for Environmental Protection and Research) [24] envisions, for the next decades up to the end of the 21st century, an increase in the average temperatures in Italy of at least 1.8°C. Reduced rainfall and higher ambient temperatures are clearly likely to cause degradation in the thermal performance and rating of cable backfills and native soils.

For Switzerland, a report from 2007 from the UFAM, Swiss Federal Office for Environment [25] claims an increase in the average air temperature of 1.5°C from 1970s levels. Since the temperature of the first soil layers is in close relation with air temperature a similar increase is expected at cable burial depths.

The average rainfall seems unchanged, while days with heavy rain have increased significantly during the last decade.

HV cable backfill technologies: little or no detailed information is available prior to the 1980s.

However, it is understood that the use of backfill became the norm for HV cables, not only for thermal reasons but also for mechanical protection purposes. The material, however, was mainly compacted sand. Weak cement mix was not used very often.

Starting from the 1970s several studies were performed to better understand the behaviour of soils under thermal gradients. The use of weak mix backfills started to be more common. Nowadays, weak mix is the main backfill material. The weak mix, or cement mortar, for 150, 220 and 380 kV cables, is covered by Terna, (Italian TSO) specifications. The grain size distribution (sieve analysis) of the sand to be used is also specified.

During the 1990s other studies were performed to better define the grain size distribution curve for the sand to be used in weak mix backfills.

3.3.12 Middle East

Most cable projects are undertaken by cable manufacturers on turn-key basis. It is left up to the discretion of these contractors to base the system design on their experience/expertise of the local regions. In the recent past, local universities and utilities have developed in-house capability to conduct route thermal surveys as well as sourcing and formulation of corrective thermal backfills. Standards for the type, quality and installation of these backfills have been developed and made available for the projects.

3.3.13 Netherlands

Many studies have been performed in the Netherlands on the subject of soil. The results from these studies can also be used for power cable installations and calculations because the physical behaviour of soil will be the same. A brief overview of some of these studies is provided in Appendix B.5.8, and compliments the literature survey in Sections 2.1 – 2.4, commencing with the 1952 PhD thesis by D.A. de Vries [26], followed in 1957 by one of the most influential publications in this field, by J.R. Philip and D.A. de Vries, "Moisture movement in porous materials under temperature gradients" [27], and an impressive body of further related work by Dutch researchers [28]-[38].

In the Netherlands, an established set of procedures for identifying bottlenecks, and conducting field surveys and laboratory tests resulting in representative thermal resistivity values has been developed. A more detailed account of the procedure is given in Appendix B.5.8.

3.3.14 New Zealand

Practices were similar to the ones in Australia. Cable trench backfill in general was a locally available 'river sand' that was assumed to have 'acceptable' thermal characteristics. It proved not to be the case on a major cable failure. Investigation of the cable failure brought awareness not only to the utilities in the 'region' but also to the cable industry as a whole. Thermal surveys of several cable routes were commissioned for 'ampacity audit'. Local suppliers of ready-mix concrete have taken keen interest and advantage of this newly introduced material and its applications.

3.3.15 North America

Years of data, collected during route thermal surveys of new and existing installations of underground power cables in North America (Canada and USA), lead to the following general observations regarding the quality and performance of backfills for various cable types. These included, directly buried, cables in concrete duct-banks, fluid-filled and pipe-type cables. On several projects, data were collected during the investigations of cable failures which were concluded to have occurred as a result of thermal runaway.

Before the 1960s it was not a standard practice of utilities, cable manufacturers or electrical contractors to conduct surveys of cable routes to determine the soil thermal characteristics. In general, a 'rule of thumb' value of 0.9 K·m/W for the TR of soils was considered 'good enough' for preliminary cable design.

Utilities would often contact the local government and research agencies to obtain any available information on the soil thermal and mechanical characteristics for the area. Weather data (temperature, moisture, precipitation, snow cover, frost penetration, etc.) were also gathered from meteorological stations and other sources.

Numerous utilities and some cable manufacturers had in-house specifications and guidelines for what was considered to be a 'good' thermal backfill. In many cases a readily available backfill material was 'concrete

sand'. It was inexpensive, easy to install, and it had low TR when moist or wet (TR of $\sim 0.6 \text{ K}\cdot\text{m}/\text{W}$). However, in a totally dry condition, this value increased to well over $3 \text{ K}\cdot\text{m}/\text{W}$.

In 1960s and 1970s, a granular by-product of aggregate producing plants, 'limestone dust', was found to exhibit excellent thermal and mechanical performance. This too was relatively inexpensive, readily available and easy to install. The main problem with any such granular type backfill is the quality control during installation. It had to be installed at a specific density ($\sim 95\%$ of standard Proctor density) and at a moisture content close to the 'optimum' value in order to achieve the desired performance. During this period, some non-standard materials such as 'blended sands' (sand mixed with clay) were also used with good success but at an increased cost. A small number of sand/aggregate suppliers specialized in formulating and delivering such materials specifically for U/G power cable projects.

In the late 1970s two research projects were undertaken. One was to develop a test instrument for in-situ measurements (i.e. for a route thermal survey) and laboratory measurement of thermal characteristics of native soils and backfills. The other project was to develop an easy to install fluidized backfill. This was a mixture of readily available materials such as sand, aggregate, flyash, cement and water which, as the name implies, was to be an initially a fluid material that could be installed by conventional techniques used for pouring concrete. The resulting material, was a self-consolidating, flowable slurry, which cured to give a good compressive strength and suitable thermal characteristics for use in cable installations. Over the last 30+ years this has been a widely used backfill material for most utilities in Canada and USA.

Sometimes, native soil is used as a 'non-classified' backfill above a concrete duct-bank, or above the fluidized backfill envelope, based on an assessment of its quality and meeting the ampacity requirement of the circuit.

The use of 'thermal grout' for HDD as well as for 'jack & bore' installations is also very common.

The carrying out of route thermal surveys coupled with soil sampling and thermal testing have also become a standard practice for most HV and EHV cable projects. Similarly, DTS (distributed temperature sensing) for cable temperature monitoring is also widely used. Most utilities have stringent quality assurance test programs for the backfills during the construction phase. This includes duct-bank concrete, granular or fluidized backfill and thermal grout.

Potential hot-spots are identified during a thermal survey, and mitigated during the construction phase by the most practical and economical solutions.

There have been some isolated incidences of hot spots (dry rings around the cable) at locations that have been excavated to repair cable or add instrumentation. The measured soil properties were significantly poorer than that which would have been expected based on the original installation specifications. In one case high solar exposure due to a vertical embankment along with high surface traffic was noted. There were also reports of dry soil conditions due to competitive vegetation root systems. Dryout areas and thermal runaway have been reported in submarine crossings especially in organic / clay materials.

A summary of a Canadian geotechnical data base is given in Appendix B.5.4.

3.3.16 South America

Relatively little information is available from South American countries on their practices on this subject. In the recent past, local cable manufacturers have acquired instruments and technology for route thermal surveys, laboratory testing and for the sourcing/formulation of corrective thermal backfills.

With the exception of major EHV projects that require high level of expertise, proven technology and products, the information is seldom shared by the cable manufacturers or the successful bidders of the turn-key projects.

3.3.17 Spain

In Spain, the usual practice for HV underground lines is to install the cables in conduits that are laid in trenches. The conduits are laid on spacers, and concrete is poured into the trench to cover the conduits. The rest of the trench is then backfilled. A value of 95% Modified Proctor compaction is specified for the backfilling material. When the native soil meets the minimum requirements in terms of degree of compaction, granulometry, etc., it is reused. Failing that, new material is brought in to backfill the trenches

over the concrete. The cables are pulled at a later stage, utilising joint bays (manholes) along the cable route. This type of methodology, similar to duct/manhole installations in, e.g., the US, allows the civil works to be separated from the cable laying operation.

As concrete has good thermal properties, specific thermal measurement campaigns are not commonly undertaken. Major connections, such as the HVDC links between Spain and France or the landfalls of submarine lines, have more detailed thermal resistivity surveys and analysis of local material.

3.3.18 Sweden

Sweden is a relatively cold and wet country in northern Europe. Sweden is also a relatively large and sparsely populated country. Cables have historically only been used for the distribution of electrical energy near the city centres. Cables have also been used in production plants. In the larger cities the transmission of electricity requires high or extra high voltage cables. In smaller cities medium voltage cables are normally used. Appendix B.5.9 summarizes the practice in Sweden more completely, with some comments on the South West Link.

3.3.19 UK

As can be seen from the literature review the UK was one of the places in which moisture migration /drying out and thermal runaway were discovered. As a consequence, consideration of these aspects is generally taken and in many cases "special backfill" (e.g. sand selected for good thermal properties or a stabilised backfill – weak sand /cement mix) are used within the 50°C isotherm.

For directly buried transmission cables either selected sand or stabilised backfill can be used, however the former is not allowed where there is any possibility of backfill movement.

Ducts or HDD are also used where necessary. On road crossings duct banks are encased in concrete where possible.

In other situations, ducts may be pre-installed and cables pulled in later. In these cases, particularly for cost sensitive installations the native soil (where suitable) may be used for backfilling the installation.

Ducts are frequently filled with a pumpable bentonite, sand cement mixture.

In some instances, fluidized backfill is used.

Installation contractors may use proprietary materials, generally subject to having carried out suitable (perhaps full scale) tests to demonstrate the values of thermal resistivity being achieved.

Beach landings for submarine cables: for submarine cable installations the landing area determines the rating of the whole circuit as generally temperatures are lower elsewhere, i.e. in the fully submerged part of the submarine cable. On the land part the cable will usually be of a different design and rating is thus usually independent of the submarine cable. HDDs are used from 3 to 4m below the low water mark to the junction with the land cable. This duct would either be water filled or bentonite filled (a proprietary mix). The ducts would be spaced and also the armour may be removed from the cable in order to achieve the required circuit rating (eddy current losses in the armour contributing to heating for AC cables). Generally special backfills are not used.

4. TERMINOLOGY AND CHARACTERIZATION

Underground cables are installed in trenches that are backfilled either with **native soil** or with a special backfill; commonly referred to as '**corrective or controlled backfill**'. These materials have to meet specified thermal and mechanical characteristics that are somewhat interrelated. Although it is not necessary to list or to conduct a whole range of geotechnical tests that are routinely performed for 'civil engineering' applications, a simple approach of visual inspection and identification will help an electrical or a cable engineer identify potentially good soils and backfill materials. Once selected, the performance qualification testing must be conducted in accordance with the local or international industry standards.

Heat transfer through the soil and backfill materials may take place by three main physical mechanisms:

- conduction
- convection
- radiation

According to Fourier's Law, the heat is transferred from hot medium to cold environment by **conduction**, so the heat flux is proportional to the temperature gradient. In native soil and backfill materials conduction occurs everywhere, although convection dominates the heat transfer in pores and voids filled with moisture or air.

Convection occurs in fluids – air, water and gases. The flow of the heat due to convection is described by complex differential equations (convection/diffusion equations) with normally nonlinear coefficients governed by the gradients of temperature and density. In native soil and backfill materials convection occurs in the pores and voids between fines and at the surface of the earth.

Radiation of heat is proportional to the difference between the fourth power of the respective absolute temperatures, i.e. proportional to $T_A^4 - T_B^4$, where, e.g., T_A is the surface temperature of a heat source (e.g. a cable) and T_B is the ambient temperature, or a surface element of another object in direct line of sight of the heat source. In native soil and backfill materials, radiation mainly takes place from the exposed soil surface. Radiation is a significant surface to surface heat transfer mechanism where cables are installed in tunnels or unfilled pipes, or ducts.

4.1 TYPES OF SOIL AND BACKFILL

Soil is a combination of materials (components) which are possibly in three different phases: solid, liquid and gas. It typically contains soil minerals, organic materials, moisture and air. All these constituents have an influence on the thermal resistivity of the soil. An overview of the thermal resistivities of these constituents is given in Table 4.1 below.

Table 4.1 Typical thermal resistivities of materials present in soils

Material	Typical Thermal resistivity [K·m/W]
Quartz	0.11
Clay minerals	0.34
Organic material	4
Stationary water	1.7
Sandy soils	0.3 (wet) – 5.0 (dry)
Clay soils	0.6 (wet) – 10.0 (dry)
Peaty soil	2.0 (wet) – 17.0 (dry)
Stationary air	40

Quartz has a low thermal resistivity of 0.11 K·m/W. The thermal resistivities of water and air are much higher than this, so the total thermal resistivity of a quartz based soil, which contains quartz, water and air, is much higher than 0.11 K·m/W.

Added value in backfill performance evaluation: soils and backfills in particular with high permeability are prone to loss of fines, wash-outs, erosion and settlement that can degrade mechanical and thermal performance. For a given backfill, the dry density has the highest impact on thermal resistivity in totally dry condition. The higher the density, lower the permeability and thermal resistivity. Cement bound sand is a typical example. These properties are shown for several materials in Table 4.2.

Table 4.2 Thermal-mechanical properties of soils and backfills

Type of Soil or Backfill	Saturated Moisture Content (wet) (%)	Dry Density (kg/m ³)	Hydraulic Permeability (cm/s)	Porosity (%)	Thermal Resistivity (K·m/W)	
					Wet	Dry
Fine-coarse gravel	3-5	2150	10 ⁻¹ to 1	25-50	0.5	2.5
Gravelly sand	7-15	2050	10 ⁻² to 10 ⁻¹	25-50	0.5	2
Fine-coarse sand	12-18	1900	10 ⁻² to 10 ⁻¹	20-35	0.5	1.7
Uniform sand	12-18	1700	10 ⁻³ to 10 ⁻²	20-35	0.55	3.5
Sandy silt	15-25	1800	10 ⁻⁴ to 10 ⁻³	30-40	0.65	2.5
Silt	15-25	1750	10 ⁻⁵ to 10 ⁻⁴	35-50	0.7	2.5
Silty clay	20-30	1750	10 ⁻⁷ to 10 ⁻⁶	35-50	0.65	2
Lean clay	20-30	1700	10 ⁻⁸ to 10 ⁻⁶	30-50	0.75	2.5
Fat clay	30-40	1600	10 ⁻⁹ to 10 ⁻⁷	35-60	0.9	3.0
Stone-dust	9-12	2050	10 ⁻⁴ to 10 ⁻³	20-35	0.45	1.1

Material density, specific gravity, porosity and hydraulic permeability are related. At lower density the material has higher voids content, higher porosity and thus higher permeability. This type of material is prone to erosion and 'wash-out' of fines. Non-cohesive granular materials such as sand/gravel/silt may exhibit fairly low thermal resistivity in wet condition but this value may increase 3 to 7 fold when in a totally dry condition.

Native Soil: native soil may consist of large particles ranging from >100 mm (cobbles) to very fine particles such as silt and clay (<0.075 mm). Based on the percentage of each particle size (as determined by sieve analysis or gradation characteristic), the soil may be described as 'granular', 'cohesive', 'uniform', 'well-graded', etc.. Figure 4.1 shows sieve analysis graphs of various soils and backfills and relates the visual description to the grain size diameter. This applies to native soils as well as man-made materials such as backfills.

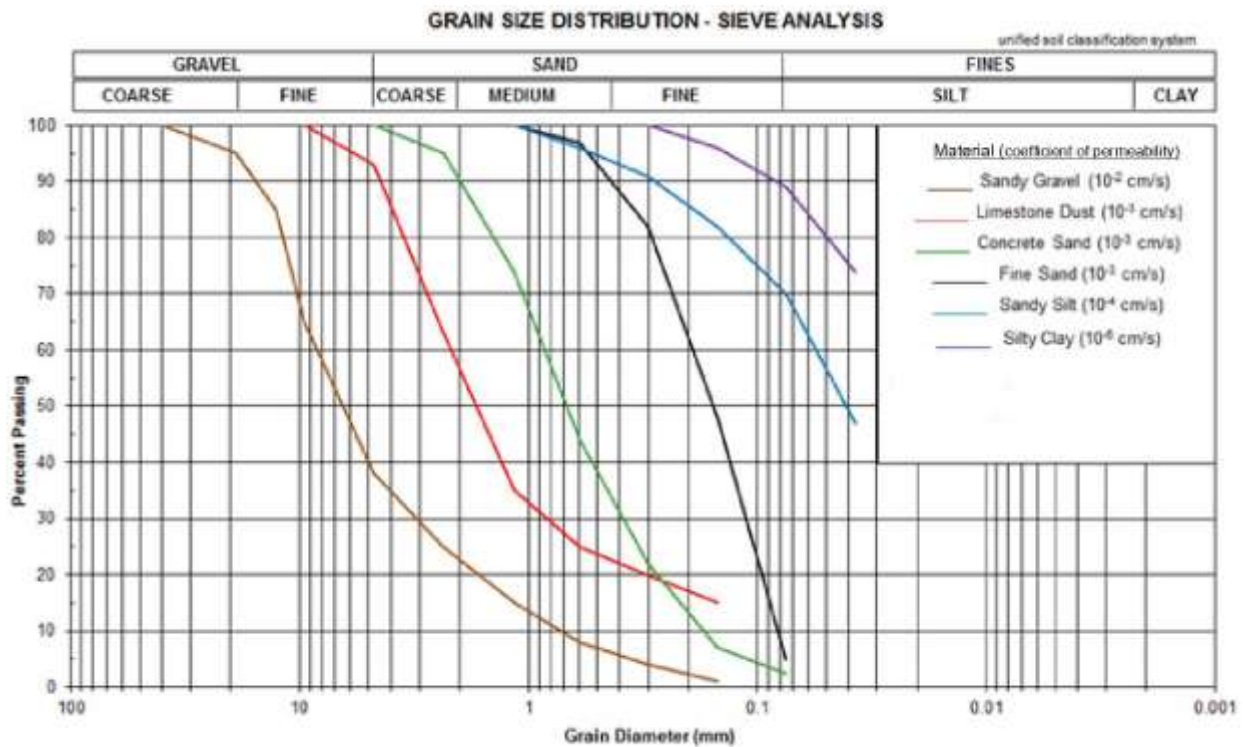


Figure 4.1 Sieve analysis of typical soils and backfills

Common visual descriptions for coarse grained (non-cohesive) soils are:

Fine-coarse Gravel, Sandy Gravel, Gravelly Sand, Fine-coarse Sand, Well-graded Sand, Silty Sand, etc..

Common visual descriptions for fine grained (cohesive) soils are:

Silty Clayey Sand, Sandy Clayey Silt, Clayey Silt, Silty Clay, etc..

If organic material is encountered, the descriptions may contain adjectives such as 'with organics or root matter or vegetation', 'slightly organic', 'highly organic', 'amorphous peat' or fibrous peat'. An example of soil classification is given in Table 4.3.

From a consistency (strength) viewpoint, soils may be described as 'soft', 'firm', 'stiff' and 'hard'. This gives an indication of the bearing capacity and also of the ease or difficulty of excavation.

Native soil is not commonly used as backfill because of the relatively poor thermal performance and problems associated with storage and re-instatement to its completely natural condition. In some countries, authorities do not permit the re-instatement of removed native soil. However, if its thermal and mechanical performance is acceptable, or where required for example for agricultural or environmental reasons, it can be installed on top of a concrete duct-bank or a fluidized backfill envelope.

Table 4.3 Example of soil classifications

Major Division		Letter Symbol	Group Name	
COARSE GRAINED SOILS CONTAINS LESS THAN 50% FINES	Gravel and gravelly soils More than 50% of coarse fraction retained on no. 4 sieve	Gravel with 5% fines	GW Well graded GRAVEL	
			GP Poorly graded GRAVEL	
		Gravel with between 5% and 15% fines	GW-GM Well graded GRAVEL with Silt	
			GW-GC Well graded GRAVEL with clay	
			GP-GM Poorly graded GRAVEL with Silt	
			GP-GC Poorly graded GRAVEL with clay	
	Sand and Sandy Soils More than 50% of coarse fraction passing on no. 4 sieve	Gravel with ≥ 15% fines	GM Silty GRAVEL	
			GC Clayey GRAVEL	
		Sand with 5% fines	SW Well graded SAND	
			SP Poorly graded SAND	
		Sand with between 5% and 15% fines	SW-SM Well graded SAND with silt	
			SW-SC Well graded SAND with clay	
			SP-SM Poorly graded SAND with silt	
			SP-SC Poorly graded SAND with clay	
FINE GRAINED SOILS CONTAINS MORE THAN 50% FINES	Silt and Clay	liquid limit <u>less</u> than 50	ML Inorganic SILT with low Plasticity	
			CL Lean inorganic CLAY with low Plasticity	
			OL Organic SILT with low Plasticity	
		liquid limit <u>greater</u> than 50	MH Elastic inorganic SILT with Moderate to high plasticity	
			CH Fat Inorganic CLAY with moderate to high plasticity	
			OH Organic SILT or CLAY with moderate to high Plasticity	
		Highly Organic soils		PT PEAT soils with organic contents
		Notes:-		
		1)	Classifications are in accordance with ASTM D2487 and ASTM D2288	

Backfills: most commonly used backfill materials are: stone dust or stone-screenings, well-graded sand with some silt, cement stabilized sand, etc. These are often referred to as "Thermal Sand". The other commonly used backfill is "Fluidized Backfill". This is a high-slump, fluidized mixture of medium aggregate, concrete sand, some fly ash and cement. Duct-bank concrete and thermal grout can also be used as backfills. In order for a material to be classified as 'Thermal Backfill', it must have relatively low thermal resistivity even when completely dry. A typical specification may call for the thermal resistivity to be no higher than 0.6 K·m/W at 'critical moisture content' and no higher than 1.0 K·m/W in a totally dry condition.

Readily available and commonly used, 'stone-dust' samples are shown in Figure 4.2, together with a sample of 'thermal sand'. Stone-dust is a by-product of aggregate processing plants that is finer than 6mm and is well-graded, with about 10-15% fines content. Thermal sand can be a natural material or it can be a manufactured product. Regular concrete sand can be blended with a small amount of silt / clay or cement to increase the fines content to an acceptable level.

Although larger aggregate size offers lower thermal resistivity, the possibility of damaging the cable jacket is much higher and therefore the particle size is limited to 4-6mm. Larger size aggregates (40mm) have been used in fluidized backfills where the separation between cables is large (200mm) and on pipe-type cable installations.



Figure 4.2 Samples of backfill materials

Non air-entrained, nominal 20 MPa duct-bank concrete will have dry thermal resistivity of $\sim 0.8 \text{ K}\cdot\text{m}/\text{W}$. This material is made up of 10-20 mm aggregate, concrete sand, cement and water. Sometimes admixtures are used to improve slump (workability), rate of hardening, etc..

Well-graded Clean Sand: In the 'early days' the most readily available 'concrete sand' was commonly referred to and used as 'thermal sand' on underground cable installations. It was later determined that its thermal characteristics were not ideal, especially in the low moisture content range. Although this sand was reasonably well-graded (particle size ranging from 0.075 mm to $\sim 5 \text{ mm}$), its primary use was in the formulation of concrete and thus it had to be fairly 'clean' ($>4\%$ passing #200 sieve size). This lack of fines resulted in low dry density ($\sim 1750 \text{ kg}/\text{m}^3$) and high porosity when compacted and with thermal resistivity of $\sim 2.0 \text{ K}\cdot\text{m}/\text{W}$ in totally dry condition.

Cement Bound Sand: One way to improve the density and thermal performance is to add a small quantity of cement or lime to concrete sand. The two commonly used mix ratios of sand:cement are 14:1 and 7:1. When mixed with appropriate quantity of water and compacted at about 95% of the standard Proctor density, it resulted in an increase in density and significantly lower dry thermal resistivity ($\sim 1.2 \text{ K}\cdot\text{m}/\text{W}$). The compressive strength of the 7:1 mix can be rather high (in excess of 5 MPa) that would make it difficult to excavate.

Uniform Sand: As the name implies this is a fine-medium grained material with only one or two sieve size particles. This type of sand does not compact well (dry density $< 1600 \text{ kg}/\text{m}^3$) and the dry thermal resistivity approaches $\sim 3.0 \text{ K}\cdot\text{m}/\text{W}$. Because of the lack of coarse size particles, it is not an ideal candidate for cement bound mix.

Thermal Sand: a well-graded sand, with particle sizes ranging from 6mm to ~15% fines (200 sieve size), of sound mineral rock is an ideal thermal backfill. This material when installed at near optimum moisture content (~10 to 12%) and at 95% of standard Proctor density (~2000 kg/m³) will have dry thermal resistivity of about 1.0 K·m/W. There may be minor variation in the thermal characteristic based on the type of rock - mineralogy (limestone, granite, dolomite, etc.).

Well-graded Soil: As described earlier, well-graded material compacts well and exhibits good thermal characteristic. For example, gravelly sand with silt/clay will compact well at a dry density of ~2100 kg/m³ and have dry thermal resistivity similar or better than a select thermal backfill. If the largest particle size in this type of soil is no larger than say 6mm, it will make an ideal backfill even for directly buried installation. Similarly, silty clay or clayey silt with high percentage of sand and gravel may behave in a similar fashion. If dry thermal resistivity of say about 1.5 K·m/W is acceptable, most well-graded native soils may satisfy this requirement when installed properly.

Curves showing the Standard Proctor test moisture-density relationship are shown in Figure 4.3 below. For a given compaction energy, the dry density of a material increases with the increase in moisture content up to a certain point. This level of moisture is known as the 'optimum' moisture content and the attained density is referred to as 'maximum' dry density. The theory is very simple – moisture acts as a lubricant that reduces the friction between soil grains, bringing them closer, reducing the volume of voids in the matrix. The ultimate result is a more dense material that is mechanically and thermally superior. Above the optimum moisture level the voids that are almost fully saturated and thus any increase in water content starts to separate the soil grains, i.e. the dry density decreases and the material becomes softer. Although the thermal characteristic may change only marginally, the mechanical strength may deteriorate significantly. From thermal viewpoint, this defines a density and moisture for optimum performance.

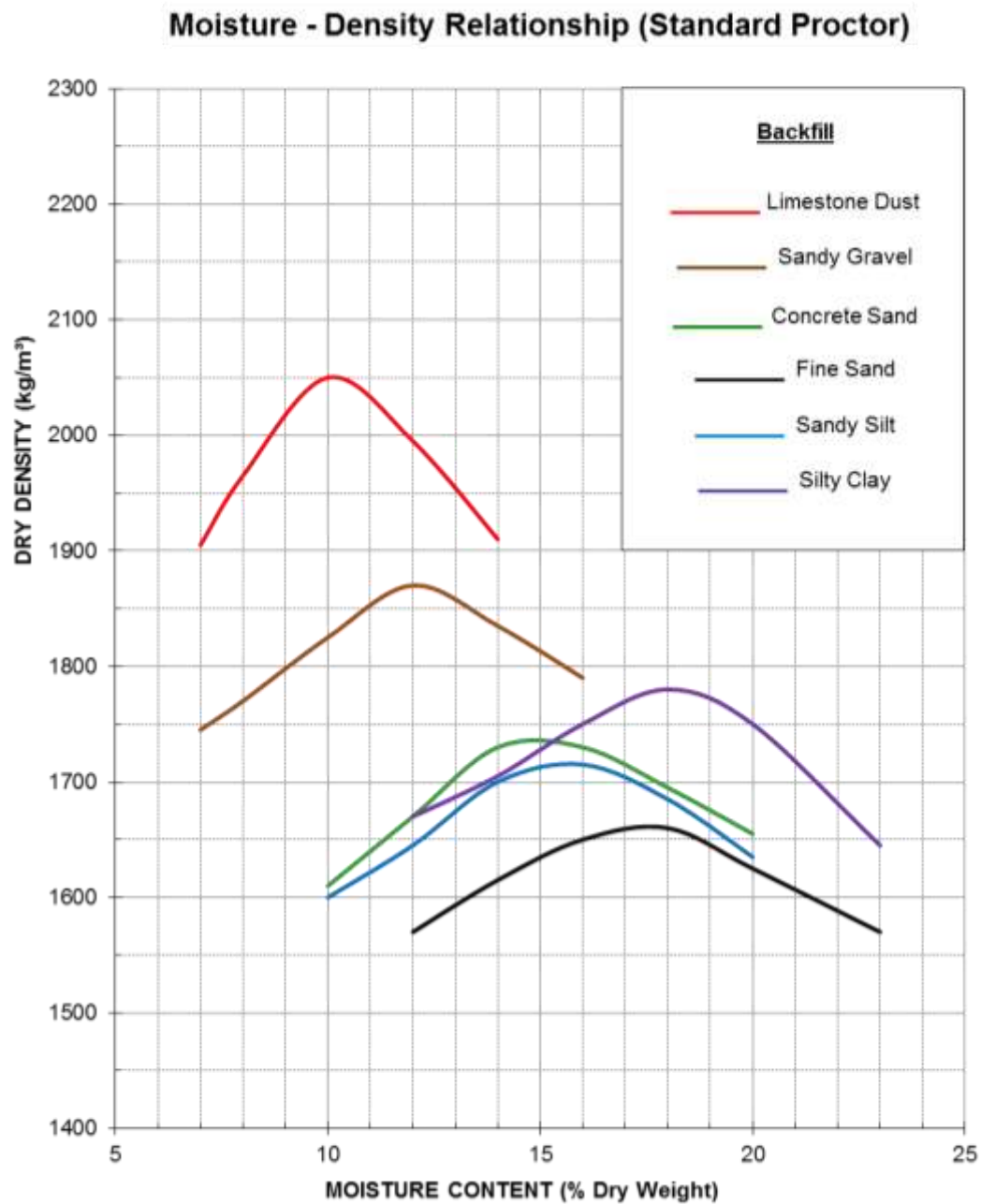


Figure 4.3 Effect of moisture content on compaction achievable for different materials

After compaction to the optimum density, subsequent variation of the backfill moisture content has a major effect on the thermal resistivity of the backfill, as shown in Figure 4.4 below. The effect of drying out on the thermal performance is much more marked for some materials than for others. In general, the critical moisture content for a material is near to the “knee” of the curve.

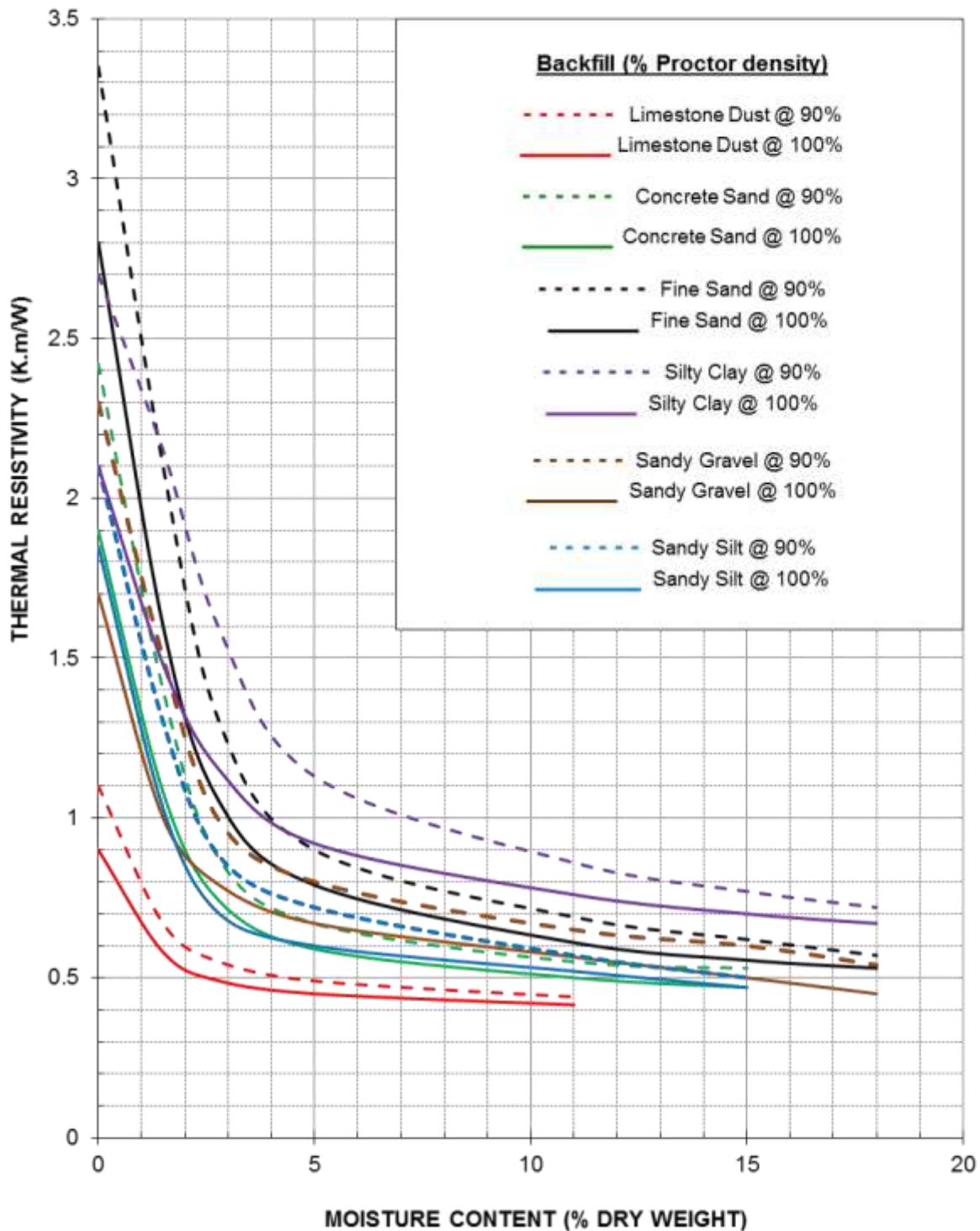


Figure 4.4 Effect of moisture content (or dryout) on thermal resistivity of different materials

For selection of a granular backfill for optimum thermal performance, it must be well-graded with a certain percentage of 'fines' as shown Figure 4.5 below. It must consist of sound (non-porous) mineral aggregate and be free of any organic matter. Limestone dust and natural sands with the above gradation characteristic make a good thermal backfill when compacted at near optimum moisture content and to achieve near maximum dry density. Sieve analysis of candidate materials can be plotted to assess suitability. A selected material must be tested to evaluate its density/moisture relationship and thermal characteristics at the proposed installed density. This is done in accordance with the Standard Proctor Compaction Test.

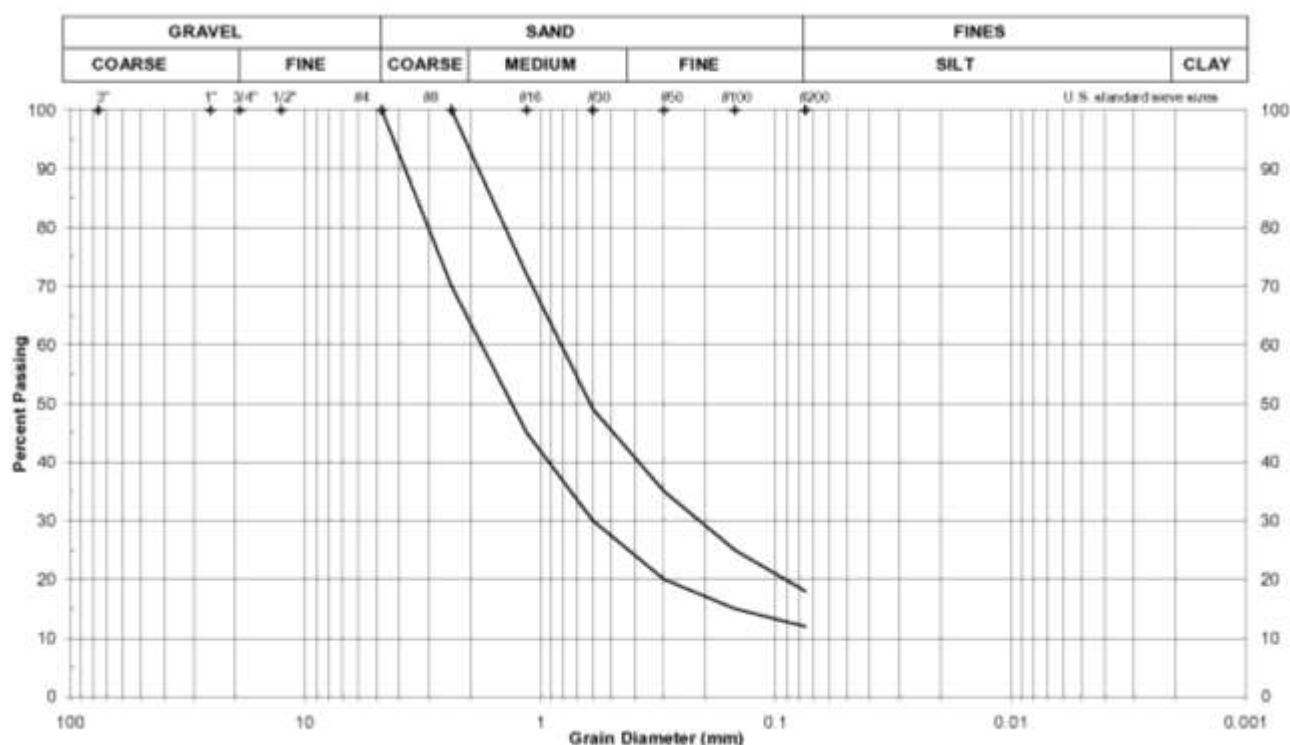


Figure 4.5 Sieve analysis envelope for an ideal granular thermal backfill (for directly buried cable installation)

4.2 TERMINOLOGY

The terminology and parameters involved in the characterisation of backfills (including soils) are listed in alphabetical order below:

Air-knife (or air-vac) Excavation: a method of excavation of cable backfill, minimising possible damage. A blunt end (nozzle) of a short length of metal pipe is connected to a high pressure air hose and the air jet is used to gently break up the material. A vacuum hose is inserted in the excavated zone to suck-up the loose soil.

Ampacity: see current rating

Atterberg Limits: the soil moisture contents delimiting the plastic (i.e. lowest moisture at which the soil is malleable) and liquid (i.e. lowest moisture content at which the soil is a viscous fluid) states of a fine grained soil.

Benthos Sampler: a torpedo-shaped drop weight used to insert sample tubes into submarine substrata sediments.

Bentonite: a highly reactive, natural clay used to formulate drilling mud used in drilling, mining, injection grouting and construction.

Cement: a manufactured powdery substance made with calcined lime and clay. It is mixed with water to form mortar or mixed with sand, gravel, and water to make concrete.

Cement Bound (Bonded) Sand (CBS): a backfill material comprising a weak mix, typically of 14:1, sand to cement mix, this makes it easier to break into for access to the cables is required, compared to a normal sand / cement mix used for building purposes. Proportions of between 7:1 and 14:1 have been used, the precise mix being chosen depending on the required thermal and mechanical characteristics.

Classified Thermal Backfill: a granular type soil that meets specific gradation and mineral soundness characteristics e.g. specific gravity, porosity & mineral type. Unclassified soils may be termed “bank run” or “pit run”.

Clay: very fine grained, cohesive soil particles, smaller than 0.002 mm.

Fat clay: a cohesive and compressible **clay** of high plasticity, containing a high proportion of minerals that make it greasy to the feel. It is difficult to work when damp, but strong when dry.

Lean clay: a **clay** of low to medium plasticity owing to a relatively high content of silt or sand.

Coarse Grained Soil: granular soils that are coarser than the #200 sieve size (i.e. sands and gravels).

Coefficient of Permeability: see permeability.

Composite Thermal Resistivity: a weighted average of the thermal backfill resistivity and the native soil resistivity, taking the backfill thickness into consideration. Sometimes it is referred to as effective thermal resistivity ($K \cdot m/W$).

Concrete Sand: sand suitable for use in the mixing of concrete (thermal properties of the sand not being defined).

Consistency: a qualitative indication of the strength (i.e. soft, firm, stiff, or hard) of fine grained soils.

Controlled Thermal Backfill: imported material that is placed around a cable to improve thermal resistivity and stability:

- a good thermal soil that has the texture to allow it to be easily compacted to a high density (well graded sands, crushed stone screenings),
- controlled density fills such as fluidized backfill and some weak mix concretes.

Critical Heat Rate: the heat rate (see definition below) below which a given soil will remain thermally stable indefinitely (2-zone-model).

Critical Temperature:

- 1) the isotherm between the zone with dry soil, high thermal resistivity, and the zone with moist soil, low thermal resistivity (2-zone-model).
- 2) the temperature where the drying-out of soil starts and the dynamic equilibrium is broken (thermal runaway)

Critical Temperature Rise: the temperature rise above the ambient temperature, which is the isotherm between the dry zone and the moist zone.

Critical Moisture (Water) Content: the point (% moisture content) on a thermal dryout curve below which the thermal resistivity begins to increase disproportionately, see Figure 7.7. This is also referred to as critical saturation degree.

Current Rating: (also known as ampacity):

Cyclic rating: the maximum current from a defined load characteristic (daily or weekly load cycle).

Dynamic (Real time) rating: current rating given in real-time based on load history and possibly load and environmental forecasts.

Emergency rating: current that the cable can safely conduct for specified periods of time; usually under defined load conditions during emergencies.

Steady state rating: the maximum allowable steady-state current under continuous loading

Critical Saturation Degree: see critical moisture content.

Dilatency: a reaction by silty soils, where moisture comes to the surface when a patty of soil is shaken (or vibrated), and moisture is drawn back when the soil is squeezed.

Directional Drilling: guided boring, incorporating an articulating drill-head, which allows a borehole to curve horizontally and vertically along a prescribed alignment.

Drilling Mud: a viscous fluid formed by mixing a small quantity of bentonite (or chemical additive) with water. It is commonly used to lubricate, fill and support boreholes and tunnels.

Dry Density: the ratio of the dry weight to the total volume (kg/m^3).

Drying Time Test: a thermal probe test extended in excess of the time to dry out, i.e. until the soil thermal resistivity begins to increase significantly above the ambient value.

Dryout Curve: when soil or backfill loses moisture either naturally or induced by the heat-front of energized cables, its thermal resistivity increases. The relationship between thermal resistivity and moisture content at a constant dry density is referred to as 'thermal dryout characteristic' and is depicted in Figure 4.4.

Dynamic Cone Penetration Test: an instrumented cone, with the tip facing down, is pushed into the ground at a controlled rate to measure the properties of a soil. See Standard Penetration Test, below, for an alternative test more commonly used for backfills.

Fine-Grained Soil: soil or backfill containing a high percentage of particles that is finer than #200 sieve size or $75\text{ }\mu\text{m}$.

Fines: particles of soil or backfill that are finer than #200 sieve or less than $75\text{ }\mu\text{m}$ and are used to improve the properties of controlled thermal backfill.

Flowable Fill: also referred to as "controlled density fill", and "slurry backfill". It is a mixture of sand, flyash and water, with a little cement that can be installed by pouring. It consolidates to a constant density and is used as a backfill in civil construction. In general, it does not meet the thermal requirements of a corrective thermal backfill. The thermal resistivity of this type of flowable fill is typically 0.6 to $0.7\text{ K}\cdot\text{m/W}$ in a wet condition at or near saturation moisture content, but in a totally dry condition this value will be of the order of 4 to $6\text{ K}\cdot\text{m/W}$.

Fluidized Backfill: a free flowing controlled density fill, consisting of stone aggregate, sand, etc.. It is usually installed around cables by pouring. This is a typical fluidized backfill; designed primarily for underground power cable applications. Its thermal, strength and flow characteristics are taken into consideration in the design. Typically, the thermal resistivity in a moist condition is in the range of 0.4 to $0.5\text{ K}\cdot\text{m/W}$ and in a totally dry condition this value is in the range of 1.0 to $1.2\text{ K}\cdot\text{m/W}$.

Fly Ash: a by-product of coal burning power plants. Classified fly ash may be used as a partial cement replacement in concrete or as a large volume fill.

Good Thermal Soil: a soil with a fairly constant moist thermal resistivity, a relatively low dry resistivity, and a sharp knee at a low moisture content in its *thermal dryout curve*. Applicable *thermal dryout curves* are indicated in Figure 4.4.

Gravel: coarse grained soil particles ranging from 5mm to 75mm . Particles larger than 75mm are referred to as 'cobbles' (75mm to 200mm) and 'boulders' (larger than 500mm). These are generic visual descriptions.

Grain Size Distribution: is a list of values that defines the relative amount, typically by mass, of particles present according to size.

Grading Coefficient (Cc): is a value which represents the grain size distribution. Well graded soils have a value between 1 and 3 .

Grout: See "Thermal Filler" below.

Heat Capacity (volumetric): the amount of heat necessary to change the temperature of a unit volume of soil by one K [e.g. $\text{J}/(\text{K}\cdot\text{m}^3)$]. Heat capacity is also sometimes referred to as volumetric heat capacity and is equal to specific heat divided by density of the material.

Heat Flux: the amount of heat emitted per unit area, [e.g. W/m^2].

Heat Rate: the amount of heat emitted per unit length, [e.g. W/m].

Initial Drying Time: the same as "Time to Dryout", see below.

Initial Lag Time: on a thermal probe test plot, with a logarithmic time axis, the initial curved portion before the straight line develops. It is the time required to overcome the probe heat capacity (or probe time constant) and probe to soil contact resistance and for the heat to be fully entering the soil. (It is sometimes referred to as the "initial transient").

Liquid Limit: the lowest moisture content at which a fine grained soil is a viscous fluid (i.e. it has a consistency of soft butter), [%]; the boundary between liquid and plastic states.

Lowest Expected Ambient Soil Moisture Content: the driest natural condition of a soil due to weather and environmental conditions, [%].

Maximum Dry Density: the highest dry density measured during the Proctor test, also known as Proctor value.

Moisture Adsorption: see moisture retention capacity.

Moisture Capacitance: see moisture retention capacity.

Moisture Content: the gravimetric water content is the weight of water divided by the dry weight of the soil solids, expressed as a percentage, [%].

Moisture Migration: the tendency of soil moisture to move away from a heat source.

Moisture Retention Capacity (also known as moisture capacitance, moisture adsorption or water holding capacity): is the capacity of a soil (or backfill) mass to hold moisture within the pore structure under negative pressure. Granular soil has high porosity and thus it can only hold surface moisture under negative pressure (suction). On the other hand, cohesive soil (clay) has high surface area per unit mass, has very low permeability and negative pore-pressure that results in higher moisture capacitance.

Nuclear Density Moisture Gauge: a device, based on the scattering of radiation from a radioactive source, which allows in-situ measurements of soil/backfill density and moisture content.

Optimum Moisture Content: the moisture content belonging to the maximal dry density of the Proctor test. This moisture content is the optimal value for compacting the soil.

Permeability: is a measure of how fast water can flow through the material under a unit pressure gradient. Coefficient of permeability is also referred to as 'hydraulic conductivity'. For a given density, the porosity of the material (voids content) is the governing factor, higher porosity leading to higher permeability.

pF Curve: also known as water retention curve. Shows the relationship of the soil moisture content with respect to the suction tension in the soil which is related to the distance of the groundwater level.

Plasticity: the cohesion or ability of clayey soils to hold their shape when kneaded.

Plastic Limit: the lowest moisture content (%) at which a fine grained soil is malleable (i.e. it has the consistency of stiff putty). Defines the boundary between plastic and semi-solid states.

Porosity: the amount of space (or voids) between the mineral grains in a soil mass, defined as the ratio of the volume of voids to the total volume.

Proctor Density:

Standard Proctor Density: result of a standard laboratory compaction test that defines the optimum moisture (%) and maximum density that a soil can attain [kg/m^3]. The compaction ability of field equipment can be related to the standard Proctor density.

Modified Proctor Density: in civil engineering applications, certain road-base and foundation materials are required to be installed at much higher density than normally attained by 'Standard Proctor' energy. Higher density ensures stability and greatly minimises any future settlement. For native soil or granular backfill used as cable trench backfill, this is not normally required because of the cost of installation and also the possibility of damaging cable jacket/insulation.

Reactive Clay: the plate-like particles of clay mineral have an ionic attraction that attracts and holds water, and makes clay cohesive.

Saturation (degree - %): the volume fraction of the soil voids that is filled with water. A saturated soil has all the voids filled with water (100 %).

Shannon and Wells Test: a laboratory test to determine the thermal diffusivity of a soil sample.

Shelby Tube: a 76 mm diameter by ~600 mm long, thin wall, steel tube with a bevelled edge that is pushed into a soil to obtain a relatively undisturbed sample.

Sieve Analysis: a dry soil is mechanically passed through a series of sieves with progressively smaller mesh size. The percentages (by dry weight) passing each sieve are plotted to give the grain size distribution (or particle size gradation).

Silt: fine grained, semi-cohesive soil particles finer than 0.075 mm (0.075 mm to 0.002 mm).

Slump Test: the amount by which a standard cone of concrete decreases in height (or slumps) when the metal mould is removed, [mm]. A measure of the workability (or flow) of concrete and fluidized backfill.

Stage Drying Test (Thermal Dryout Characterization): a moist soil sample is dried in stages with a thermal resistivity determination at each stage.

Standard Penetration Test: the number of blows of a heavy hammer, free-falling to advance a split spoon sampler. It gives an indication of soil density or consistency.

Specific Heat: the specific heat is the amount of heat, in Joules (J), required to raise the temperature of unit mass (1kg) of material by one Kelvin (K). The symbol C_p is used to represent specific heat and the unit is J/(kg·K).

Texture: a qualitative description of the soil grain sizes and their distribution, using accepted adjectives to describe the amount of each component (i.e. silty clay; fine sand with a trace of gravel).

Temperature Sensor: a device for sensing temperature which can be connected to a suitable measuring instrument. A commonly used temperature sensor for underground cable application is Type 'T' (copper-constantan) thermocouple. It has a wide range of operation (-50 °C to +200 °C) with a resolution of ~0.1 °C and can be used for spot measurement as well as for long-term and continuous data collection. Several thermocouples of different lengths can be bundled together to form a "thermocouple tree". This can be installed in a vertical borehole to obtain a profile of ground temperature as a function of depth. Other types of sensors such as thermistors, RTDs, temperature transducers can be used. These have different ranges of operation and resolution.

Thermal Backfill: see "Controlled Thermal Backfill" above.

Thermal Conductivity: the inverse of thermal resistivity, see Thermal Resistivity below.

Thermal Bridges: these are the last thin layers of water clinging to the solid particle contacts, as a soil dries, providing good heat conduction paths through the soil; also referred to as capillary bridges.

Thermal Diffusivity: an index of how rapidly a soil will undergo temperature change. Thermal diffusivity is defined as the ratio of heat conducted to heat absorbed. Therefore diffusivity, δ = conductivity / heat capacity (units m²/s). Thermal diffusivity is necessary for cyclic load calculations.

Thermal Dryout Curve: a plot of thermal resistivity vs. soil moisture content for a given soil at a specified dry density.

Thermal Filler (Thermal Grout): a flowable slurry with good thermal properties that can be used to fill voids and small spaces between cables and/or ducts, e.g. fluidized backfill, sand-bentonite slurry, sand/cement slurry and cement/flyash slurry.

Thermal Probe (Thermal Needle): a slender metal cylinder, containing a heater and a temperature sensor that approximates a line heat source for use in measuring the thermal resistivity of soils and backfill materials.

Thermal Probe Test: a test for measuring the thermal resistivity of a soil or backfill in which a thermal probe is inserted into a soil with minimum disturbance. Constant power is applied to the heater and the temperature-time data is monitored. The soil thermal resistivity may be calculated from the slope of the straight-line portion of the test curve (time on a log scale).

Thermal Properties Tester: a computer controlled instrument that runs the thermal probe test with specified test parameters, analyses the results, calculates the thermal resistivity and diffusivity and gives an indication of the test validity.

Thermal Resistivity: the ratio of temperature gradient to heat flux density (the inverse of thermal conductivity). The measure of the resistance to heat flow, ρ [$K \cdot m/W$].

Thermal Runaway: the heat from a cable causes soil moisture to migrate away from the cable. This causes the soil resistivity to increase, which results in higher heat generation from the cable and subsequently more moisture migration. If this condition persists, the soil around the cable will eventually dry out. If this process is self-limiting to a stable situation this is called partial drying-out. In the case of a persisting thermal instability this may lead to a thermal runaway, i.e. overheating of the cable.

Thermal Stability of Soil: when the soil thermal resistivity remains fairly constant. In a thermally stable condition, cable heat induced moisture migration is balanced with a return flow of water due to capillary suction.

Time to Dryout: when a thermal probe, emitting an appreciable heat rate, is inserted in a moist soil, the time required for the soil thermal resistivity to begin to increase significantly above the ambient value, is the time to dryout and is an indication of the onset of soil drying.

Time to Dryout of Soil: the initial drying time (from a drying time test) divided by heat source diameter squared, t/d^2 [days/cm²]. Mainly used in the USA. It is a constant for a given heat rate [W/cm], thus the initial drying time determined using a thermal probe can be extrapolated to the initial drying time for a large diameter cable.

Tremie: a method of placing concrete or thermal grout in boreholes, casings (or tunnels) to ensure complete and voids free filling. Concrete or grout is pumped through a tube (grout pipe) that is slowly withdrawn from the borehole. This is implemented in HDD installations of long length, in order to minimize the development of back-pressure during pumping.

Two-Zone Model: a simplified method to calculate temperatures when there is partial dryout (moisture migration) in the cable backfill or native soil. It is assumed that the backfill is dry inside a critical isotherm delineated by a critical temperature, temperature rise or heat flux, and outside the critical radius the backfill is moist.

Unified Soil Classification System (USCS): a standard means of grouping soils based on sieve analysis and Atterberg limits.

Uniformity Coefficient (C_u): is the ratio between the dimensions of the imaginary screen openings for which 60% (d_{60}) and 10% (d_{10}) of a soil is able to pass. Thus $C_u = d_{60}/d_{10}$. It is also used to express the uniformity of sand, which is the material between 0.063 mm and 2 mm.

Vibracore Sampler: a vibrating tool that is used to insert sample tubes into submarine sediments to collect relatively undisturbed sub-bottom samples.

Volumetric Ring Method: taking an undisturbed soil sample with a ring with a known volume.

Water Holding Capacity: see moisture retention capacity.

5. MOISTURE MIGRATION

The thermal resistivities of soil and backfill are the most important characterization parameters of a cable trench, and have decisive effects on the current rating of the cable. In many locations, these parameters will remain relatively constant, but in certain locations they may also undergo significant short- and /or long-term changes. Seasonal variation in moisture, water table depth, and long-term changes in the cable environment may change the ambient moisture conditions (and thus thermal resistivities) around a cable. Another important factor is the possible formation of dry zones around underground power cables due to the heat flux from the losses in the cable and the consequent thermal and moisture gradients. The latter is known as moisture migration, and is the subject of this chapter.

5.1 OVERVIEW OF THE PHYSICS

5.1.1 A qualitative introduction to moisture migration

Moisture migration refers to a net moisture movement in a porous media away from a heat source when a set of critical conditions are exceeded. Warm surfaces tend to dry, and water vapour tends to condense on cool surfaces. Heat tends to flow from hot to cold and water tends to migrate from wet to dry, which happens, for example, when we dry dishes, that is, before the tea-towel gets soaked.

Qualitatively then, it is relatively simple to appreciate that in a porous medium, such as an underground cable backfill, heat will flow away from a heat source (in the form of heat conduction in solids, convection and conduction in liquids, and convection, radiation and conduction in gases). Moisture vapour will transfer from a warmer to a cooler surface, and liquid water tends to flow back to the drier region. As long as the outward vapour and liquid transports are balanced by the return hydraulic (liquid) transport, the net thermal resistivity (resistance to heat transfer) remains approximately constant. If these processes are in equilibrium, we may say that the medium is thermally stable. However, when the outward transfer of moisture dominates, the capillary moisture bridges between backfill particles reduce and eventually disappear. The break in these capillary (thermal) bridges corresponds to a number of critical conditions: critical moisture content, critical temperature rise or gradient, and critical heat flux. The process has been discussed in many of the references in Section 2, in particular, [11], and many of the papers in the tables in Appendix B, e.g., [20] and [39] – [45]. Reference [46] provides a comprehensive overview of the phenomena and their modelling.

For a given soil condition (density, moisture content and porosity) the magnitude of the temperature gradient induces other changes, such as vapour pressure (corresponding to temperature gradients) and hydraulic pressure (corresponding to moisture content gradients). These in turn will influence moisture migration.

5.1.2 Proximity of the water table

One movement of 'wet to dry' is the possible movement of moisture from the water table into the cable environment. The proximity of the water table strongly influences the moisture retention in the soil at cable burial depths, and is a function of the hydraulic permeability of the soil and its gradation characteristics. Fine grained, cohesive soils such as silt and clay have much larger particle surface area per unit weight and therefore, these soils have higher moisture capacitance.

Hydraulic pressure in the soil is commonly referred to as 'pore pressure'. This is a function of the degree of saturation and the overburden (depth below surface).

5.1.3 Porosity, voids and moisture retention

Voids are always present in soil or backfill material. The amount of water that occupies the voids is a function of many external factors, related, among others, to pore pressure and permeability, ambient temperature and atmospheric conditions.

In fact, there are three main mechanisms for moisture retention. Moisture may be contained in the pores (voids) between solid particles and this will define the 'degree of saturation'. It may be contained within the

pores of the soil grains and this is referred to as 'absorption'. A third water retention mechanism is the water that 'clings' to the surface of particles, referred to as 'adsorption' or 'moisture capacitance'.

The higher the water content in the voids, the higher the degree of saturation and the lower the thermal resistivity. In the worst case, when all the water is driven out of the voids, the material is defined as 'totally dry' and has the highest thermal resistivity. The relationship is depicted in thermal dry-out curves (the thermal resistivity as a function of moisture content for a given dry density).

When cables are buried in soils or backfill materials with low water content and high thermal resistivity, heat dissipation is compromised and may result in increased cable temperatures or, if known and accounted for, limit the cable rating.

5.1.4 Precipitation

Precipitation obviously has a direct effect on ambient moisture content in the cable vicinity, and seasonal rainfall correlations have been explored by, e.g., [17].

Granular soil or backfill has relatively high porosity and permeability and thus, rain water (precipitation) can re-wet or saturate this type of backfill quite quickly. This will lower the thermal resistivity significantly. On the other hand, if the material is not well compacted, the possibility of wash-out or migration of 'fines' is also high, which is discussed in Section 8.2. This can lead to higher porosity, an increase in thermal resistivity as well as mechanical instability (settlement). If the cable system is located within the bounds of a fluctuating water table, the long-term performance may be adversely affected.

5.1.5 Moisture behaviour in different operating conditions

Over the long-term, the system load may increase due to gradual increase in energy consumption and the connection of more customers and/or distributed generation, increasing the risk of moisture migration, which may be problematic if not designed for.

Emergency operation imposes significantly higher loads than normal operating conditions, which will exacerbate any inadequacies in the backfill and thermal design of the installation. Also mentioned in Chapter 8, along with some cautionary tales, are the possible consequences of changing the time of peak loading. In particular, air-conditioning loads in central business districts are tending to shift peak loading from mid-winter, which is cold and moist, to mid-summer, which may be dry and hot, with severe consequences on temperatures and rating. Depending on the duration and frequency of emergency operations, and the cable surface temperatures reached during such events, the backfill materials may deteriorate more rapidly, mainly by faster moisture migration leading to higher thermal resistivity of the backfill and also by inducing premature chemical reactions at higher temperatures between aggregates and other chemicals present in various forms, some of which may create an aggressive chemical environment around the cables (see Section 7.5).

The shape of the load profile that a cable connection is subject to has an obvious impact on ampacity. With regard to moisture migration, it should be noted that under unstable conditions moisture migrates away from a heat source at a higher velocity than it returns, whereas in a stable situation these velocities will be equal. This must be considered with respect to emergency loads and also the increasingly stochastic nature of cable connections associated with renewable generation sources, and points to the need for algorithms that can model moisture migration in real-time.

5.1.6 Summary of physical parameters and processes influencing moisture migration

To summarise, the major soil and physical parameters influencing moisture migration are listed in Table 5.1.

Similar parameters related to the native soil at the backfill /soil boundary will also affect moisture movement. For example, saturated native soil will help maintain a much higher moisture level in the backfill (moisture equilibrium based on differential moisture retention capacity).

Table 5.1 Processes influencing moisture migration

Primary driver	Behaviour	Cause	Affecting factors
Temperature gradient	Liquid water tends to move from warmer to cooler regions	Surface tension of warmer water is lower than cooler water, thus weaker retention, thus mobility of warm water to cold region	Impeded by the hydraulic gradient resulting from this temperature gradient induced process and influenced by the permeability of the soil to liquid
	Water vapour tends to move from warmer to cooler regions	Higher vapour pressure in warmer region than cooler region, thus water tends to evaporate from the warmer surfaces and condense on the cooler surfaces	Impeded by resultant hydraulic gradient, permeability of the soil to vapour
Hydraulic gradient	Water tends to move from wet to dry	Capillarity – adhesive forces between water molecules and solid particles and cohesiveness between water molecules	Permeability of soil to liquid water increases with moisture content, but decreases with drying, leading to long rewetting times (although permeability of soil to vapour increases with drying, vapour transport does not occur from cooler to warmer surfaces...)
Gravity	Drainage of cable region	If it were not due to adsorption, the tendency for vapour to condense on a surface and capillarity (the 'wicking' of water from the water table through the capillary channels between solid material in a porous media to the drier region and retention of water due to surface tension) cable regions would rapidly dryout unless water constantly percolates from above due to rainfall...	Suction tension, distance to water table, permeability of soil to liquid water transport
Evaporation and vegetation	Removal of moisture from the ground surface may be exacerbated by plant and tree roots in the cable vicinity	Lowering of moisture content in cable surroundings	Rainfall, rewetting from water table
Enclosures	Prevent or slow down moisture movement between immediate cable environment and surroundings	e.g., the permeability of concrete is much lower than most porous backfill materials meaning that net moisture movement away from the cable surface must overcome higher hydraulic forces. On the other hand, if the region does dry out, rewetting will take even longer...	Permeability and sealing of cable enclosures, initial moisture content of the medium inside and outside the enclosure.

5.2 MOISTURE MIGRATION MODELLING

During the eighties, there were many papers, conferences and discussions concerning moisture migration. The Philip/de Vries-model (PdV-model) is the most generally accepted physical description of the complex processes, qualitatively described above, that constitute heat and moisture transfer. The PdV-model [45] describes the transport of water, water vapour and heat as functions of the temperature and moisture content gradients, and of gravity, depending on some external parameters, e.g., the distance to the groundwater level. The arising nonlinear partial differential equations are governed by a set of 5 diffusion coefficients, which can vary by up to several orders of magnitude in relation to the temperature and moisture content. It took KEMA some years of extensive experimental work to evaluate these parameters for only three types of sandy soil [44]. It should be noted that, in collaboration, KEMA and Heidemij performed a theoretical study and years of extensive experimental research on the subject of heat and moisture migration around power cables. Reference [44] contains the results of the physical mechanism which will lead to a dried-out zone around the power cable in sandy soil. Experiments in both the laboratory and the field showed that there is a relation between the properties of sand (retention capacity) and the critical conditions where the drying-out of sand around the power cable starts. The start of the drying out depends primarily on four factors. Namely, the heat flux generated by the power cable and the temperature of the cable sheath are also important, as well as soil properties such as the dry density and moisture content.

The subsequent discussions (e.g. during the KEMA-conference in 1985 and later) revealed that there are groups, e.g. in the USA, which do not accept the simplifying two-zone model (described below, i.e. dry soil inside an isotherm at a critical temperature rise, wet soil outside) for partially out-drying soils. Their (profound and correct) argument is that the time-dependent transport mechanisms in soil are not primarily governed by temperatures, but by the gradients and thus by the heat flux density. Indeed, some tests for the classification of soils with respect to "thermal instability" are defined in terms of a "time to drying-out" of a soil as a function of the heat flow. This may explain why the question regarding an assumed critical temperature was not answered by the USA and Canada in the CIGRE WG B1.35 questionnaire.

As mentioned, the equations that model the processes described above, usually Philips and De Vries [45], are complex, and have a number of diffusivity coefficients that are difficult to obtain, although these equations have been implemented, either analytically in a one-dimensional form, or in a FEM solver. Computation times are heavy. It is for this reason that a simplification of moisture migration is commonly used by cable rating engineers. It is described in steady-state form and in terms of a modified cyclic rating factor in the IEC standards [1][2], and has been implemented in a real-time formulation by [47]. The two-zone model assumes that the backfill or soil inside a critical distance, usually related to a critical isotherm, is dry, and outside moist.

The critical isotherm corresponds to a critical temperature rise, for a given backfill/soil, at a given density and in-situ moisture content, and its position with respect to the cables of interest will vary if moisture content or loading vary.

Brakelmann [22] and Pilgrim et al [43] have independently shown that under some conditions, and these are the conditions that are of concern for underground cable installations in adverse situations, two-zone modelling is a good approximation to the more complex reality qualitatively described in Section 5.1. Indeed, many engineers who have made thermal probe tests have noticed a dry ring around the probe when running prolonged tests at high powers, for example Figure 5.1, which shows a probe setup that utilises end insulation (that normally covers the transparent Perspex surface) to prevent longitudinal heat flux. The test samples were cleaved longitudinally in a) and b), and end insulation was removed to reveal this phenomenon after a lab test in c).

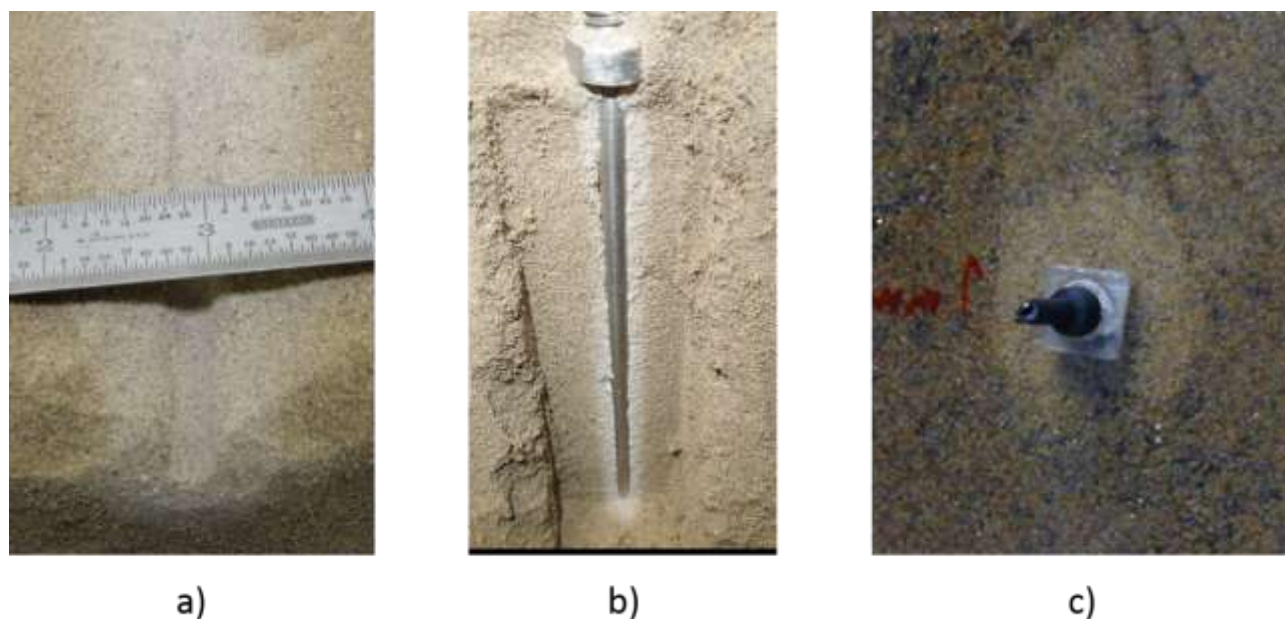


Figure 5.1 Visible drying after tests for moisture migration.

5.3 FURTHER COMMENTS AND AN ILLUSTRATION

This section will review and discuss the implications of some of the detailed measurement and analytical work done in the past regarding moisture migration, as well as present the results from a recent analysis using the thermal potential concept by a member of the working group, Professor Heinrich Brakelmann [48].

The two-zone model has been widely adopted – see the CIGRE/Electra-publications from 1986-1992 in Appendix B – because of its simplicity and manageability. Indeed, as mentioned above in Section 5.1, the Philip/de Vries-model has been employed independently in a one-dimensional analysis [22] and in a FEM simulation [43], to show that two-zone behaviour delineated by a critical isotherm may actually happen in nature, if somewhat disadvantageous soil parameters and conditions (e.g. remote groundwater level, low ambient moisture content) are present. IEC 60287 [1] has also provided a general steady state rating equation based on the two-zone model, taking into account the critical temperature rise and the thermal resistivities of the wet and dried-out soil. The same standard gives a formula for the limitation of the cable external temperature with respect to the critical temperature.

This all means that, for rating calculations based on the two-zone model, drying-out, however seldom it may happen in practice, should be taken into account during the design of the cable trench. Moisture migration may be the result of an initial thermal instability, but will limit itself by the reduced temperatures and heat flux densities at the border of the arising dry zone. Accordingly, partial drying-out should not be seen as an unlimited, disastrous process culminating in cable failure, but should rather constitute an expected transition to a limited dried-out zone, properly considered in the design phase, and which should therefore not lead to unallowable cable temperatures (“thermal runaway”). For this to be true, of course, the assumed thermal properties of the dry zone and the surrounding wet zone, i.e. the dry TR, the wet TR and the critical temperature rise, must be selected on the safe side.

The crucial advantage of the two-zone model is that it allows current rating and thermal field calculations in a quite simple way, even for complicated configurations in a cable trench, such as groups of different cables with different load cycles, other heat sources, crossings, etc. Some basic theory to this is given in [48], which shows, by means of a simplified geometric model, that restrictions of the cable losses with respect to either a critical heat flux or a critical temperature for drying-out, are likely to lead to extreme and unnecessary current rating reductions. The alternative is to take into account a limited drying-out of the soil, thus utilising the thermal limits (permissible conductor temperature) of the cables. The introduction of a so-called “thermal potential” enables the analysis of nonlinear thermal fields, even for complicated structures [48].

This is illustrated in Figures 5.2 and 5.3, where for the two-zone-model with a critical temperature rise of $\Delta\theta_{cr} = 21.7$ K (load factor $m = 0.80$), permissible currents as well as the conductor temperatures are shown. In Figure 5.2, first the rated current I_m for a homogeneous, moist soil ($\rho_m = 1$ K·m/W) without any restrictions is considered, thus giving the highest values. With the current rating I_{pd} , partial dry-out of the soil is taken into account (conductor temperature 90 °C); furthermore, the rated current I_d for a homogeneous, completely dry soil ($\rho_s = 2.5$ K·m/W) and finally the rated current I_{cr} for restricting the temperature rise at the cable surface to the critical temperature rise of $\Delta\theta_{cr} = 21.7$ K, i.e. to a cable surface temperature of $\theta_{cr} = 36.7$ °C, are shown.

The two-zone model enables the admissible cable temperatures to be fully utilised. Taking partial drying-out into account, i.e. designing the cable by means of the two-zone model, leads to permissible currents I_{pd} that are reduced by some 10...15 % compared to the currents I_m for a thermally stable, moist soil. Just the opposite happens if surface temperature is restricted to avoid moisture migration. This would result in seriously reduced currents, even lower than those which result from the assumption of a completely dried-out soil.

Figure 5.3 shows the highest conductor temperatures of the cable with partially dried-out soil, described by means of the two-zone model. It illustrates that if the cable is wrongly designed, i.e. for the thermal conductivity of the moist soil, the occurrence of moisture migration will lead to an extreme overheating of the cable (> 150 °C), i.e. to a so-called "thermal runaway".

On the other hand, if the cable is designed for the thermal conductivity of a completely dry soil, or if the cable surface temperature is restricted to the critical temperature θ_{cr} of the soil, the conductor temperatures remain lower than 50 °C, so that the thermal limits of the cable are very much under-utilised.

Key to figure 5.2 and 5.3:

soil with a critical temperature rise of $\Delta\theta_{cr} = 21.7$ K; $\rho_m = 1.0$ K·m/W; $\rho_d = 2.5$ K·m/W;

$\theta_{amb} = 15$ °C; load factor $m = 0.80$ [48]

380-kV-cable (2500 mm² copper conductor) in flat formation with distance Δs between cable centres.

The results are:

I_{pd} is the rated current for the partially out-drying soil (conductor temp. 90 °C);

I_m is the rated current for a homogeneous, moist soil (ρ_m) without any restrictions

I_d is the rated current for a homogeneous, dry soil (ρ_d)

rise at the cable surface to $\Delta\theta_{cr} = 21.7$ K; I_{cr} is the rated current for restricting the temperature

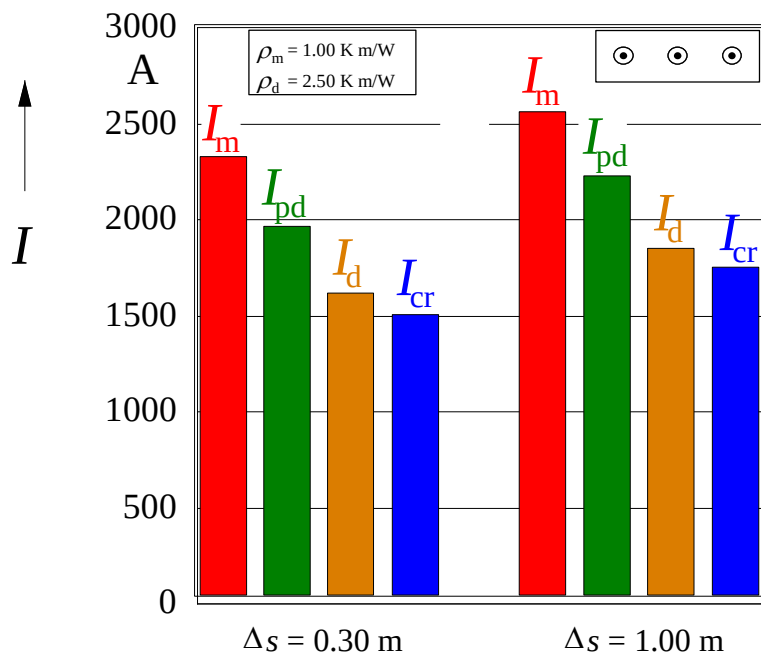


Figure 5.2 Permissible currents for different approaches

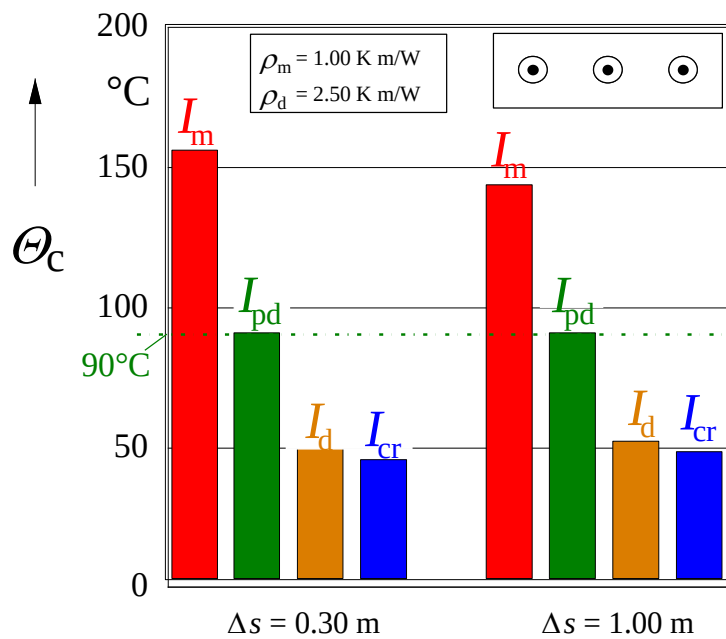


Figure 5.3 Related to Figure 5.2, but showing the highest conductor temperatures for the four different currents, considering a partially out-drying soil (see Figure 5.2)

5.4 SOME EXPERIMENTAL RESULTS

This section briefly summarizes some of the experimental results from the moisture migration work carried out by KEMA in [44], with some comparisons with [48].

Some results summarizing KEMA's field measurements are shown in Figures 5.4 and 5.5, reproduced with kind permission from [44]. With cable scale heat sources – simulating one single cable or three cables in flat formation, Figure 5.4 – temperatures and heat flux densities were measured for two sandy backfills, "sand M" and "sand K".

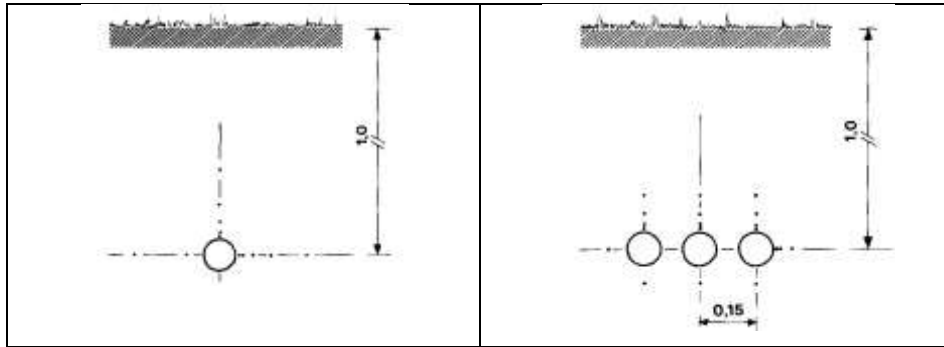


Figure 5.4 Cable test circuits from KEMA testing [44]

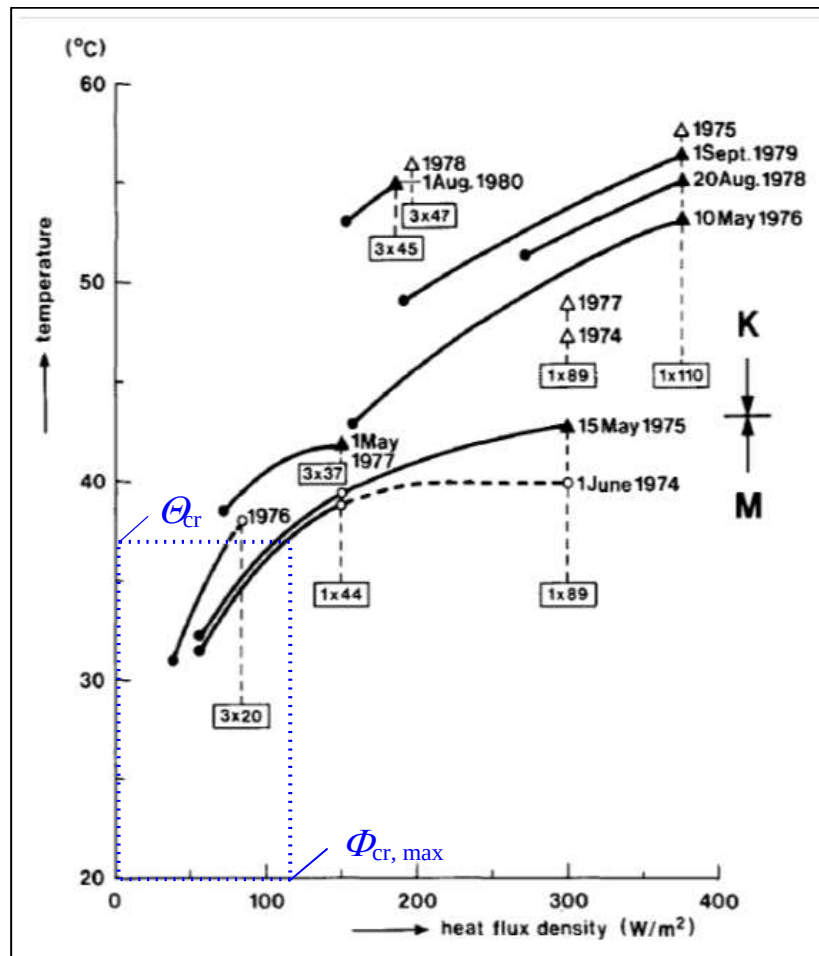
Figure 5.5 shows the drying-out conditions in sand K and M measured in the cable test circuits between 1974 and 1980, especially the conditions at the cable sheath with and without moisture migration, respectively, as well as the conditions at the boundary surface between dry and moist soil. The driving parameter was the heat dissipation of the cables which was continuous and held constant for long periods of time.

KEMA observed that drying-out around the cables will initiate at a specific heat flux density, when the cable sheath temperature exceeds a critical value. This critical temperature and the necessary heat flux density, however, are not invariable factors but depend on the initial moisture content of the soil. KEMA further noted that for the (thermally poor) sand M the critical temperatures at the cable surface and at the moist/dry boundary surface have values between 31 °C and 43 °C, corresponding to temperature rises between 19 K and 33 K. The safe limit for the heat flux density during warm and dry seasons at which no risk of drying-out is present can be derived from Figure 5.5 to be approximately 50 W/m² for sand M, and for the (thermally better) sand K, approx. 150 W/m².

Another statement by KEMA [44] is most interesting:

"The prevailing temperature in the moving boundary surface between dry and moist soil has no constant value: the temperature as well as the heat flux density decrease as the dry zone extends. This process may be attributed to the gradual decrease of the average moisture content of all the surrounding soil, as a consequence of a low precipitation level, increasing evaporation at the soil surface and/or extracting of moisture by plants."

In Figure 5.5, some information has been added (in blue) from [48]: namely the critical temperature of $\theta_{cr} = 36.7$ °C and the resulting maximum critical heat flux density of $\theta_{cr, max} = 110$ W/m² of the examples in [48]. The range of these values seems to be in good agreement with the results of KEMA, when we consider the three curves for sand M with cable losses of 3x20 W/m, 3x37 W/m and 1x44 W/m, which show critical temperature rises ($\theta_{amb} = 10...12$ °C) in the boundary surface between dry and moist soil (points ) and at the cable sheath (point ) between 19 K and 33 K respectively, as well as heat flux densities of 40...60 W/m². One must keep in mind that the sandy backfill "M" investigated by KEMA had rather poor thermal characteristics.



- ▲ condition at cable sheath when moisture migration set in
- △ condition at cable sheath in a year when no moisture migration occurred
- conditions at the boundary surface between dry and moist soil in the year when the dry zone reached its maximum extent
- cases in sand M where drying-out started immediately after the cable load was switched on [44]
- dissipation of cable(s) in W/m (continuous)

Figure 5.5 Results from [44]: drying-out conditions in sand K and M measured in cable test circuits between 1974 and 1980 (blue: additional information from [48])

Finally, the KEMA results confirm the assumption that drying-out of a soil/backfill is a self-limiting process, which can be taken into account in the design of cable installations, if the parameters of this process are chosen on the safe side.

In Figure 5.6, for a given heat flux, the onset of moisture migration is evidenced for periods with elevated soil temperatures and low preceding precipitation; the drying-out frontier being indicated by the kink in the temperature dependence. Comparing the curves for 1st Aug. '76 with 1st Sept. '76 and 10th June '77 with 5th July '77 it can be seen that in both cases the temperature and the heat flux density decrease with time as the drying-out frontier moves away from the cables. It is sobering to observe that heat flux densities as low as 35 W/m² may be sufficient for moisture migration to initiate.

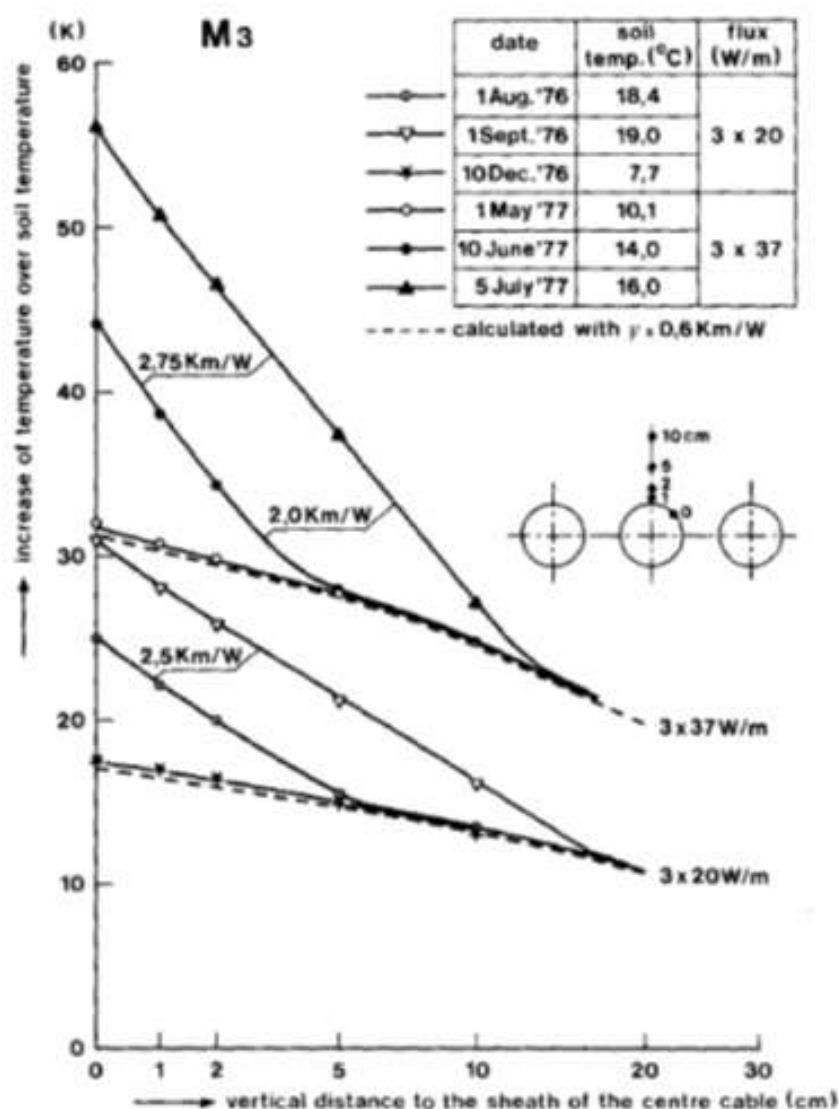


Figure 5.6 Results from [44] (20 W/m corresponds to a heat flux of 35.4 W/m²)

5.5 CRITICAL PARAMETER OBSERVATIONS

As should now be evident, when soil dries out its thermal resistivity will become much higher. The thermal resistivity of dried out sand can be easily 5 times higher than that of wet sand. Ideally, a backfill should be chosen that has low dry values of thermal resistivity and is less likely to dry out. If drying out cannot be avoided (and as the previous section suggests, it might be needlessly punitive to avoid moisture migration), moisture migration should be taken into account in the cable system design.

IEC 60287-1-1 [1], based on the two-zone model for moisture migration developed by Cox and Coates [20] and discussed in Section 5.3 of this brochure, utilises an equation with two extra parameters: 'v', the ratio of the thermal resistivity between dry and moist soil and $\Delta\theta_x$, the critical temperature rise of the soil. The critical temperature is the critical temperature rise of the soil added to the ambient temperature. This is the boundary between the dry and moist zones. The values used for the critical temperature and the thermal resistivity of the dry zone differ from country to country, and also between industries and voltages levels. For instance, the values used in the Dutch guideline [49] for XLPE insulated cables (6-30 kV) are a critical temperature rise of 30 K and a thermal resistivity of 2.5 K·m/W, whereas in Germany it depends on the daily load factor m, resulting in a temperature rise of 15 K for continuous loading ($m = 1.0$) and 32 K for

$m = 0.50$. Other countries keep the cable surface temperatures below 50 °C, presumably to avoid moisture migration, although unfortunately, moisture migration has, in certain installations, been found to occur at much lower temperatures. The critical temperature rise can be approximately ascertained by, for example, a long duration probe test (1-2h) at high losses, corresponding to emergency cable rating.

In Figure 5.7 the conductor temperature is calculated for different loads. The boundary between the dry zone and the moist zone is the 30 °C isotherm. The thermal resistivity of the dry zone is varied.

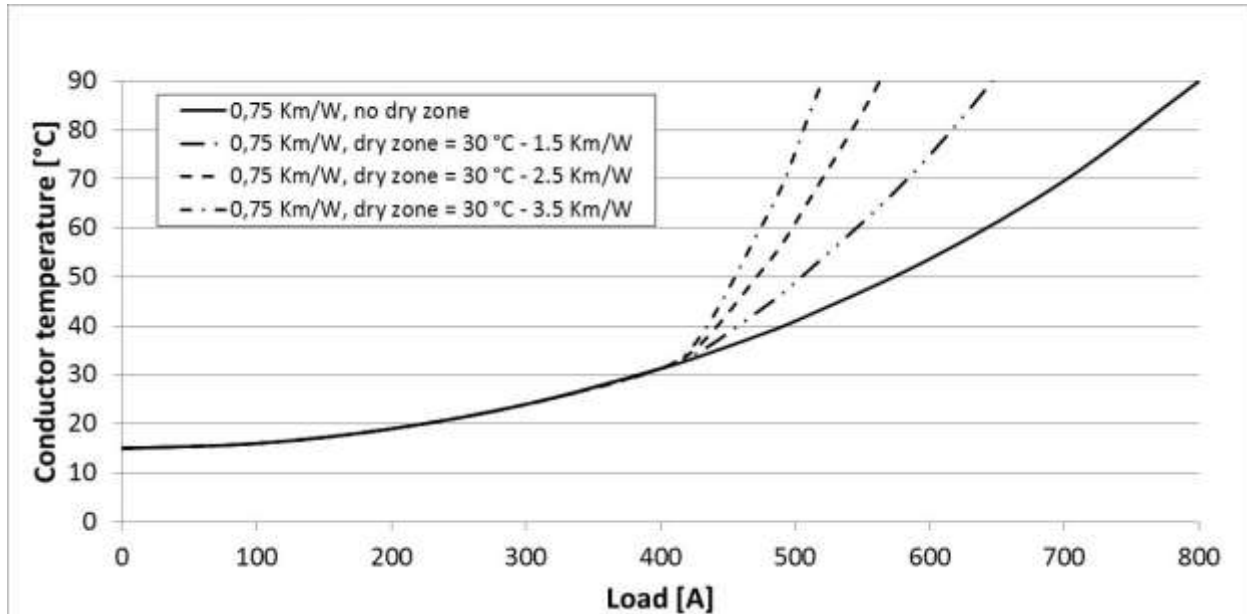


Figure 5.7 Influence of drying out of soil on the conductor temperature

There is also another definition of the critical temperature. This is the temperature where the drying-out of soil starts and the dynamic equilibrium is broken; more water will be evaporated in the neighbourhood of the cable than will flow back after condensation. This is a more complex approach and the following parameters have a significant influence on the drying-out phenomenon:

- water retention capacity
- degree of compaction
- specific minimum water content
- the heat flux density at the cable jacket

This phenomenon has been described in detail in a paper of J. Vermeer [50]. The procedure and examples are described in two paper of G. Koopmans [51] and [52]. Measurement methods required to ascertain the relevant parameters are detailed in Section 7.2.

5.6 APPLICATION OF MOISTURE MIGRATION MODELS

There are two main configurations for cable trench installation:

- A special thermal backfill is installed around the cable, and natural soil surrounds this (although this may not be quite correct, since the backfilled soil in the trench may have other properties than the surrounding undisturbed soil).
In this case, for the trench design, the dry TR of the special backfill will be assumed, i.e. it is assumed to be completely dried-out, and for the surrounding soil will be assumed to be moist.
- There is a small zone of bedding material around the cables, i.e. a sand or sand-mix, and apart from that the "normal soil". In this case, mostly the properties for the bedding material are assumed to be close to those of the soil. Conservative two-zone moisture migration modelling means that drying-out in a zone surrounding the fully loaded cables is acceptable, if it is accounted for.

6. INSTALLATION OF BACKFILLS

6.1 GENERAL

To achieve the design current rating, within the maximum normal operating temperature of the cable, the heat generated by the cables must dissipate through the surroundings.

For buried cables this may be the native soil or, as it is usually the case, the native soil together with a backfill with defined properties and additional elements relevant to the cable installation technique.

The way to install backfills around cables depends on the cable installation design and the type of backfill selected.

For underground cable system design, a route survey of the native soil is conducted for environmental, mechanical and thermal characterization. Soil boring, sampling and in-situ testing is conducted at several locations, and up to the maximum burial depth of the cables.

For native soils, their thermal and mechanical properties, such as density, critical moisture content and thermal stability, are further investigated in the laboratory. If properties are not satisfactory, as is often the case, different backfill materials can be selected. Sometimes the selection can be stipulated by the construction technique, environmental conditions, national or local authority requirements or material availability.

A short review of the backfill installation methods is given below in accordance with the cable system installation design and techniques.

Section 6.4 dedicated to the backfill compaction, one of the most important issues during the backfilling installation, but in many cases neglected or not properly done.

For further details on cable installation techniques can be found in Cigre TB194 [53].

6.2 BACKFILL INSTALLATION

Underground cables can be installed in air (using cleats), directly buried or in buried structures. Installations which generally use a backfill material are as follows:

- Directly buried
- Inside ducts or conduits, and duct banks (HDPE, PVC, steel pipes, etc.)
- Inside troughs
- In structures realized with trenchless technologies

In the following situations backfills are less likely to be utilized as the installation is generally in air:

- In tunnels, dedicated or not
- On bridges

hence this review does not consider the last two items.

6.2.1 Direct buried cables

For direct buried cables the native soil can be used as backfill, if found to be suitable and permitted by local authorities (see Figure 6.1).

Cable trench excavation is normally performed using backhoes or trenching equipment, such as chain trencher or disk cutter. The excavated native soil can be left aside the trench and, after cable laying, reinstated as the backfill. The native soil has to be properly compacted to eliminate any future settlement.

Since standard construction equipment and native soil are used, this method offers a very cost effective solution.

The limiting factors are the thermal and mechanical characteristics of the native soil. For example fine gravel with sharp edges may damage the cable outer sheath during compaction (however an additional or special cable outer sheath, or a light armour can solve such problems), or thermal stability may not be achieved if the required design level of compaction is not met.



Figure 6.1 Backfill with native soil

The re-use of the excavated soil may not be possible or allowed due to local legislation or particular restrictions. In this case the excavated material has to be disposed and a selected backfill needs to be installed. In cases where the trench runs in or across roads the backfill may be subject to special requirements (stability, strength, water permeability, etc.).

Native soil is the backfill material also in the case of cable ploughing: a wedge of soil is lifted with a ploughshare, and the cable is then guided into the plough and continually installed below this soil wedge. The soil wedge then falls back and encloses the cable. Soil compaction can be done afterwards. This type of installation is common offshore or in wetlands, and has been used for several high voltage submarine cables projects (see Figure 6.2). Onshore cable ploughing is seldom used, requiring suitable soil features and morphology (almost flat areas, no existing services to be crossed). In practice it can be used in agricultural areas.



Figure 6.2 Cable plough for near shore burial

Other techniques for offshore installation such as water jetting or suction dredging are used to bury submarine cables. The native soil will then settle with the natural sedimentation around the cable.

Most of time the thermal and mechanical properties of the native soil are unsuitable to achieve the required cable system ampacity, and/or to provide a good mechanical protection and thermal stability. A selected backfill is then installed, at least around the cables, where such properties are most important.

Historically the first type of backfill used has been sand (see Figure 6.3). A first layer of sand is created on the trench bottom, and after the laying of the cable/s the trench is backfilled with sand up to a certain level above the cable. Work crews are necessary to evenly distribute the sand, to level it and to compact it. Protective slabs and/or monitoring tapes are then laid on top of the selected backfill and the native soil is used to complete the trench backfill up to the undisturbed terrain level. A certain amount of native soil has then to be disposed of.



Figure 6.3 Sand backfill

The trench excavation requires some care to create a regular shape, reducing the amount of sand to be used up to the design volume. Sand is placed in layers (lifts) with a reduced thickness, around 15-20 cm, to be able to compact it up to the required level (See paragraph 6.4). Each layer of sand must be wetted to an optimum moisture content to facilitate proper compaction. Attention should also be paid to voids created at the edge of the cable trench when shoring boards are removed.

To improve the thermal performance, sand with well-defined grading characteristics and grain size has to be used. Such sand may be from selected sources only, or it can be obtained by mixing different types of sands or crushed materials from quarries.

In this case sand can be installed from truck mixers, but care must be taken to keep an optimum moisture content and a good distribution of the components.

To improve the thermal characteristics, and up to a certain extent the mechanical performance of the backfill, sand-cement mixes were introduced in the 1950s.

The ratio between the sand and cement is typically 14:1 by weight and so these mixes are called weak mixes or cement bound sand (CBS). A 7:1 mix may be used for a mechanically stronger material.

The installation procedure for CBS is almost the same of the sand backfill, but instead of dumpers, truck mixers have to be used to put the aggregate into the trench while maintaining a good mixing of the components (see Figure 6.4).

As for the sand backfills, the mix must be evenly distributed and well compacted

Additional materials, even polymeric, can be added to the basic components to improve selected performance characteristics (thermal stability, strength, etc.).



Figure 6.4 Cement bound sand backfill installation

If acid sulfate soils have been identified, polyethylene sheeting surrounding the cable bedding can be used to reduce the long term interaction of the sulfates in the native soil with the cement within CBS backfills, see Figure 6.5.



Figure 6.5 Polyethylene sheeting enclosing cable installation (sealed around backfill) in high acid sulfate environment.

A third type of backfill are fluidized backfills: they are a mix of sand, cement, water, fluidizing agents and other components. Cement mixers or cement pumps have to be used to pour them into the trench (see Figure 6.6). This class of backfill is self-levelling and does not require compaction, while trench excavation and preparation must be a bit more accurate, especially at the ends and where other services have to be crossed.

Time has to be allowed for backfill solidification before completing the trench reinstatement.



Figure 6.6 Fluidized backfill installation

6.2.1.1 Direct buried MV cables for renewables projects

On most projects for power production from renewables, MV cables are used to connect the generators with dedicated substations or to the grid.

The sites are often remote (rural) and total cable trench lengths in excess of 50 km are common, especially for wind-farms. Short routing, speedy installation procedures and techniques are employed to reduce the investment costs and increase project profitability. The reinstatement of the original site conditions are required to reduce the environmental impact, as well as lowering the impact on farming activities if the cable route crosses crop fields.

The cables are directly buried in native soil; laid in flat formation or in trefoil configuration. In most cases, the native soil is simply reinstalled over the cables and minimal compacting is performed to the top surface layers. This results in a poor thermal environment for the cables which should be taken into account in the system design, see Figures 6.7 and 6.8.



Figure 6.7 MV cable trench



Figure 6.8 Backfilling

Use of corrective thermal backfills has been considered on projects where hardpan or bedrock is encountered at very shallow depths, or where native soils cannot be used as a result of very high thermal resistivity or because of the presence of cobbles and boulders.

It must be considered that the loading pattern of such cables is very different from distribution cables. Historically, typical MV cable loads in urban and suburban locations follow a reasonably predictable diurnal cycle, with higher loading during the day (and perhaps peaks in the morning and evening, depending on the demand mix). MV cables from generators such as wind turbines can have very rapid transitions between zero and full load and vice versa, even daily, and long periods of sustained full load.

This aspect together with the above mentioned “low cost” installation has an important impact on the reliability of such cable systems.

6.2.2 Cables installed in ducts or conduits and duct banks inside trenches

Ducts or conduits are normally used in urban areas, offering the possibility of carrying out civil work independently from the cable installation. Also, the flexibility of cable maintenance or replacement with minimum disturbance of local traffic and economic activities are an advantage of this type of installation.

A duct bank comprises multiple ducts installed in the same trench and encased in either concrete or fluidized backfill, allowing more circuits or different services along the same route (see Figure 6.9). This solution is standard in several countries.

Ducts for cable systems may be used for short sections, for example road crossings, if restrictions from Authorities and/or planning reasons do not allow open trenches for days or weeks.



Figure 6.9 Multiple Duct bank

Ducts and multiple-duct banks can be installed by conventional open trench excavation. Ducts can also be used with trenchless technologies (see the dedicated paragraph below).

In the first case, once the trench excavation is completed, a layer of bedding material is installed on the trench bottom, usually sand or lean concrete.

Then ducts are installed with different methodologies, according to the duct or conduit material:

PVC / FRP (fibre reinforced pipe) conduits are rigid, and they are laid on the trench bottom in sections of a few metres length and then joined with couplers or using the bell ends of the sections. Curves are realized with preformed elements. Special gaskets, glue or solvent (for PVC) are used to seal the duct joints. Spacers may be used to properly separate the conduits of the cable circuit

HDPE conduits are flexible and two installation methods are possible: either in short lengths, typically 6 or 12 m for the conduits to be used for HV power cables, or in longer lengths pulled inside the trench in the same way as the cables. This last solution has the disadvantage of requiring an area for conduit section pre-assembly and the preparation of the pulling line with rollers, winch, bend limiters, etc.. However in some circumstances it is unavoidable, such as when crossing several existing services or where it may not be

possible to assemble conduits within the trench. The conduit lengths can be joined in several ways, with butt fusion, electro fusion, slip on couplings or mechanical connections. Again, spacers may be used to separate the conduits. In case of duct banks multiple conduits are used, installed in different configurations as required.

Where ducts can be used for single or double cable circuits, native soil, sand or cement bound sand can be used as backfill. It should be noted that compacting effort during installation of these backfills must be limited in order to avoid damaging the ducts. When the spacing between ducts is tight (less than 100 mm) it may not be possible to use native soil or a granular type backfill to achieve a void free installation; the only option may be to use a fluidized backfill (See Figure 6.6). However in the case of road crossings, where very good mechanical strength is required, concrete is generally used.

In this case structural concrete is used. This provides good protection and stability of the whole duct system. The ducts are installed with spacers, to maintain their relative position, and anchored to avoid the buoyancy effect when the concrete is poured.

For long ducts no grout is used for the space between the cable outer sheath and duct inner wall. The reduced cable thermal properties (air in ducts) are offset by the future options to replace the cable in case of repair or upgrading.

In some markets, once the cable is installed, a special grout, fluid enough to fill the gap between the cable and the duct wall, is used to provide a favourable thermal environment to the cable system.

The grout is pumpable and is selected in accordance with the duct length involved and the duct characteristics (diameter, bends, profile, pumping pressure, volumes of the batches to be pumped, etc.). Basically the grout can be a bentonite based mix, with sand and water, or a cement based mix, again with sand and water. Additives may be used to increase pumpability and/or mix stability, avoiding the sedimentation of the components and allowing a uniform consolidation without shrinkage. Ducts filling must then be very accurately designed for the injection phase, which may last several hours.

From a practical point of view, it is necessary to make a calculation of volume and verify that the injected grout exactly matches the calculated volume, otherwise a leak or a "big bubble" of air has to be suspected. A good air and water purging is essential for a homogeneous backfill.

After the injection both ends of the duct are sealed to prevent moisture / water loss.

For short ducts, for example road crossing in between two stretches of directly buried cables, the gap between the cable and the duct wall may be grouted depending on local practice.

6.2.3 Cables installed in steel pipes

Steel pipes have a limited use for cable installation, essentially being selected where there is a special requirement, for example road/railway crossings or bridges, or where the pipes are the bore casings of trenchless installation methods. The main advantage of steel pipes such as protection and ruggedness is offset by a potential of corrosion, increased costs and eddy current losses.

The backfill around steel pipes can be fluidized backfill or granular thermal backfill (typical in the case of open trench installations), or native soil plus the drilling mud in the case of bore casings. Inside the steel pipes, the cables may be installed with or without plastic inner ducts. Most installations will utilize inner plastic ducts. The space between the steel pipe and duct is usually filled with grout although it can be left empty.

Granular type backfills such as 'stone-dust', 'thermal sand', 'cement-bound-sand' were commonly used as the backfill. A 150-200 mm thick layer of this select backfill is laid at the bottom of the excavated trench, and compacted at the specified density and moisture content. The cable pipe(s) are laid on top of this layer and spaced appropriately. The trench is then backfilled with the select material to the specific height. In order to meet the specified thermal and mechanical parameters, the material is installed in thin layers not exceeding 200 mm. Appropriate equipment such as small vibratory roller or plate-type compactors (jumping-jack) are used to compact each layer. During the installation phase it is important to conduct quality assurance testing of the backfill material for its gradation characteristic, density, moisture content and thermal properties.

6.2.4 Pipe type cables

A particular case of the above is represented by the pipe type cables, where the steel pipe is an integral part of the cable system, representing the metallic sheath of the cable system.

These high voltage cable systems are quite common in North America (USA, Mexico and Canada) and have been used for high voltage lines in Europe (Germany, UK, France and Russia) as well as Asia (Japan and Taiwan). Basically the two types of pipe-type cable systems are High Pressure Fluid filled (HPFF) and High Pressure Gas Filled (HPGF). Three phase cable systems with insulation made of impregnated paper or paper polypropylene laminate are contained in steel pipes of various sizes (150-250 mm diameter are common).

Pipes of standard lengths are welded together, bent and inspected prior to installation. Pipes are coated with special insulation materials such as mastic, polymer-concrete, or extruded polyethylene for protection against abrasion and damage during installation as well as to electrically insulate the pipe to facilitate cathodic protection and prevent corrosion.

Unlike other cable systems, HPFF pipe-type cables have a rather smooth distribution of the temperature throughout the route, minimizing the presence of "hot spots". This is possible thanks to the circulation of the insulation fluid through the oil feeding plant and heat exchangers or by shuttling fluid back and forth in an oscillating operation.

The most common type of installation is in trenches where a single or multiple pipes (each pipe containing all three phase conductors) are installed in selected backfill material.

If the native soil thermal parameters are acceptable and if it can be installed without causing any damage to the pipe insulation, native soil can be used as the backfill. However, in most cases this is not done because of the impracticality of storage and reinstatement of the native soil in its original state or because selected backfills permit higher ratings.

The backfill material is the same of for steel pipes, see Figure 6.10.



Figure 6.10 Fluidized backfill poured over a pipe type cable

6.2.5 Cables installed inside troughs

A trough is a (generally) prefabricated U-shaped covered housing, see Figure 6.11. The cable route is built installing the trough sections end to end.

Where troughs are used for cable installation, mechanical protection is provided by the trough walls and reinforced concrete lids, so there is no need for a backfill with particular mechanical properties. The trough backfill can be sand, weak mix or fluidized backfill.



Figure 6.11 Cable Trough

In the case of filled troughs the filling is installed up to the trough cover for the best thermal performance.

6.2.6 Cables installed with trenchless technologies

In case of difficult soil conditions or environmental, construction or regulatory constraints (e.g. crossing of railways, highways, rivers, coastal sand dunes) conventional open trench techniques may not be used.

A number of construction techniques are available to build the cable route, from dedicated small bore to medium or large tunnels, without effecting the ground surface. These techniques can be grouped under the definition of Trenchless Technologies. It is possible to include among them Horizontal Directional Drilling (HDD), pipe jacking, micro-tunnelling with tunnel boring machine (TBM), auger boring, pipe ramming, impact moling and pilot tube.

Generally ducts or pipes are pulled inside the bore and the cables are subsequently pulled inside the ducts or pipes.

The considerations for the grout between cable outer sheath and duct inner wall are those outlined in Sections 6.2.2 and 6.2.3. In most cases, this annulus is not filled-in with grout because of the difficulty in achieving a void free installation. If the length of the ducts is short (less than 100m) and if the difference in the diameter of the conduit and cable is large enough, filling may be possible.

For the backfill outside the cable duct, each technique brings particular aspects to be considered.

The Horizontal Directional Drilling (HDD) technique is widely and commonly used. The soil is first displaced by the drill string. In general, a mixture of water, bentonite and additives is used during the drilling process to lubricate and to stabilize the walls of the pilot hole; at this stage the drill is left in the ground. In a second step, the pilot hole is enlarged with a back-reamer which is attached to the drill. More than one back reaming operation may be carried out, with the size increasing each time. In the final stage an outer casing, possibly containing a number of internal ducts, is installed by pulling behind the reamer. Drilling mud is again used for lubrication and stabilization. Cables are then pulled into the casing or ducts.

The knowledge of the actual condition of the soil around the outer casing, and hence its properties and stability, is quite difficult. Depending on the type and strength of the native soil (granular or cohesive, soft, firm or stiff) and factors such as the difference between the bore diameter and that of the casing, type and

consistency of the drilling mud, etc., the geometry, mechanical and thermal characteristic of the 'soil layer' around the case will change.

If cable ducts are installed inside the casing, the space in between can be filled with sand/cement grout. This grout must be designed to meet specific parameters such as compressive strength, rate of hardening, heat of hydration, thermal properties, etc. The installation must be voids-free.

Where HDD is used for landings of submarine systems, the cables are pulled inside the casing and generally no backfill fills the annular space. Filling with grout can only be done when the end nearest to the sea is accessible, for example at low tide. The ends are sealed after filling with grout. The selection of thermal grout is based on criteria similar to those for any HDD installation.

Tunnel boring machines, microtunnel equipment, auger boring and pipe ramming do not displace the soil around the bore. The excavated material is removed back to the drive shaft or pit. Over time the native soil is then settling back around the bore casings made of steel, HDPE or reinforced concrete.

Concrete can be used to fill the volume inside the bore casings in the case of short HDD installations or, for long lengths, sand/cement grout can be used.

Both impact moling and pilot tube techniques displace the material around the bore, with a local change of soil density and moisture. Again it is difficult to fully determine the extent of the changes, as it depends on native soil characteristics and bore size. Both techniques are used for small and relatively short bores.

Casings for cables may also be installed with recently introduced 'semi-trenchless' techniques, where the soil excavation is very limited and the backfill around the installed casing is the compacted native soil.

6.3 BACKFILL INSTALLATION FOR SUBMARINE CABLES

6.3.1 Offshore sections

In the past, off shore sections of submarine cable systems were simply laid on the seabed, with some exceptions. Today the design of all submarine systems, based on accurate desk top studies and detailed marine survey, foresees a complete protection of the cables. Unprotected sections are in general limited to areas of very rough seabed conditions or in deep waters.

Cable burial is the common protection method. Various equipment exist to bury submarine cables, such as plough, water jetting tools, trenchers, disc cutters or a combination of them. In all cases, with the natural sedimentation, the native soil becomes the backfill material.

In some specific sections or cases the cable is protected with a rock berm or concrete/tar mattresses. The rock berm is made of selected crushed rock, being the main driver for the berm design the cable protection against fishing tools or anchors. Concrete/tar mattresses are used as well for protection, especially at crossings with existing services. Both systems are used also for cable stabilization in areas with strong sea currents.

It is common practice to carry out TR measurements as part of the seabed route survey. TR values found are typically between 0.7 and 1.0 K·m/W. These values compare with values from IEC 60287-3-1 of 0.7 K·m/W for "very moist soil" and 1.0 K·m/W for "moist soil". The low and fairly constant water temperature of the offshore section provides a good environment for the cables.

6.3.2 Near shore and landing sites

The landing sites are the most demanding areas as far as thermal conditions of the cable are concerned (high temperatures, high burial depth and, in general, soil high thermal resistivity) and so they are the main driver for submarine cable design (significant changes may be encountered in case of submarine cables to offshore platforms).

The landing sites approach in terms of design issues and hence as backfilling can be split in two, having the Sea Land Joint (S/LJ) pit, as the boundary. Usually the S/LJ is placed as close as possible to the water mark.

For the land cables up to the S/LJ pit, the standard land cable approach is used most of the time, considering the trench design with backfilling, mechanical protection, etc. In some cases the mechanical protection is different to the standard one according to actual soil and environmental conditions.

From the S/LJ towards sea, a case by case design is used according to several factors, such as morphology of the area (sandy beach, rocky area), the length of the route from the S/LJ to the position where a cable laying vessel can stay safely for the laying operations, the slope of the area, sea currents, etc.

The near-shore part of the route of submarine cables as well as the transition where submarine cable(s) come on land and reach the S/LJ, are critical zones where significant fluctuations in the water level (tidal) can have detrimental effect on the installation. Wave action, ice build-up, iceberg scouring, tidal currents, build-up and movement of sand bars, organic and marine life growth, vessels traffic etc. are the main factors to be taken into account.

Based on both previous practice and studies, some key points are always considered beforehand:

- Most important is the mechanical protection of the cables: the landing points are always selected to be, as far as possible, sheltered from waves, currents and away from dangerous human activities.
- Second issue, which is becoming more and more relevant over time, is the reduction of possible impacts to on shore or near shore activities, such as leisure activities, fishing, or natural features (reserves, bird nesting areas). Usually this not only constrains the time window for construction and repair activities, but also to specific (and often, from construction view point, unfavourable) periods of the year.
- Third issue is the repair constraints: while faults in land section may have, in general, permits issues only, faults in the marine part shall have also severe weather issues, with the need of performing the works in a restricted timeframe.

For the above reasons the use of special backfill in the near shore and landing site is not common at all. If the cables are installed in trenches the native soil is used as backfill only if it meets the thermal requirements. If there is high risk of washing out of fines as well as infiltration of organic fines into the backfill, the trenches can be lined with filter-fabric (geotextile) to encapsulate the installation and help minimize the risk. Sometimes the cables have an additional protection of cast iron shells, cement bags or mattresses, and the backfill is left to native soil settling.

However trenchless technologies such as HDD are used commonly to install casings across the tidal area from a point below the low water mark-up to the S/LJ. The reduced environmental impact of the HDD works, the good protection provided, and the possibility of planning the cable installation works with fewer constraints and in a safer and faster way, gives to the trenchless technologies a great advantage to the conventional ones.

The casings are usually polyethylene pipes, to avoid the issues of corrosion and eddy current losses of steel. The pipes start as close as possible to the S/LJ (or, vice-versa, the S/LJ is located where the HDD works start on the land side), and end below the low water mark. The methodology used is similar to HDD works on land, with some additional care used to control and avoid any pollution at sea, see Section 6.2.6.

Cables are pulled inside the casings and generally no backfill fills the annular space. In some cases the last part of the pipe could be in air, for example if the S/LJ is made where there is no permanent water, worsening the thermal conditions around the cable. If such length is significant to the point of impacting the entire submarine cable design, special arrangements can be considered to improve the situation. This may include filling the gap between the cable sheath and the pipe with a grout of high fluidity and low thermal resistivity. In this case sealing of the pipe-end is necessary. The most common method is to create a plug using high-strength, non-shrink concrete. If at high tide the pipe is fully filled, the sealing can be closed at that time, preventing the sea water to flow back. Otherwise the pipe can be filled with bentonite based grout and then sealed. The selection of the grout and the installation is not easy, since it must be taken into consideration the possibility to remove the cable in case of replacement, and the risk of grout flowing out of the pipe sea end, with the resulting water pollution.

6.4 BACKFILL COMPACTION

In general, soil and backfills can be described as composite materials made up of solid particles (soil grains), air voids and water (in liquid or vapour form). Depending on the quantity of water in a soil/backfill mass, it is considered wet, moist or dry. From the mechanical viewpoint, soils with high moisture content have low

bearing capacity (soft) and are prone to settlement and shrinkage. The case of soils with a high void content is similar.

From a thermal viewpoint, soil grains have a thermal resistivity of 0.12 to 0.5 K·m/W, water is ~ 1.7 K·m/W (considering no movement) and air is ~ 40 K·m/W. Therefore, in order to lower the thermal resistivity, the increase of the density of a material with removal of air is most important: Since most of the heat conduction is through the soil mineral particles and their contacts, one must ensure a soil mixture that maximizes these contacts - that is, a high density soil. Well graded soils are denser because smaller grains can efficiently fill the spaces between the larger particles, increasing the density which results in improved thermal and mechanical performance.

Compaction is a process (static, dynamic or hydraulic) for improving the density of soil or other construction materials. Appropriate compacting equipment is selected depending on the type of backfill (native soil or granular type) and trench dimensions, see Section 6.5.

A backfill reaches a maximum density at specific optimum moisture content; as determined in the laboratory by the standard Proctor compaction test (see Section 7.2.6). At higher or lower moistures a lower density is attained. This behaviour is indicated by the standard Proctor laboratory test as shown in Figure 6.12.

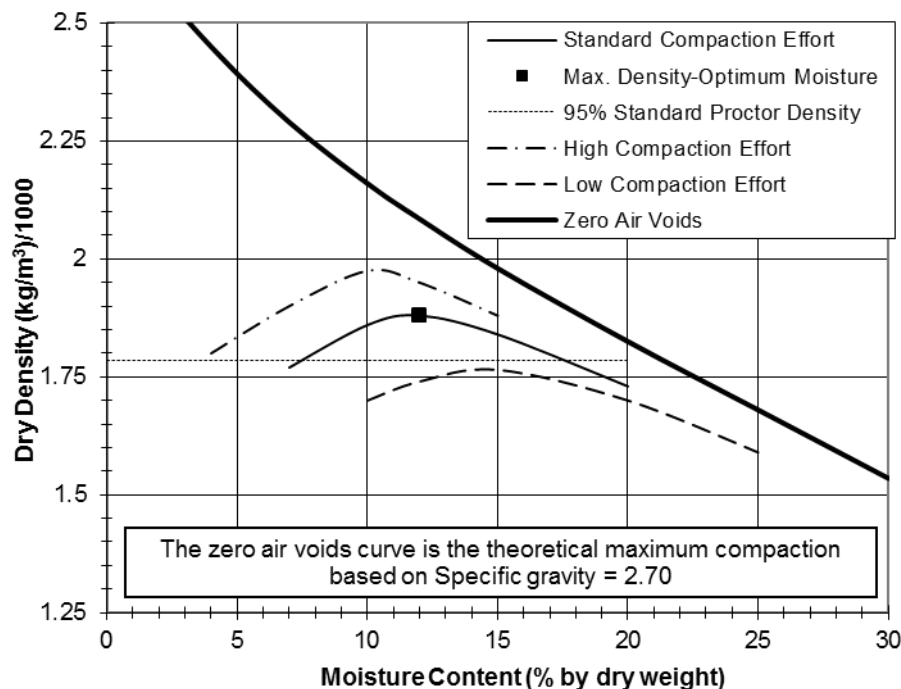


Figure 6.12 Effect of moisture content on ease of compaction

Once the density reaches an optimum level a further increase in moisture reduces the ability to compact the soil and hence reduces the final density that can be achieved.

For most of underground power cable applications, the native soils (if used) are installed at the 'field in-situ' moisture content and at $\sim 95\%$ of the maximum dry density, obtained in the Standard Proctor test.

Corrective backfill materials are normally installed at the 'optimum' moisture content and at 95% of the maximum dry density. Use of a selected backfill at this density will demand extra care during installation around cables/accessories due to possible damage. Techniques used for reinstatement during road repairs may not be suitable. One possibility is to use a 'fluidized' backfill with similar thermal mechanical properties to minimise such problems.

A good thermal compacted backfill is specified by a narrow range of grading based on a sieve analysis, and should have a wet thermal resistivity of 0.35 to 0.45 K·m/W and a dry resistivity of 0.90 to 1.10 K·m/W.

Although the cost of a properly designed and installed thermal backfill is somewhat higher than using unclassified material (i.e. native soils, bedding sands), cable rating increases are significant and can amount to substantial revenue gains for a utility over the life of a cable. It is usually not cost effective to modify the native soil by using additives to improve its thermal properties.

Corrective backfills must meet several criteria, including thermal efficiency and engineering performance compatible with road base material (mechanical support, no settlement, resistance to erosion and frost heave).

Following the above considerations it is clear that proper installation of a granular backfill is crucial to meet the above criteria. In order to meet the required density, contractors must use suitable compaction equipment, ensure optimum moisture, compact in thin lifts, and use several passes over the same area.

Thermal degradation in sand backfill has often been evident. In most cases, this could have been minimized or eliminated if the backfill was installed at near optimum moisture content and at higher density.

The civil/construction industry has adopted certain standards and guidelines for installation of materials for long-term performance. For example, the foundation of a building has to be strong enough to carry the design load without any future settlement and the base and sub-base materials under streets and highways must be able to carry heavy loads without deformation, settlement or erosion.

Similar performance is expected of the backfills used to bury underground utilities (gas, water, sewer, power, telecom, etc.). In cold climates, these backfills have to perform well under freeze-thaw cycles as well. In all these applications, the key factors for long-term and stable performance are the use of appropriate materials and to follow procedures for quality installation.

6.5 SOILS/BACKFILLS COMPACTION TOOLS AND METHODS

The compaction devices commonly used for trench backfilling are plate vibrators, vibrating rollers and dynamic impactors (jumping jacks). The vibratory type's compactors are more efficient with granular backfills while the impact types are more effective with clayey soils. Small compactors such as plate vibrators or jumping jack are easy to manoeuvre and achieve good compaction in narrow areas, and therefore are preferred over larger roller types that tend to bridge over loose spots at depth. Flooding and hydraulic filling will not yield the same result as backfill may segregate. Ramming or drop weights are not acceptable because of poor control and the possibility of damage to cables.

Compacting in thick layers may be faster but it leaves low density fill material beneath a dense crust. This is why rigorous inspection is required. For smaller, trench-type compactors the lifts should not exceed 150 to 200 mm, especially close to the cables where more care and effort are needed to ensure maximum density. Thicker layers may be permitted as the backfill reaches the ground surface. Usually stock-piled backfill is not at the optimum moisture content. Generally it is too dry, and moisture must be mixed in or sprayed on each lift before compacting. For granular, non-cohesive soils the optimum moisture is about 8 to 12% and the soil appears quite wet. For a compaction of ~95% of the maximum dry density, obtained in the Standard Proctor test, the dry density could be between 1800 and 2000 kg/m³.

Depending on the soil/backfill material (its particle size distribution, porosity, etc.) the optimum compaction level could be achieved by using a different compaction effort (e.g. 2 passes of a compactor with vibrator plate could be enough for some backfills and several could be necessary for other). This is another reason to investigate the behaviour of the corrective material prior to its use.

7. TEST METHODS FOR BACKFILL PROPERTIES AND STABILITY

This chapter summarises the standards, most used technologies and techniques for measuring and determining the thermal properties of soils and backfills. Materials were also assessed in this chapter for their suitability for use as thermally stable backfills.

As described in paragraph 4, thermal behaviour of a soil is dependent on the following physical properties:

- the composition of the soil (type of particles)
- the grain size distribution
- the moisture content
- the dry density

To evaluate a backfill which can be used in power cable design and engineering studies, the physical properties mentioned above should be known.

During the year the thermal resistivity can change (e.g. variation of moisture content), the designer should be aware and take into account this phenomenon by measurement or calculation of the thermal properties of the backfill.

7.1 STANDARDS AND STANDARDS ORGANIZATIONS

This section summarizes a selection of internationally accepted standards. There are additional national standards which are often derived (partly) from the international standards. There also company standards which describe soil surveys or measurements, some are mentioned in the chapter.

One should understand that the various standards are not aligned. Particularly for definitions and classifications there could be differences between standards. For instance the definition of fines: the ASTM standard defines fine particles (silt and clay) as particles smaller than 75 μm but the ISO standard defines fine particles as particles smaller than 63 μm . When presenting the results of a measurement or recommendations for a certain backfill material one should also specify the standard which is used for the measurement or recommendation to prevent any misinterpretation.

ISO 14688 Geotechnical investigation and testing - Identification and classification of soil

The standard consists of two parts the ISO 14688-1 and ISO 14688-2. The standard describes the soil identification and classification based on the material and mass characteristics of the most common soil for use in the field by an experience person.

Part 1 of the standard describes the identification and description of the soil. For the identification, it contains a flowchart and definitions of the particle size fractions. Further it covers how to deal with composite soils, plasticity, organic content and peat. The standard also prescribes methods for identification and description of soil and the minimal information requirements for reports.

Part 2 of the standard describes the classification of soils. The classification principles established in the standard of grouping soils into groups with approximately the same composition. The soils can be classified according to the particle size distribution (grading), plasticity, organic content and genesis, independent of the moisture content and compactness of the soil. Also, some other principles of classification are described and in the appendix, is there an example of classification of soils, based on the particle size distribution (grading).

ISO 17892 Geotechnical investigation and testing - Laboratory testing of soil

The ISO 17892 covers several laboratory tests of soils. The test of the different soil properties are describes in separate parts of the standard, e.g. the determination of the water content is describes in ISO 17892-1 and the determination of particle size distribution is describes in ISO 17892-4. The structure of the standards is comparable. First, a section about the terms and definitions then a section with the used equipment in the

tests and the test procedure, followed by the test results and test reports. In the appendix are the calibration, maintenance and checks describes with sometimes some additional information.

IEEE 442 Guide for soil thermal resistivity measurements

This guide covers the measurement of soil thermal resistivity. A thorough knowledge of the thermal properties of a soil will enable the user to properly install and load underground cables.

The purpose of this guide is to provide sufficient information to enable the user to select useful commercial test equipment, or to manufacture equipment which is not readily available on the market, and to make meaningful resistivity measurements with this equipment. The designs for both laboratory and field thermal needles are described in this guide.

The thermal needle probe or 'transient thermal needle' method described in the Guide is the one most frequently employed. It is versatile and gives results of high accuracy in a relatively short time frame (few minutes).

EN 933-1 Tests for geometrical properties of aggregates - Part 1 Determination of particle size distribution - Sieving method

This standard describes the method and procedures to perform a sieve test to determine the particle size distribution.

EN 933-2 Tests for geometrical properties of aggregates - Part 2 Determination of particle size distribution - Test sieves, nominal size of apertures

This standard specifies the size and shape of the apertures

EN 1097-5 Tests for mechanical and physical properties of aggregates - Part 5 Determination of the water content by drying in a ventilated oven

This standard describes the method with an oven to determine the water content of a soil in the laboratory.

EN 1097-6 Tests for mechanical and physical properties of aggregates - Part 6 Determination of particle density and water absorption

This Standard specifies different methods for the determination of the particle density and water absorption of among others soil.

EN 1097-10 Tests for mechanical and physical properties of aggregates - Part 10: Determination of water suction height

Reference test methods for determining the water suction height of an aggregate in direct contact with a free water surface.

EN 13286-1:2003, Unbound and hydraulically bound mixtures - Part 1 Test methods for laboratory reference density and water content - Introduction, general requirements and sampling

This standard describes the requirements of the laboratory test to determine the density and water content.

EN 13286-2 Unbound and hydraulically bound mixtures - Part 2 Test methods for laboratory reference density and water content - Proctor compaction

This standard describes the proctor test, which gives the maximal density the soil can be compacted by the optimal moisture content.

EN 197-1 Cement Part 1 Composition, specifications and conformity for common cements

This standard covers the main constituents of cement, the composition and notation of the cement, the requirements of cement regarding mechanical, physical, chemical and durability properties and the conformity criteria of cement. In the appendix, a table is inserted with the used CEM Cement types per CEN member countries.

EN 206 Concrete; specification, performance, production and conformity

This standard covers the classification, requirements and specification of concrete as well the verification methods for the classification. Further it contains the production and delivery of concrete and the conformity control and conformity criteria.

EN 450-1 Fly ash for concrete part 1 definition, specifications and conformity criteria

There are different kinds of fly ash depending on the type of coal and type of boiler used in the coal power plant. This standard covered siliceous fly ash for used as type II addition for concrete produced conform the standard EN 206. This standard describes the requirements for the chemical and physical properties and quality control procedures of the fly ash

ASTM C177 Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded-Hot-Plate Apparatus

This test method covers the measurement of heat flux and associated test conditions for flat specimens. The guarded-hot-plate apparatus is generally used to measure steady-state heat flux through materials having a "low" thermal conductivity and commonly denoted as "thermal insulators." Acceptable measurement accuracy requires a specimen geometry with a large ratio of area to thickness.

ASTM D653 Standard Terminology Relating to Soil, Rock, and Contained Fluids

Unit Weight Determination

ASTM D698 Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort

Moisture-Density Relations of Soils and Soil-Aggregate Mixtures Using 5.5 lb (2.49 kg) Rammer and 12 in. (305 mm) Drop

ASTM D854 Standard Test Methods for Specific Gravity of Soil Solids by Water Pycnometer

These test methods cover the determination of the specific gravity of soil solids passing a sieve by means of a water pycnometer.

ASTM D1452 Standard Practice for Soil Exploration and Sampling by Auger Borings

Soil Investigation and Sampling by Auger Boring

ASTM D1556 Standard Test Method for Density and Unit Weight of Soil in Place by Sand-Cone Method

Density of Soil In Place by the Sand-Cone Method

ASTM D1557 Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort

Moisture-Density Relations of Soils and Soil-Aggregate Mixtures Using 10-lb (5.54-kg) Rammer and 18-in. (457.2mm) Drop

ASTM D1586 Standard Test Method for Standard Penetration Test (SPT) and Split-Barrel Sampling of Soils

Penetration Test and Split-Barrel Sampling of Soils

ASTM D1587 Standard Practice for Thin-Walled Tube Sampling of Fine-Grained Soils for Geotechnical Purposes

Thin-Walled Tube Sampling of Soils

ASTM D2166 Standard Test Method for Unconfined Compressive Strength of Cohesive Soil

Compressive Strength, Unconfined, of Cohesive Soil

ASTM D2216 Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass

Water Moisture Content of Soil, Rock, and Soil-Aggregate Mixtures

ASTM D2487 Standard Practice for Classification of Soils for Engineering Purposes

Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)

ASTM D2488 Standard Practice for Description and Identification of Soils

Description and Identification of Soils

ASTM D2937 Standard Practice for Description and Identification of Soils

Density of Soil In Place by the Drive-Cylinder Method

ASTM D2974 Standard Test Methods for Moisture, Ash, and Organic Matter of Peat and Other Organic Soils

Moisture, Ash, and Organic Matter of Peat and Other Organic Soils

ASTM D4220 Standard Practices for Preserving and Transporting Soil Samples

Preserving and Transporting Soil Samples

ASTM D4318 Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils

Liquid Limit, Plastic Limit, and Plasticity Index of Soils

ASTM D4611 Standard Test Method for Specific Heat of Rock and Soil

Specific Heat of Soil and Rock

ASTM D4643 Standard Test Method for Determination of Water (Moisture) Content of Soil by Microwave Oven Heating

Water (Moisture) Content of Soil by the Microwave Oven Method

ASTM D4648 Standard Test Methods for Laboratory Miniature Vane Shear Test for Saturated Fine-Grained Clayey Soil

Laboratory Miniature Vane Shear Test for Saturated Fine-Grained Clayey Soil

ASTM D4914 Standard Test Methods for Density of Soil and Rock in Place by the Sand Replacement Method in a Test Pit

Density and Unit Weight of Soil and Rock in Place by the Sand Replacement Method in a Test Pit

ASTM D5334 - 14 Standard test methods for determination of thermal conductivity of soil and soft rock by thermal needle probe procedure

This test method presents a procedure for determining the thermal conductivity (λ) of soil and soft rock using a thermal needle probe. This test method is applicable for both intact and reconstituted soil specimens and soft rock specimens. It is suitable only for homogeneous materials.

The test procedure is applicable to dry or unsaturated materials over temperatures ranging from <0 to $>100^{\circ}\text{C}$, depending on the suitability of the thermal needle probe construction to temperature extremes. However, care must be taken to prevent significant error from: (1) redistribution of water due to thermal gradients resulting from heating of the needle probe; (2) redistribution of water due to hydraulic gradients (gravity drainage for high degrees of saturation or surface evaporation); (3) phase change of water in specimens with temperatures $<0^{\circ}\text{C}$ or $>100^{\circ}\text{C}$. These errors can be minimized by adding less total heat to the specimen through either minimizing power applied to the needle probe and/or minimizing the heating duration of the measurement.

ASTM D5856 Standard Test Method for Measurement of Hydraulic Conductivity of Porous Material Using a Rigid-Wall, Compaction-Mold Permeameter

This test method applies to one-dimensional, laminar flow of water within laboratory-compacted, porous materials such as soil. The hydraulic conductivity of porous materials generally decreases with an increasing amount of air in the pores of the material. This test method applies to porous materials containing little or no air. The test method is designed to minimize the amount of air in the test specimen. However, this test method does not ensure complete saturation of the test specimen with water.

ASTM D6836-2 Standard Test Methods for Determination of the Soil Water Characteristic Curve for Desorption Using a Hanging Column, Pressure Extractor, Chilled Mirror Hygrometer, and/or Centrifuge

The standard describes five methods (A-E) for determining the soil water characteristic curve. These test methods cover the determination of soil water characteristic curves (SWCCs) for desorption (drying). SWCCs describe the relationship between suction and volumetric water content, gravimetric water content, or

degree of water saturation. SWCCs are also referred to as soil water retention curves, soil water release curves, or capillary pressure curves.

ASTM D6913 Standard Test Methods for Particle-Size Distribution (Gradation) of Soils Using Sieve Analysis

Test Methods for Particle-Size Distribution (Gradation) of Soils Using Sieve Analysis

ASTM D6938 Standard Test Methods for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)

The test method described is useful as a rapid, non-destructive technique for in-place measurements of wet density and water content of soil and soil-aggregate and the determination of dry density.

ASTM C136 Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates

Sieve Analysis of Fine and Coarse Aggregates

ASTM C150/C150M Standard Specification for Portland Cement

The specification covers eight types of Portland cement.

ASTM C618 Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete

Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete.

ASTM C177 Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded-Hot-Plate Apparatus

This test method covers the measurement of heat flux and associated test conditions for flat specimens. The guarded-hot-plate apparatus is generally used to measure steady-state heat flux through materials having a "low" thermal conductivity and commonly denoted as "thermal insulators."

7.2 MEASUREMENT METHODS OF PHYSICAL PROPERTIES

7.2.1 Test instruments for thermal measurements

Although the guarded hot-plate, or Lee's disc, method (ASTM C177) is regarded as the most accurate method of determining thermal resistivity, it has limitations when it comes to testing materials in different forms, at different densities and with different moisture contents and cannot be applied in the field (in-situ). The transient thermal probe (thermal needle) method is much more practical and is equally applicable for field and laboratory measurements. Hence the latter method is used almost universally for this purpose.

There are commercially available test instruments, for which details are given in IEEE 442, which use the transient thermal probe method.

Typically, these test instruments automate the measurement of thermal resistivity (or conductivity), specific heat and thermal diffusivity.

They provide programmable power to the thermal probes, read temperature sensors, heater current and voltage, and compute the thermal parameters in real time. Such an instrument, which is used in conjunction with a laptop computer, and a range of thermal probes are shown in Figure 7.1.



Figure 7.1 Automatic transient thermal needle method instrument, laboratory test samples and thermal probes

Further details of methods to determine thermal resistance are given in Section 7.3 below.

7.2.2 Grain size distribution (Sieve analysis)

Sieve analysis is the determination of the various size particles within the soil. This is a laboratory test. The grain size distribution is determined through a mechanical sieve analysis. See Figure 7.2, for an example.

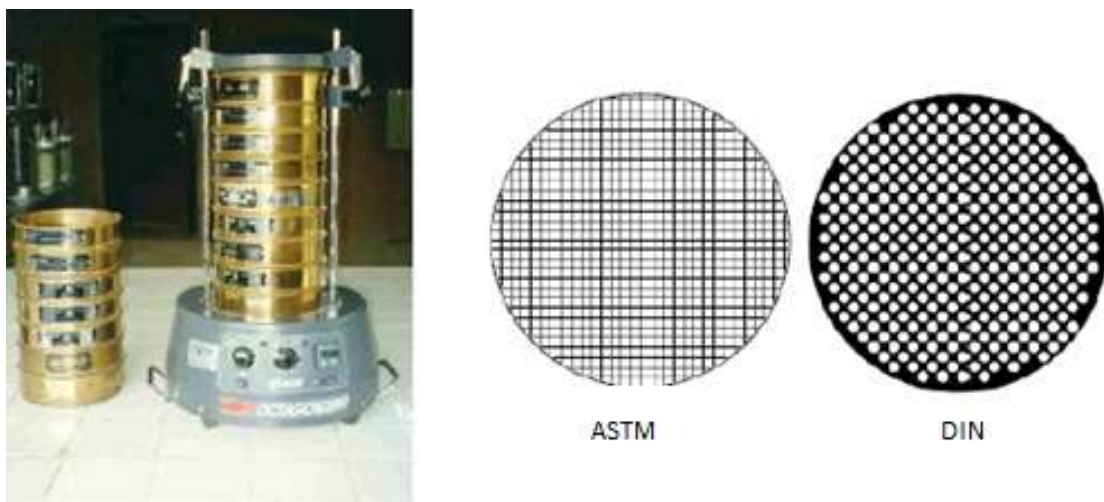


Figure 7.2 Example mechanical sieve shaker and types of sieves according to ASTM and DIN Standards

The grading curve represents this distribution as the cumulative weight fractions, which after passing through several standard sieves of different mesh size (diminishing size) remain on the sieves. A test procedure is described in the standard ISO 17892-4 Determination of particle size distribution

An example of sieve analysis test results according to the ASTM standard is presented Figure 7.3.

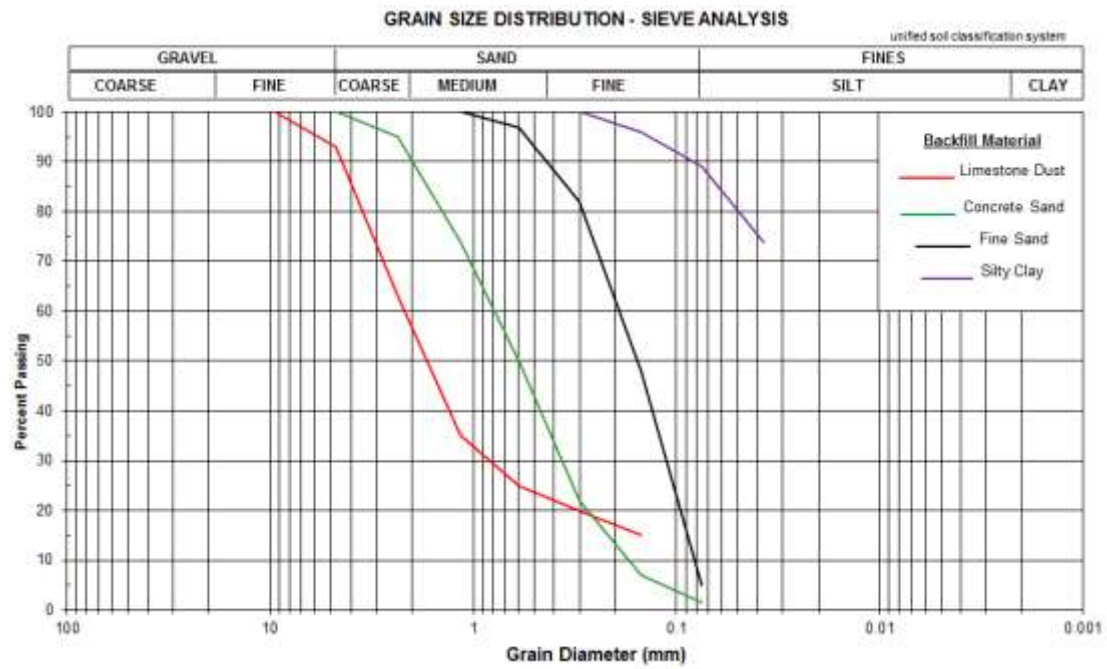


Figure 7.3 Example of sieve analysis results (ASTM)

The sieve analysis gives information about:

- the percentage of fine grained soil, also called fines, i.e. particles with size less than 0.063 mm, according to international standard ISO 14688. Note, ASTM C136 defines fines as less than 0.075 mm.
- the gradation of the soil, expressed in the uniformity coefficient,
- the type of sand/gravel, expressed in the sand/gravel median,
- the maximum particle size, i.e. particles larger than a few mm, is specified in order to minimize the possibility of damaging the cable jacket during installation of a granular type backfill (e.g. the maximum size allowed in Belgium is 4 mm or 10 mm in the case of Canada).

Once determined the grain size distribution curve, two coefficients that may be used to make a better description of the particle size distribution of a soil are:

The uniformity coefficient C_U is the ratio between the dimensions of the imaginary screen openings for which 60% (d_{60}) and 10% (d_{10}) of a soil is able to pass. Thus $C_U = d_{60}/d_{10}$.

The more uniform the soil is, the more difficult it is to compact the soil. This will give more voids between soil particles and give a poor thermal performance.

Curvature coefficient: (grading): $C_c = (D_{30})^2 / (D_{10} \cdot D_{60})$

Depending on the regularity of the curve, two types of granulometry can be defined:

- Discontinuous granulometry: The curve shows peaks and flat zones that show that several sieves don't retain material. It implies little variation of the particle sizes. These types of soils are poorly graded (e.g. beach sand).
- Continuous granulometry: Almost all the sieves retain material and the curve takes a soft and continuous shape. Those soils are well graded.

The shape of the curve (grading) can be divided in four groups [54] and is shown in the table 7.1.

Table 7.1 Details for granulometry grading curves

Shape of grading curve	C_U	C_c	C_{Usand}
Multi-graded (well graded)	>15	$1 < C_c < 3$	>2.2
Medium-graded (moderate graded)	6 to 15	<1	1.8 to 2.2
Even-graded (poorly graded)	<6	<1	<1.8
Gap-graded ¹⁾	Usually high	Any (usually <0.5)	Usually high

1) Gap-graded means that certain grain sizes are absent, which is often an artificial composite soil

An example of a sieve curve of a sandy soil is given in figure 7.4.

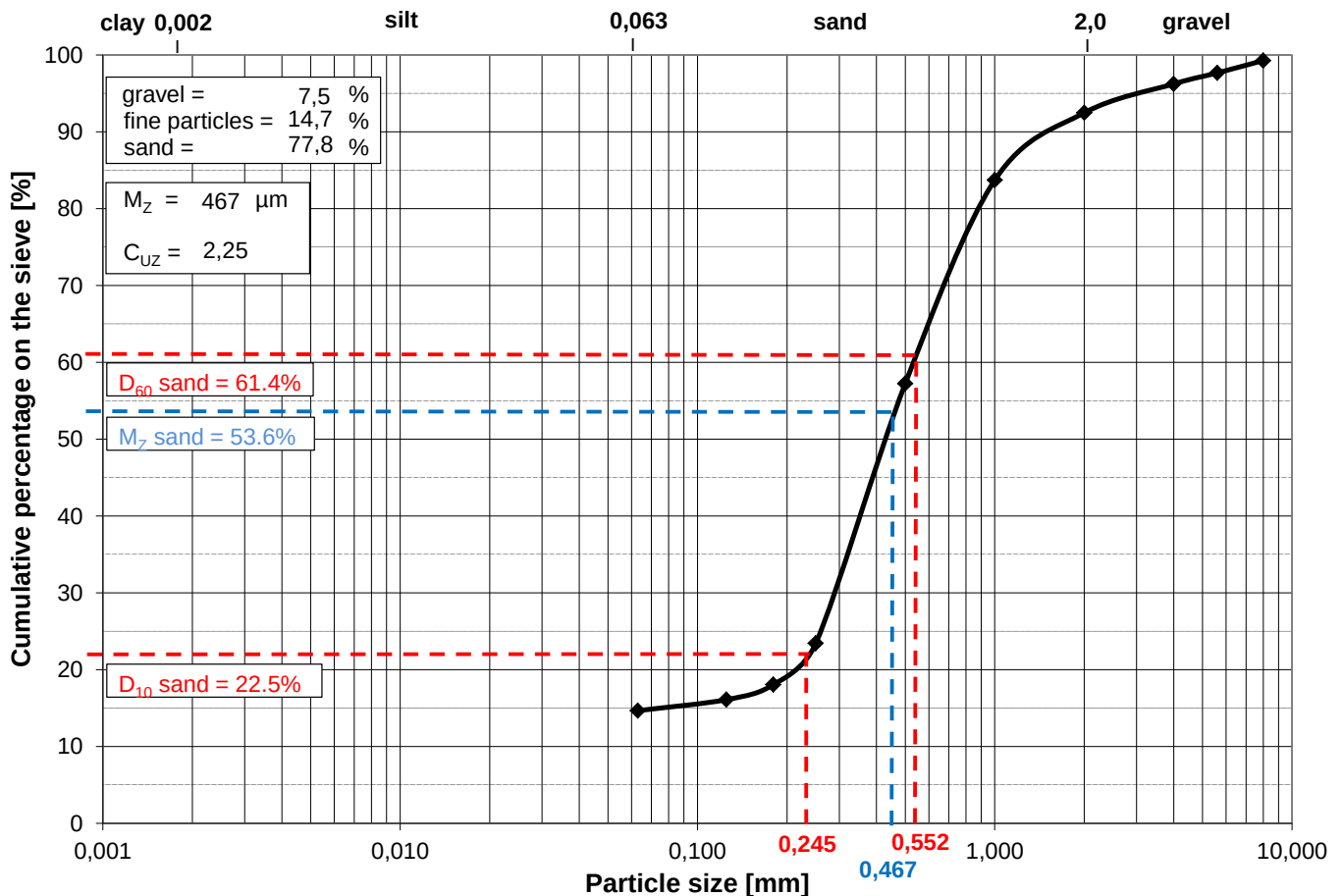


Figure 7.4 Grain size distribution curve (ISO) – Uniformity and grading coefficient

The uniformity coefficient sand (C_{Uz}) represents the ratio of sand between the sieve diameter which passes 60% of the sand and the sieve diameter through which passes only 10% of the sand. In this example D_{60} sand is 0.552 mm and D_{10} sand is 0.245 mm, which give a C_{Usand} of 2.25 (good gradation). The median of sand is 0.467 mm which means it is very coarse sand.

7.2.3 Dry density

The dry density or the specific weight is a physical property of the soil, used to evaluate how much soil is in a defined volume. Different measurement methods are used in laboratory and in the field.

The dry density is measured in laboratory by the ratio between the weight of a dry sample and the sample's known volume.

In field, the two most used methods are:

- The volumetric ring method in which a soil sample is taken with a ring of a defined volume. The samples are weighed and dried. The ratio between the dry weight and the volume gives the dry density. A test procedure is described in the Standard ISO/TS 17892-2.
- The nuclear gauge method, according to is used to measure the water content and the dry density of a soil, according to Standard ASTM D6938. Special calibrations are needed for soils with high organic content; Standard ASTM D2974, "Organic Content".

7.2.4 Moisture content measurements

For the determination of moisture content in soils several methods are available for laboratory and field measurement. The most commonly used measurement method in laboratory is the determination of the

moisture content by mass difference of soil in wet state and dry state. It is considered that the loss in weight is due to the loss of moisture.

A test procedure is described in the standards ISO 17892-1 and ASTM D2216

For field measurement, several methods could be used:

- Nuclear density and moisture gauges. This method requires a special permit for using a nuclear device.
- The volumetric ring method. A soil sample is taken with a ring of a defined volume. The samples are weighed and dried. The ratio between the wet weight, dry weight and the volume can be recalculated to the moisture content.
- Electromagnetic techniques, based on the difference in the dielectric constant between water (around 80) and soil (around 3). Methods commonly used are:
 - DC resistance measurement. This method is limited by its sensitivity to soil salinity and temperature changes.
 - High frequency capacitance probe
 - Broadband ultra-high frequency transmission line and time domain reflectometry techniques
 - Microwave techniques
- Chemical techniques. Although the method is not very accurate, it is easy to use in the field. This is the only method which does not require knowledge of the soil type and a specific calibration before measurement.

7.2.5 Moisture-density relationship (Proctor curve)

The goal of the compaction test is to determine the relation between the soil dry density and the moisture content, known as the Proctor curve.

7.2.6 Standard Proctor

This test defines the optimum moisture content of the soil at which the maximum dry density can be attained for a given compaction energy, the Optimum Proctor. A test procedure is described in the standard EN 13286-2 "Un-bound and hydraulically bound mixtures - Part 2: Test methods for laboratory reference density and water content - Proctor compaction", or in ASTM D698: "Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort".

An example of the Proctor curve obtained for typical backfill sand is presented in Figure 7.5. The maximum Proctor density for this backfill sand is 2120 kg/m^3 at a moisture content of 7.6%.

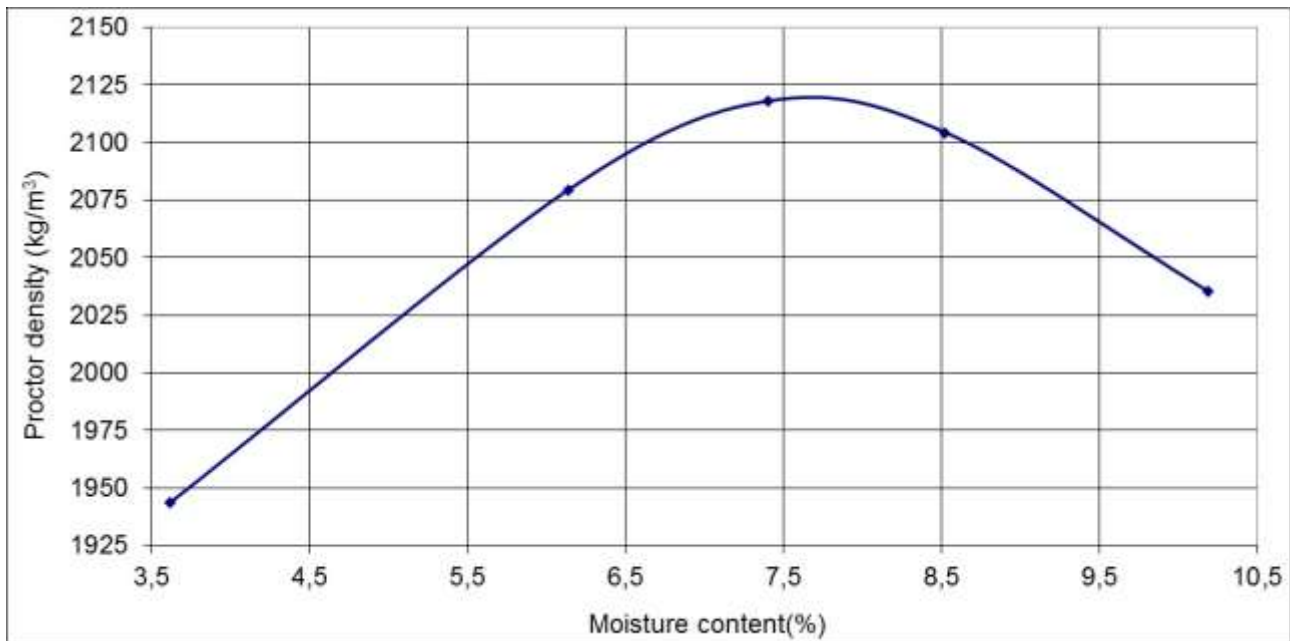


Figure 7.5 Example of a Proctor curve

A soil compacted with the optimum water content will have the maximum density and therefore will have the best thermal characteristics that the soil can attain.

7.2.7 Modified Proctor test

Besides the standard Proctor method, the modified Proctor test is also used to establish the relationship between density and moisture content for soils and backfills. The difference between the two test methods is the amount of energy given per unit volume of soil to obtain the optimum moisture content and Proctor density for that material.

In North America the standard is ASTM D1557 in Europe the standard is EN 13286-2.

7.2.8 Permeability test (also referred to as 'hydraulic conductivity')

Permeability is a function of the porosity (void ratio) of a material and thus it is related to the specific gravity and dry density. It is a soil property that indicates the ease with which the water will flow through the soil and depends of the size, shape of the soil grains and the density or porosity. A test procedure is described in the standards, e.g., [ISO/TS 17892-11], [ASTM D5856-15]. In the laboratory this can be measured by 2 different methods – 'constant head' and 'falling head'. It is not common to make field measurements as it requires a special setup and is time consuming. See Section 4.1.

The determination of the permeability factor is commonly known as the 'coefficient of permeability' with units cm/s. Some typical values for materials such as gravel, sand, silt and clay are given in Table 4.3, which also includes dry density, porosity and thermal resistivity.

7.2.9 Water retention capacity or moisture retention capacity (pF curves)

The lowest moisture content of the soil can be established using the moisture retention capacity (moisture capacitance) of the soil, which is typically reported in a moisture retention curve or pF curve. The pF curve shows the relationship of the soil moisture content with respect to the suction tension in the soil. The purpose of this test is to determine the moisture suction tension versus the degree of saturation. The shape of pF curves is a function of the grain size distribution, the soil composition and density. The pF-curves are determined for a range of moisture contents. Determination by means of gravity, or by gas pressure, is described in [39]. In the first method gravity is used as an external force to adjust the suction tension and in the second an external gas pressure is used to adjust the suction tension. Test procedures are described in standards EN 1097-10 and ASTM D6836-02.

P.E. Rijtema has measured the pF curve for 20 different kinds of soils and published the results in the paper [55]. The curves are presented as pictures in the Dutch book "Infiltratie en herverdeling van water in grond" on pages 85-90 [56].

Figure 7.6 shows a pF curve for the same soil with different compaction grades with respect to the Proctor value. The soil is compacted with 100% of the Standard Proctor value and with 90% of the Standard Proctor value.

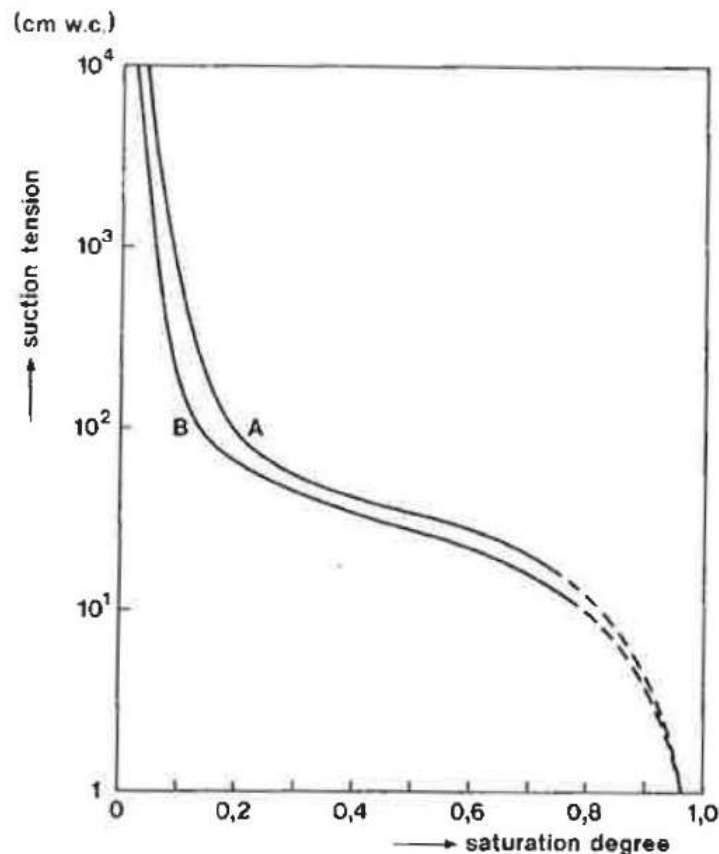


Figure 7.6 Example PF curve with 100% compaction (A) and 90% compaction (B)

It becomes clear from the graph that the compaction has influence on the suction tension of the soil and thus on the moisture content of the soil. The x-axis shows the saturation degree and the y-axis shows the suction tension in the centimetre water column (cm w.c.) in a logarithmic scale.

External factors like ground water level, precipitation and evaporation can be translated into a pF value (suction tension) at cable depth as follows:

- saturated pF = 0.0 (1 cm w.c.)
- moist pF = 2.0 (100 cm w.c.)
- dry pF = 3.0 (1,000 cm w.c.)
- very dry pF = 4.0 (10,000 cm w.c.)
- extremely dry pF = 4.2 (15,000 cm w.c.)

(The abbreviation w.c. stands for water column).

The shape of the curves could highlight the soil / backfill with a better moisture retention capacity (this could be due to a content of fines in the backfill). A higher moisture retention capacity is, of course, beneficial.

7.3 THERMAL MEASUREMENTS

Heat transfer through solids, porous materials and particulates takes place primarily through conduction and convection.

Thermal properties namely, thermal conductivity (or resistivity), thermal diffusivity and specific heat or volumetric heat capacity are required in heat conduction studies. If published values for these properties are not available or are not applicable, accurate measurements have to be performed by taking into account the conditions in the field regarding the use of the materials.

Thermal resistivity is used for steady state current rating calculations and in the case of cyclic loading calculations the thermal diffusivity is also required.

7.3.1 Determination of the thermal resistivity and thermal dryout curve

The literature review highlights the various methods used for assessing the thermal properties of soils (resistivity, diffusivity and heat capacity), where the other physical soil characteristics are assumed to be known. One of these methods was proposed by Makowski and Mochlinski. This method assesses the thermal resistivity using the soil composition, its water content and its dry density, using a nomogram proposed by M.W. Makowski and K. Mochlinski [57]. The determination of the nomogram is based on the equations of Kersten and Gemant, who determined thermal resistivity according to the soil composition and a sieve analysis – sand, clay, etc. The nomogram allows the determination of the thermal resistivity only for sandy soils and mixtures of sand and clay. This method assumes that soil particles with a diameter between 2 mm and 0.002 mm are mostly sand with the thermal characteristics of silica and that the smaller particles are clay.

Several formulae can be found in the literature. Most of these formulae were empirical, and are applicable for some soil types within a specific moisture content range.

The soil can be characterized in a laboratory for different compaction levels, water content and temperature (this has marginal impact), and for different power levels. The thermal dryout characterization of soils and backfills can be conducted in the laboratory or determined on the base of studies. For instance, referring to Section 5.2 in this brochure, in the Netherlands a study was performed over several years to clarify the drying out phenomena [42].

Thermal resistivity measurements can be made by steady state or transient methods.

Steady state techniques, such as: the guarded hot plate (ASTM C177), divided bar, or 'rhometer', require a constant thermal gradient to be established in the test sample. The equipment is cumbersome, the test duration is lengthy, and extensive thermal insulation, calibration and equilibration are required. Furthermore, in moist samples, moisture migration may be a serious problem. These methods are not practical for in-situ thermal testing. Other methods that employ cylindrical, spherical or surface heat sources are theoretically viable but are also not practical for field measurements. For example, methods such as 'pulse technique', 'parallel probe', 'buried sphere' and 'surface heat source' have been applied, but with limited success.

Transient, line heat source methods, allow thermal resistivity to be calculated from the initial heating data. Therefore, the tests can be carried out fairly quickly; lasting only a few minutes. A detailed theoretical analysis is more complex and several assumptions must be introduced. These test methods, based on cylindrical shapes, include: two probes, single probe-constant power, single probe-ramped power, and single probe-heat pulse. The single probe-constant power method has been most widely implemented because it presents the least complicated analysis and equipment requirements. Basically, a slender cylindrical probe, containing a heater and temperature sensor, is inserted into the soil to be tested. Constant power is applied to the heater and the temperature-time data is monitored. From this curve the thermal resistivity can be calculated, as long as certain theoretical assumptions were met in the test procedure. Recent research has standardized the technique and instrumentation to give results within an acceptable experimental error.

The dry-out curve represents the thermal resistivity of a soil/backfill sample as a function of the water content. To determine the dry-out curve of a soil sample, the thermal resistivity is measured from the wet state to the dry state, either on site or in the laboratory:

- 1) If a backfill sample can be obtained in an undisturbed condition, then the sample is dried out progressively until it is fully dry. During the drying out process the thermal resistivity is measured several times.
- 2) The measurement may also be made in the laboratory if a suitable undisturbed sample cannot be obtained (or for operational reasons):
 - a) For backfill : the sample will be prepared with the Proctor value and optimal moisture content, and then dried out progressively. The thermal resistivity and moisture content is measured/determined several times during this process.
 - b) For native soil: the soil sample is prepared with in-situ characteristics of dry density and actual moisture content and then dried progressively in an oven. The thermal resistivity and moisture content is measured/determined several times during the drying process.

After determination of the dry out curve the critical moisture content can be determined using a graphical method as shown in Figure 7.7. In the case shown the critical moisture content is approximately 3.3%.

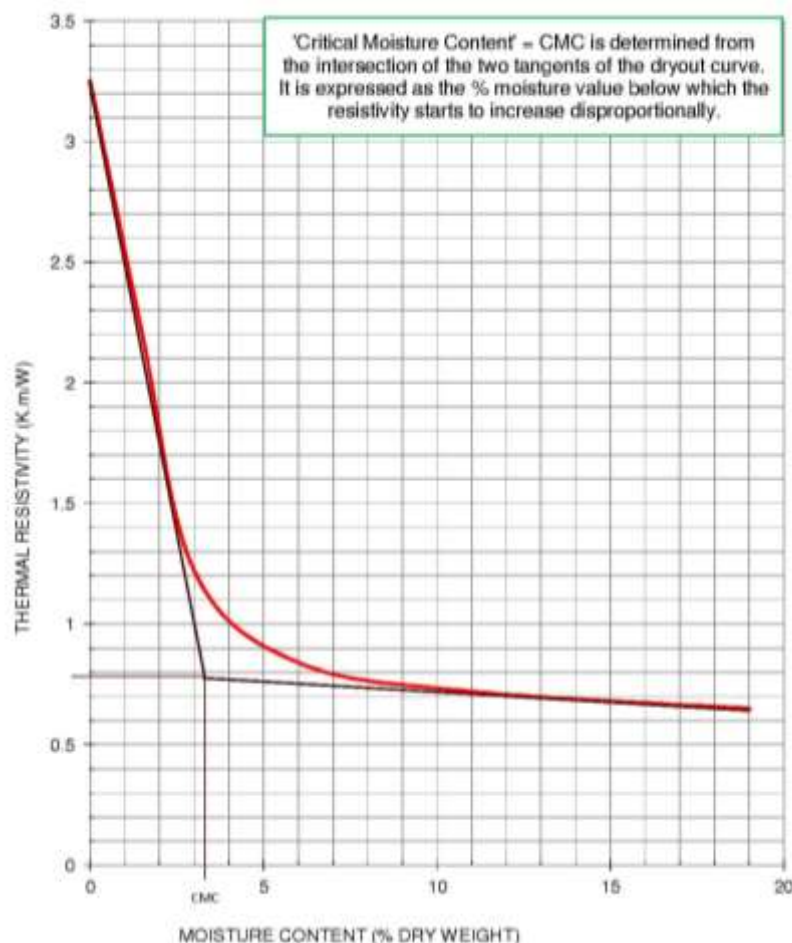


Figure 7.7 Determination of critical moisture content from dryout curve

Reference [58] describes a similar graphical method in which the critical moisture content is assumed to occur when the thermal resistivity has increased by 10% above the asymptotic value for high moisture content. It also describes the laboratory measurement of the critical saturation degree using a series of water migration tests. These tests give more exact values than the graphical determination, but are also very time consuming and subject to instabilities during the test period. For these reasons, a graphical method, as per Figure 7.7, can be used to approximate the critical water content. There are two methods to determine the critical moisture content on the thermal dryout curve. One uses the intercept of the two tangents and the other extends the horizontal line from the in-situ moisture content. There is no scientific

means to validate either method. The two tangent method gives a higher margin of safety because it gives lower moisture content and thus a higher thermal resistivity value.

According to [58], the critical saturation degree is 10 % higher than the asymptotic value of the dry-out curve (see Figure 4.4). However, the 10 % value is arbitrary; designers should consider that for values smaller than the asymptotic value, the deficiency in inter-particle water content will cause an increase in thermal resistivity.

7.3.2 Determination of the thermal diffusivity

According to IEC 60853-2, the thermal diffusivity can be determined according to the recommended methods in Appendix D (of IEC 60853-2).

Numerical values for the diffusivity are proposed in the standard IEC 60853-2 in order to allow the ampacity calculation for cyclic ratings. The standard gives the following three options:

When no data is available regarding the thermal resistivity at all, the suggested value of the thermal diffusivity is $0.5 \times 10^{-6} \text{ m}^2/\text{s}$. This is a soil with approximate values of thermal resistivity of $1 \text{ K}\cdot\text{m}/\text{W}$ and moisture content of 7% of the dry weight.

When only the thermal resistivity is available, values for the thermal diffusivity are given in the table 7.2 below, taken from the Standard.

Table 7.2 Values of thermal diffusivity to be used if only the thermal resistivity is known

Thermal resistivity [K·m/W]	Thermal Diffusivity [m ² /s]
0.5	0.8×10^{-6}
0.6	0.7×10^{-6}
0.7	0.6×10^{-6}
0.8	0.6×10^{-6}
0.9	0.5×10^{-6}
1.0	0.5×10^{-6}
1.2	0.4×10^{-6}
1.5	0.4×10^{-6}
2.0	0.3×10^{-6}
2.5	0.2×10^{-6}
3.0	0.2×10^{-6}

When the thermal resistivity, dry density and moisture content are known, the thermal diffusivity can be calculated with the following equation, Appendix D [45]:

$$\delta = \frac{10^{-3}}{\rho_T \cdot d \cdot (0.82 + 0.042 \cdot \eta)}$$

Where:

- δ = thermal diffusivity of the soil [m²/s]
- ρ_T = thermal resistivity of the soil [K·m/W]
- d = dry density of the soil [kg/m³]
- η = moisture content [% dry weight of soil]

In the case of composite materials and porous solids with gas or liquid filled pores, the measured thermal properties may vary significantly depending on the state of filling of the pores.

7.4 MECHANICAL MEASUREMENTS

For native soil the two important parameters of interest are: compressive strength (stability) and settlement (rate of consolidation).

During a preliminary route survey, a geotechnical investigation is performed by means of soil boring, sampling and in-situ testing. The Standard Penetration Test (SPT) is conducted to give an indication of the soil strength and density. Soil samples collected at appropriate depths during this survey are analysed in the laboratory for various parameter measurements. If rock or hardpan is encountered, core samples are taken for similar evaluation.

Mechanical and thermal measurements of cable trench backfill materials should include: sieve analysis, density-moisture relationship (standard Proctor test) and thermal dryout characterization at the applicable installed density. The thermal characteristic is a function of density and moisture content.

7.5 CHEMICAL MEASUREMENTS

From a thermal viewpoint, acid, alkaline, sulfate or salt based water has no measurable impact on thermal resistivity or diffusivity. However, some salts/chemicals can, in rare cases, have adverse effects on the mechanical and electrical properties of soils and backfills. For example, carbonate based soils and backfills will weaken and degrade when in contact with low pH (acidic) value water. Similarly, certain types of rock and concrete (duct-bank, manhole, protective slabs) will degrade when in contact with high alkaline or saline water. Run-off rain water contaminated with oil, salt and other chemicals can percolate through layers of soil/backfill and over the long-term will affect its quality. Spaulding of concrete floors, walls and slabs is a common occurrence when attacked by acid rain and de-icing salt water.

Commercially available test instruments can be used for these measurements of pH and salinity index.

Field tests can be conducted which will usually involve two pH tests. One is a field sample with deionised water (pH_F), and the other is "field oxidised" pH (pH_{FOX}). The pH_{FOX} measures the pH in 30% hydrogen peroxide and is conducted because the pH_F test does not detect acidity bound up in unoxidised sulfides. The H_2O_2 used in the pH_{FOX} test is quite corrosive, so appropriate protection is required, see Table 7.3.

Table 7.3 pH Classification of Soils

pH_F	Description	pH_{FOX}	Description
<4	Actual acid sulfate soil	<3	Strongly indicates acid sulfate soils
<4.5	Extremely acidic, most likely due to actual acid sulfate soils – but can be due to the soil being highly organic, or from prolonged fertiliser use	3-4	Possibly acid sulfate soil, but less certain
4.5-5.5	Very acidic soil, however it is not conclusive that low pH is due to acid sulfate soils.	4-5	Possible small amount of sulfides present
6-7	No actual acidity – common in undisturbed acid sulfate soil. Marine-influenced samples can have a pH_F up to 8	>5	With a minimal difference to pH_F , this is unlikely to be acid sulfate soil.
A typical unoxidised (potential) acid sulfate soil may have results of $pH_F \sim 6.5$ and $pH_{FOX} \sim 2$. The sample will usually react explosively to the peroxide. A typical oxidised acid sulfate soil may have results of $pH_F \sim 3.5$ and $pH_{FOX} \sim 1$. The reaction to peroxide is more variable.			

8. ANTICIPATED NATURE OF CHANGES IN SOIL & BACKFILL PARAMETERS OVER TIME

8.1 CHANGES IN INSTALLED CONDITIONS

The pattern of the annual cycles of ambient temperature and moisture content fluctuations normally stays relatively unchanged for decades, unless some modifications or changes in installation conditions have taken place during operation of the asset.

The effects of changes in the installation conditions are given in Table 8.1:

Table 8.1 Changes in installation conditions: factors affecting thermal resistivity/resistance

Reduction of thermal resistivity/resistance	Increase of thermal resistivity/resistance
Less exposure of the surface to solar radiation e.g. shade due to grown vegetation or constructed buildings leading to increased soil moisture content.	Construction in the immediate vicinity of installed cables, which may involve piling, vibrations, paving and altering native ambient and backfill surrounding of the cable
Rise of ground water level in the area of installed cables due to natural and artificial processes, leading to higher water content in the backfill and hence better thermal conductivity.	Installation of additional sources of heat, e.g. underground power cables, steam pipes, etc. resulting in a reduction of moisture content of the backfill material.
Long term natural and artificial process in the area of installed cables, which reduce amount of air pockets, pores and voids in the ground, increasing compaction improving heat transfer.	Development of root systems from growing trees etc. in the area of installation.
Sealing of the trench surface e.g. by asphalt cover, thus reducing the loss of moisture, changing the thermal resistance to the environment.	Sealing of the trench surface e.g. by asphalt cover, thus hindering the ingress of moisture, changing the thermal resistance to the environment and possibly increasing the ground temperature due to solar radiation
	Higher exposure of the surface to solar radiation due to changes in surface cover and its thickness (grass, concrete road surface or asphalt cover) or to removed trees etc. reducing the moisture content.
	Result of storms, floods and ground water flows washing away fines from the backfill materials
	Change (increase) of burial depth due to dumping materials on top of the cable trenches increasing overall thermal resistance.

The expected lifetime and reliable performance of underground cables is strongly dependent on the installation conditions. Any changes to the initial installation conditions may affect cable operating temperature. The following examples show how these changes can affect cable performance.

8.1.1 Direct buried cables

Expansion and contraction of the cables installed in poorly compacted backfill may create “air pockets” at bend positions resulting in elevated thermal resistances. As a result, the cable conductor may exceed maximum operating temperature, which may lead to an insulation failure.

Fibre optic cables attached to the power cable jacket or embedded inside the power cables may act as Distributed Temperature Sensor (DTS) and may be used to detect hot spots before any irreversible damage to the cable insulation occurs.

Construction activities in the vicinity of the installed cables may disturb thermal backfill and affect efficient cable heat transfer at this location. Moreover, in case of directly buried cable, disturbance of the surrounding backfill may lead to vertical and horizontal individual cable movement, which in severe cases may affect electrical losses, heat transfer and the influence of individual cable phases on each other. The same effect can take place as a result of water mains and sewer pipe breaks, storm and flood washout.

Moisture depletion in the vicinity of trees has also been noticed during several investigations. The depth of penetration of tree roots was much deeper than had been commonly assumed. This is an important fact to be taken into consideration in the selection of the cable route.

Electrical (I^2R) losses may in rare cases be marginally affected by the displacement of phases with respect to each other. This changes the distribution of the current in the conductors due to changes in the proximity effect and, depending on sheath bonding arrangements, the change in mutual inductance between phases may have also an effect on the sheath currents and losses. However, these effects are only likely to be significant if significant proportions of the cable connection are so affected. Changes in the heat losses in the cables will have a proportionate effect on the temperature rise of the cables, which would compound any adverse effects due to a rise in the thermal resistance of the cable environment caused by such displacement.

8.1.2 Feeders installed in concrete-encased duct bank

This type of installation is much more reliable and protected from most of the issues described for directly buried cables. They are widely used in the North America region. No deterioration of the thermal performance is to be expected due to ageing of concrete.

Concrete encasement has good thermal performance characteristics. The thermal resistivity of the (non air-entrained) concrete is considered to be as low as 0.80 K·m/W, the precise value depending on the aggregates used and the moisture content. The excellent thermal stability of concrete encasements means that deterioration over time is unlikely.

Other benefits of concrete encased duct banks are as follows:

- May not require periodic thermal resistivity measurements along the cable route
- Reliable cable configuration over long period of time (especially for multi-circuits installation)
- Protection from tree root systems and resulting dry spots along the cable route
- Ability to use spare ducts or adding DTS system or a water-cooled line
- Physical protection of the cables from external damage during construction activities
- A predetermined minimum clearance to other heat sources, crossing or which will be crossing the cable route

Disadvantages are:

- The thermal resistance of the air spaces
- The close proximity of the other cables

In the unlikely event of heavy construction activities or earthquakes, duct banks may be disturbed or broken. This may have some effect on the originally designed thermal performance.

8.1.3 Deterioration of grounding arrangement

There are various grounding schemes used for distribution and transmission class cables – single-point bonding, solid (both ends) bonding, multiple point bonding, cross-bonding, or a combination of them. Some changes at grounding connecting points may lead to circulating currents or high resistive connections, which will have same overheating effect as described above. To prevent these unwanted consequences the grounding arrangement should be tested periodically to ensure correct functioning.

8.2 MIGRATION AND LOSS OF FINES

In a granular type backfill such as 'stone-dust', 'stone screening' and 'cement bound sand', fine particles (typically less than 63 μm) normally adhere to coarser particles of sand and gravel. This matrix of fine particles is very stable when in a moist and compacted form. Depending on the density, the permeability (hydraulic conductivity) of this material can vary between 0.01 to 0.1 mm/s. This material is prone to erosion and 'loss of fines' when under high hydraulic gradient or cyclic changes in water level.

Thus the likelihood of loss of fines is dependent on installed density. If the installed density is high, the possibility of such occurrence is low except for instances when under 'flash flood' that can cause serious and rapid erosion. If the installed density is low, the material will have relatively high porosity and high permeability. In this state, the 'fines' can migrate or washout rapidly under conditions of fluctuating water level. This will make the material mechanically unstable and lead to settlement. This will increase its thermal resistivity; especially at lower moisture levels.

Wash-out of backfills and loss of fines in granular type backfills was encountered at one location near the shoreline. This was attributed to 2 factors - daily fluctuation in the water table and relatively porous sub-base material (aggregate) below the cable backfill. It has been documented by others that backfill/soil erosion can occur very easily if water permeates through the top layer of the ground surface into the backfill that was poorly installed (low density and with high voids content).

Wash-out and loss of fines can be mitigated with the use of 'geo-textile' or 'filter-fabric' encapsulating the backfill envelope. These are woven fabric mats (available in rolls) made of Nylon, Kevlar, PVC and HDPE, etc. that has high tensile strength. These are used frequently in civil industry in areas such as bogs and swamps to support load and vehicle movement (static and dynamic load). A select filter-fabric will support the backfill envelope at the design elevation without sinking and will allow movement of water through the encapsulated backfill envelope. This will result in the lowest possible thermal resistivity of the cable environment. The mesh size of the geotextile fabric must take into consideration the particle size distribution of the backfill material and the native soil.

For good thermal characteristic (low thermal resistivity), the select backfill must be well-graded and have about 15% fines (material passing #200 sieve or <0.075 mm). This is essential in order to achieve a high density and low porosity.

8.3 CHEMICAL CHANGES

Generally, little in the way of chemical change or deterioration is to be expected in backfills derived from the cable excavation (original soil) since the material should have been in chemical equilibrium with the environment for a long time already.

However, in backfills containing cement there might be some internal factors related to the production of the cement that could initiate degradation. Chemical reactions in cured cement are due to free lime (CaO) excess or magnesium oxide (MgO) excess. This should not be a risk in new installations as the content of both products is limited in cement manufacture, but it could be a problem in old installations.

8.3.1 Sulfate attack

It is known that the presence of relatively high concentrations of sulfates in soils may degrade sand / cement backfills. The most common sulfates present in soils are calcium sulfate (CaSO_4), magnesium sulfate (MgSO_4), potassium sulfate (K_2SO_4) and sodium sulfate (Na_2SO_4).

Agricultural works may contaminate soil with sodium and ammonium sulfates $(\text{NH}_4)_2\text{SO}_4$, whilst water may be contaminated by sodium and magnesium sulfates. The presence of industrial or faecal waste, or chemical by-products can also lead to sulfate attack.

Acid sulfate soils are soils which contain iron sulfide minerals. Most of them were formed after sea level rose and flooded the land, and sulfate in the seawater mixed with land sediments containing iron oxides and organic matter. The resulted chemical produced large quantities of iron sulfides in the waterlogged sediments. When exposed to air, these sulfides oxidize to produce sulfuric acid, hence the name acid sulfate soils. They are generally only found below 10 metres above mean sea level in areas such as coastal wetlands, mangroves, tidal lakes, swamps, waterlogged or scalded areas, inter-dune swales and coastal sand dunes.

As an indication where action may be required, Spain specifies under regulation the use of sulfate resistant cement for concrete when the concentration of sulfates is equal or higher than 600 mg/L in waters and 3000 mg/kg in soils.

The presence of sulfates in soils is also linked to minerals which contain the $(\text{SO}_4)^{2-}$ radical which are quite common in the earth's crust. Nevertheless, the presence of sulfates in soils is not so frequent because of its washing ease, as they have relatively high solubility and so their effective accumulation can only be produced in specific conditions (abundance of sulfates in the immediate environment, and arid or semi-arid climates).

The attack of sulfates in cement is produced when sulfates concentration is relatively high and can also be due to the presence of sulfates in underground, superficial or sea waters. Sulfates can penetrate easily in cement mixes because of their high solubility. Sulfate attack causes the loss of mechanical strength due to the reaction between salts and the tricalcium hydrated aluminate of cement. The reaction causes a molecular modification resulting in increased volume increase and cracking. This will facilitate ulterior entrance of salts, dissolved in water, which will multiply the process.

The major concern of this phenomenon is related to softness of concrete that can be a problem when it has a structural purpose.

Laboratory fully dried thermal resistivity results for backfills aged by acid sulfates will be poor, however due to the low lying nature of the land, a fully dried out situation is unlikely.

8.3.2 Other acid impacts

Beyond the above, consideration of the chemistry relating to backfills exposed to a highly humic and acidic environment, (e.g. peat) leads to the following possibilities:

In general cement has low resistance to the attack by acids. Only soils with a very high content of organic colloidal material have the acidity necessary to attack cement. In general the aggressiveness of acids occurs by their presence in underground water. Most common acids that can be found in solution in underground water are uric acid present in peat deposits and carbonic acid coming from dissolved carbon dioxide. In some cases marshy waters might contain sulfuric acid as a result of the bacterial decomposition of sulfurous components. The severity of the attack of acids will depend on the type and concentration of the acid, whether it is organic or inorganic (with the exception of the phosphoric acid, which reacts with the lime in cement forming calcium chlorides, nitrates and sulfates). Although the occurrence of natural water with a pH lower than five is very unusual, anomalous values of pH may be found in industrial environments or in areas with chemical contamination. In general acidic ground waters are less common than those containing sulfates. There may also be some reaction between cement and organic acids if they are present.

The preceding comment might also apply to crushed limestone.

8.3.3 Clay reactions

Clay platelets can bind to humic compounds in a process called flocculation. This would change the clay structure and possibly the thermal conductivity. (Process is well known to gardeners, clay soil can be lightened by the addition of humus. Clays are very variable in their chemistry depending on their original formation conditions). What applies to clays would also apply to crushed shale, another mudstone.

8.3.4 Other chemical changes

Sewer leaks also might be an issue. Although no such problems were reported in the literature, sewage involves hydrogen sulfide, which can oxidize to sulfuric acid. Attack of concrete sewer pipes at the top of the air space is apparently well documented. One speculates that a similar process might be able to take place in soils with high sulfide content from other origins also. These would also apply to crushed limestone backfills.

Another issue that might be of importance is where a cable route passes through an old industrial site having severely polluted soil. The pollutants could be acidic, highly alkaline or organic. Usually complete excavation and replacement of such materials is recommended to protect cables. Deterioration of imported backfills should also be considered a possibility in such situations depending on the nature and mobility in the groundwater of the pollutants.

Another type of attack that can affect backfills containing cement is that originated by pure water (water with dry residue under 150 mg/l), which may be produced from melting water or desalination plants, for example. This type of water has the capacity to dissolve the calcium hydroxide of cement, increasing the porosity.

8.4 LONG TERM CHANGES IN ENVIRONMENTAL AND CLIMATIC CONDITIONS

8.4.1 Environmental conditions

Ambient temperature, ground water level, precipitation, thermal properties of constituent materials in backfills and native soils, degree of compaction, porosity and effective burial depth are all factors that influence the operating temperature of underground cable systems. The main concern, however, is thermal resistivity and how it varies with temperature and moisture. The change in the thermal resistivity of dry or fully saturated soil/backfill as a function of a normal range of temperature (10 °C to 70 °C) is negligible. For example, the thermal resistivity of uniform sand in dry conditions ($\sim 3.3 \text{ K}\cdot\text{m}/\text{W}$) is the same at 25 °C or at 50 °C. However, the thermal resistivity is very strongly dependent on moisture content and, for a given intermediate (neither fully dry nor fully saturated) moisture level, the change in resistivity may locally be affected by moisture migration, starting from the interface between the hottest cable surface and the backfill. This too, can cause a dramatic reduction in cable ampacity, or if not taken into account, severe overheating of the cables. Moisture migration is described in Chapter 5.

Most backfills are to some extent thermally stable, but with improperly graded/compacted sand and crushed rock backfills, the difference between wet and dry TR can be significant (e.g. wet/dry from 0.6 to more than $3.0 \text{ K}\cdot\text{m}/\text{W}$), and dry-out can occur at lower temperature rises than are generally assumed (e.g. $< 15 \text{ °C}$ above ambient). Unfortunately, this is likely to apply to some of the old cable installations. The definition of 'thermally stable' must be understood, because any porous material can be driven to become unstable – a function of load, time and other factors. As mentioned above, the difference in wet and dry thermal resistivity of soil can vary by a factor of 4, or even as high as 8. For a 'good' thermal backfill, however, this factor is 2.5 or less. This type of material is classified as 'thermally stable' solely because of the narrow range of change in resistivity from best to worst moisture condition. Decisive for a good thermal backfill is not the narrow range, however, but an assured low extreme value of thermal resistivity in the dried-out situation.

If steady-state ratings (continuous load) are of interest, then the thermal resistivity relationship to moisture content and critical heat flux is all that is needed for a given backfill/native soil. The two-layer-model with critical isotherm is an often used calculation method that should err on the safe side. If cyclic or real time rating or temperature prediction is required, then the effect of dryness on heat capacity or diffusivity is also of some interest (John Millar used an adaptation of Van Wormer's formula [59] for this phenomenon in [47]), as is the time to dry out once the critical temperature is reached, and the time for moisture return (which is very long). Simple formulae, quite similar to the rating equation in IEC 60287, are available for cyclic loads and the two-layer-model with critical isotherm, derived by a combination of the methods of Neher / McGrath [21] and Cox / Coates [20].

Environmental / installation conditions can sometimes change radically over the long life of underground cables, such as an increase in the effective burial depth of a cable by putting infrastructure over it (e.g. motorway on-ramps). The moisture return path from the ground water level can be interrupted by

infrastructure under cable routes (e.g. underground car parks, service tunnels), and in some northern regions, a large amount of the annual precipitation comes in the form of snow, which is usually trucked out of urban and suburban centres, preventing rewetting of the cable environments in the spring thaw.

Although not strictly an environmental cause, another underlying driving factor can be the type of loading a cable is subject to. Load flows are becoming more stochastic and dynamic with the increase of renewable generation, demand side management, etc., i.e. the loading 'environment' of some existing cable connections is changing significantly. Advice should be sought from cable and accessory manufacturers as to the kind of impact this is likely to have on cable life. Over the past 15 years, the renewable energy sector (wind and solar) has grown significantly world-wide. Use of 'native soil' as backfill (without proper compaction) on their 35 kV connections has given rise to a significant number of premature failures.

The use of a reclaimed landfill site as part of a cable route can result in higher ground temperatures. As the waste biodegrades methane and heat are given off. Over time the heating effect of the landfill decreases as more material becomes stable. So as far as long term performance of backfills and soils, this situation would improve. This timeframe will vary depending on the composition of the landfill. A working group member has observed methane issues in an oil monitoring pit on a section of cable over reclaimed landfill. This is despite the landfill being covered some 30 years ago. The DTS installed on this cable shows ambient ground temperatures during summer on the reclaimed site are approximately 2-6 °C warmer compared to similar grassed unshaded areas elsewhere. There was no variation observed in winter temperatures between sites. Pockets of trapped methane gas may also result in a locally raised thermal resistance of the ground.

8.4.2 Climatic conditions

Seasonal temperature variations can be significant (annual variation is approximately sinusoidal with the peak lagging from mid-summer - depending on latitude).

Seasonal moisture variation and stochastic moisture variation may be even more significant, especially in sites that drain well. Climate change related issues, such as increasing temperature (marginal), flooding, etc., may also have an impact. Climate change has been reported to have a direct effect on ambient temperature and moisture content in Victoria, Australia [7], with an increase in temperatures at cable burial depths. A major consideration is the change in the time of peak loading to different seasons with different climatic conditions, and the previous reference is only one of such examples found in the literature review in Section 2.

Long dry and warmer periods are likely to have an adverse effect on cable ratings, mostly due to increases in thermal resistivity and an increased propensity for moisture migration, although any rise in ambient temperatures at cable burial depths will also have some effect.

9. DETERMINATION OF LONG-TERM BACKFILL DEGRADATION

As the responses to the questionnaire circulated by WG B1.41 elucidate (see Chapter 3), cable failures because of degradation effects in the cable trench are rather unusual. This leads to the consequence, that only in very special cases are utilities interested to examine the actual properties of the cable trench after installation, i.e. during operation.

For existing power cable installations examples of situations where assessment of the backfill is required are:

- when the installation conditions may have changed, and with them the current rating of the existing cable system, which might be reduced from the original design level,
- if the operational conditions have changed or are to be changed, so that an upgrading of the cable circuit is desirable,
- if more powerful and accurate methods for the determination of the current rating have become available since the original design was carried out, necessitating accurate data for the thermal environment of the circuit in order to reassess the rating accurately.

9.1 SURVEY OF BASIC AMBIENT PARAMETERS – DESKTOP ANALYSIS

Some general information and basic parameters of the cables' surroundings can be collected without opening the cable trench.

Where available, historical meteorological data and statistics for e.g. air temperature, precipitation, ground temperatures and possibly ground water levels near the cable trench should be obtained.

If route inspections have been carried out previously, preferably on a regular basis, then these should be reviewed to understand whether there have been significant changes to the ground surface conditions and the environment within the vicinity of the cable route. E.g. if the nature of the ground surface has been changed (street construction increasing the burial depth, sealing by asphalt covering changing the heat loss from the surface etc.) or if nearby vegetation has grown in such an extent that the moisture content of the soil has been affected (which may be particularly a problem for large trees).

Further information can be collected e.g. by vertical drilling near to the cable trench, which will disclose the properties of the soil beside the trench, its composition and compaction as well as its arrangement in layers, and the groundwater level, see the following section.

The installation of other infrastructure in the vicinity of a circuit such as: other cables, gas or water pipes, steam pipes for district heating etc. may have a severe impact on the cables. On the one hand the civil work for these measures – nearby to the cable trench or crossing it – may change the local structure and properties of the cable trench. On the other hand, the infrastructure element itself may have thermal (e.g. steam pipes) or mechanical (streets, railways etc.) impacts on the cable trench, thus changing the operational conditions. Thus a thorough registration of new installations near to the cable route, both parallel and crossing power cables, steam pipes etc. is very important for the assessment of the status of an installation.

It is often the case that the responsibility for such activities with civil work possibly impacting or even endangering the cable trench may be not assigned to the TSO or utility, but to other authorities. To enable a general survey about such activities and to secure an adequate coordination is a difficult challenge of asset management, especially in big cities.

9.2 IN-SITU INVESTIGATIONS

9.2.1 Practical evaluation of thermal properties of the backfill of a cable circuit

Note: in undertaking the following procedure it is essential that the precise location of the cables is known and that precautions are taken to prevent any damage to the cables due to the sampling procedures.

9.2.1.1 Route survey

In cases where there is no risk to the cables under investigation or other services which could be installed in the vicinity then a thermal survey of the soils along the cable route may be performed in order to confirm or define the required design parameters – ambient temperature and soil thermal resistivity. Soil boring and sampling is conducted at select locations; using a soil drill-rig, see Figure 9.1.



Figure 9.1 Drilling for soil samples

If cohesive soils are encountered, undisturbed samples can be collected in nominal 76mm diameter by 500mm long steel tubes (Shelby tubes) or in split-spoons, see Figure 9.2. For non-cohesive granular soils, bulk samples of auger cuttings are normally taken.



Figure 9.2 Shelby sampling tubes

In some cases, samples are collected from a test pit excavation. During soil boring/sampling activity, in-situ thermal resistivity and ground ambient temperature measurements are taken at appropriate depths. Samples

of backfill and soil are used for the measurement of moisture content, density and for thermal dryout characterization. Additional tests such as sieve analysis, organic content, compaction and chemical analysis may be conducted if necessary.

In addition to samples taken by boring or drilling the cable trench may be opened locally for a detailed inspection of the actual situation. When excavating down to locate and expose the cables under investigation the most common (and considered the safest) method is to use an air knife excavation system (see Section 4.2). In this situation water jetting (water knife) is not recommended as it will change the moisture content of the materials and hence may hinder the investigation. Pot-holing with the air knife and vacuum removal of material and the air knife are shown in Figure 9.3.



Figure 9.3 Potholing with air knife and vacuum

Opening a local trench enables the collection of information about the current state of the backfill and soil, the taking of samples of soil and backfill for laboratory analyses and the direct measurement of thermal resistivity and ground temperatures at down to the cable depth. The latter need to be measured quickly as they will begin to change as soon as the trench has been opened.

Thermal resistivity measurements are conducted in the trench using needle probes and an automatic measuring instrument, see Section 7.2.1

The same method is also used to test samples returned to the laboratory. In addition thermal dryout characterization of native soil or backfill material is conducted in the laboratory; on samples reconstituted at the 'in-situ' or at proposed 'installed' moisture and density. Several measurements are made in stages; with the moisture content ranging from the 'wet' to totally dry condition. The sample is brought to thermal and moisture equilibrium at each test stage before test run. Critical moisture content and applicable thermal resistivity values can be selected from the dryout curve, taking into consideration the field (in-situ) moisture content.

A series of thermal dryout curves for various soil types and backfills are given in Figure 4.4.

A typical time-temperature graph of thermal resistivity test is give below in Figure 9.4. The initial data from 1 minute to 10 minutes gives the thermal resistivity at the test moisture content. If the test is continued for longer period under the same power level, the rate of increase in temperature increases. The point of intercept of the two slopes (knee of the curve) is the time at which the soil moisture migration starts to increase the thermal resistivity. This is the onset of soil drying at the soil/probe interface.

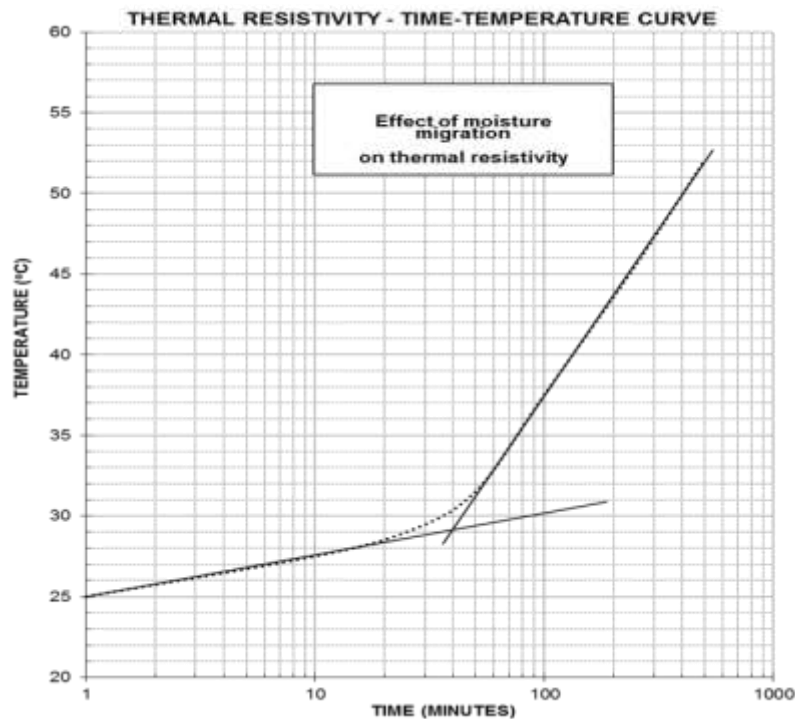


Figure 9.4 Effect of moisture migration on thermal resistivity

When the trench is opened it may also be possible to install temporary or semi-permanent sensors (thermocouples etc.) in the vicinity of the cables to get further information during operation, see Figure 9.5.



Figure 9.5 Typical test pit excavation to expose protective slab above cables, make in-situ measurements and collect samples of soil and backfill for laboratory analysis

For this purpose, explorative trenches along the route can be opened perpendicular to the cable trench. Such explorative trenches first allow a visual inspection of the cross section of the cable trench, e.g. to find out if the backfill is wet or dried-out and if its status (structure, compaction etc.) is as expected.

Samples of soil and backfill can be taken to enable thermal, mechanical and chemical measurements to be carried out in the laboratory.

The open trench offers the possibility of installing thermal sensors in the soil and backfill in the vicinity of cables. For examples of field installations see Figures 9.6 to 9.8. By this means, a retrofit monitoring system can be realised. Data obtained at the site by regular monitoring over a period of time would provide knowledge about the actual values at a specific time and also their variations. For example the range of values during wet and dry periods in the seasonal variations could provide information of the maximum and minimum values of thermal resistivity. It can also provide quantitative data of the effect of topography and surface conditions (e.g. sealed surfaces, vegetation, open fields). Soil ambient temperature as well as cable temperatures could also be monitored. Note that the response time of thermocouples situated on the cable surface may be significantly slower than that of a cable-integrated DTS-system. However optimistic readings may be obtained if the replacement backfill is better than that removed. Regular monitoring is becoming easier with the continuing development of data loggers for field monitoring, sensors, the Internet and other communications technology.



Figure 9.6 Ampacity Audit - In-situ measurement of temperature and thermal resistivity in native soil and trench backfill above cables; samples of soil and backfill were collected for laboratory analysis

Carrying out such an investigation by opening the existing cable trench is delicate and may be associated with severe problems such as the risk of damaging the cables or of endangering personnel by approaching cables under voltage. However, if there is in a need to open the trench for some other purpose, maybe for maintenance work on the cables or joints, there may be an opportunity to install thermal probes and to obtain information about the state of the cable trench.



Figure 9.7 Installing temperature sensor and thermal probe in backfill above cables



Figure 9.8 Type 'T' thermocouples installed on pipe type cable for real-time monitoring normally connected to data logger or SCADA system

9.2.2 Installations with integrated monitoring systems

The best chance to get information about the thermal situation and possible changes in the trench without opening is if thermal probes or thermal monitoring systems are already installed. As shown in CIGRE TB 640 [4], the latter is becoming a more usual practice for new installations.

The increasing use of such Distributed Temperature Sensors (DTS) allows the measurement of cable and/or soil temperatures along the cable route by means of optical fibres, which either are integrated into the cable construction or installed outside, in the neighbourhood of the cables in the trench. Measurements can be made over great lengths with a precision down to 1 K and with average measurements at 1 m intervals.

Such DTS systems are frequently coupled with a dynamic rating system (DRS). Modern DRS will have a module that adjusts some system parameters in their thermal model (e.g. soil thermal resistivities or ambient temperature) to match the measured values and, by this, to allow prognoses of ratings and temperatures. By means of such measurements, information about the cable's surrounding can be derived, so that potential long-term degradation of the installation can be identified if the derived system parameter change with time.

If such a system shows that a cable system is overheating along a section of the cable route, the operating company may need to open the cable trench and to make visual and other examinations of the trench situation and of the soil properties in particular, as described above.

The findings have often indicated that the thickness of backfill layers was not consistent and that the quality of backfill installation was rather poor (low density as a result of poor compaction). Often, it could not be confirmed if the material was 'moisture conditioned' when installed; nor if any quality assurance program was implemented.

Detailed soil / backfill sampling, and in-situ and laboratory measurements show thermal and moisture gradients as a function of the radial distance from the interface. This gives a true picture of the thermal environment for comparison with the original design parameters.

9.2.2.1 Type of measurements and test protocols

If samples of soil and backfill have been taken from an explorative trench, thermal, mechanical and chemical measurements can be made in the laboratory. The analyses of these samples with respect to density, compaction, moisture content, thermal conductivity etc. require the test methods, measurements and test protocols which are described in Chapter 7.

10. CONSEQUENCES OF PERFORMANCE CHANGES ON CABLE SYSTEMS

In this chapter the consequences of changes (impacts) in backfill thermal and mechanical performance have been analysed separately. No significant electrical impact due to the degradation of backfills has been reported.

10.1 THERMAL IMPACTS

The thermal resistance of the surrounding medium of an installed cable is mainly composed of the soil and the backfill thermal resistance. It is referred to as the parameter T_4 in IEC 60287-1-1 and expressed in K·m/W [44]. T_4 depends on many factors, but can represent about 3/4 of the entire thermal resistance of the cable system. So, any change in the thermal characteristics of the soil/backfill or immediate surroundings of the cable may have a major impact on the cable system performance.

An increase in the thermal resistance of the soil/backfill will lead to an increase in the temperature of the cable system for any given load.

Table 10.1 shows an example of the calculated effect of an increase in the thermal resistance of soil/backfill on cable rating/temperature. This is a 225kV XLPE cable system with 1600 mm² copper segmental conductors and an aluminium foil sheath and a PE oversheath. The cables are laid in flat formation, direct buried, with a soil reference thermal resistivity of 1 K·m/W and ambient temperature of 13°C. The sheaths are single point bonded. The axes of the cables are 1.3 m below the ground surface and the phases are 0.2 m apart.

Table 10.1 An illustration of the thermal impact effect

		Reference case	Increased value of T_4
T_4		$T_4 = 1.43 \text{ K·m/W}$	$T_4 + 20\%$
No drying out	rated current for $\Theta = 90 \text{ °C}$	$I = 1564 \text{ A}$	1453 A
	conductor temperature for $I = 1564 \text{ A}$	$\Theta = 90 \text{ °C}$	105°C
Drying out (2 zone model; $\Theta_{\text{critical}} = 60 \text{ °C}$)	rated current for $\Theta = 90 \text{ °C}$	$I = 1470 \text{ A}$	1366 A
	conductor temperature for $I = 1470 \text{ A}$	$\Theta = 90 \text{ °C}$	119.5 °C

Table 10.1 illustrates the dependency of the rated current versus the thermal resistance T_4 . Other examples are given in Chapter 5. An increase of 20% in the thermal resistance T_4 results in a net increase in conductor and insulation temperatures. As a consequence, the current rating has to be decreased by about one hundred amperes in order to continue to operate at a hottest conductor temperature of 90 °C. On the other hand, if the cable system is operated with a higher external thermal resistance than initially planned (+20% in this example), we note the sharp dependence of the (calculated) temperature reached by the conductor on the thermal analysis method that is selected. Considering a basic soil model (no drying out) the conductor overheating will be 15 °C in the example in the table, while it will be double with a 2-zone soil model of soil dryout (moisture migration).

Although very rare (and when coupled with such factors as high loading, surrounding soil of high thermal resistivity with high susceptibility to moisture migration, high ambient temperature and/or proximity to

external heat sources), the high temperature increase in dielectric losses in oil-paper insulation, when the insulation is operated at temperatures in excess of the rated value, has occasionally been suspected to cause or exacerbate the thermal runaway of cable systems. This was rumoured to be a contributory factor in the failure of the final oil-filled cable in the 1998 Auckland, NZ, CBD power failure, although inadequate backfill for long dry summer periods was the main causal factor of the rated temperature being exceeded [16].

Another famous cable failure, in Belvedere London in 1962 [12], led to a serious reappraisal of moisture migration and cable rating practice. Once again, this was not a case of backfill degradation, but knowledge gained the hard way, where it was found that moisture migration could occur at much lower temperature rises than previously thought.

In one instance the moisture content was determined to be low (less than 5%) and thermal resistivity was in the range of 1.2 to 2.5 K·m/W; as a function of the radial distance away from the cable/backfill interface. In another instance the backfill material used was 'cinders' with thermal resistivity of higher than 3.0 K·m/W. In the first case, it was suggested to de-rate the circuit, but when it was not implemented, the cables failed as a result of thermal runaway.

Over the long service life of a cable, many changes to the cable thermal environment may occur, including corruption of the original backfill when other services are added, and possible contamination of the backfill from external sources or even the cable itself, which may or may not affect the thermal performance. Such occurrences would warrant opening the trench for investigation and possibly decontamination. Indeed, if a trench has to be opened for whatsoever reason, the existing and/or replacement backfill TR values should be tested to ensure that the thermal performance of the cable environment is satisfactory [60].

A completely dry zone in the immediate vicinity of the cables was evident in most observed cable thermal failures. The thickness of the dry-zone layer was an indication of the magnitude and duration of the load imposed on that particular backfill. Sand-backfill has shown the highest moisture loss and highest thermal resistivity. However not all dry rings are an indication of an impending thermal runaway.

Some installation techniques are more sensitive than others: a pipe or conduit laid with the HDD technique and not properly filled with grout during the installation creates a hot spot that cannot be "corrected". While this is a case of faulty installation, not backfill degradation, such a hot spot will remain in place throughout the lifetime of the cable system, increasing the local temperature or possibly limiting the current rating of the underground link if it is known about or detected (a derating study has been reported).

Arrhenius' law is largely used as an ageing accelerating factor for insulating materials. A shorter life time of the cable system can be expected as a consequence of a higher temperature. The lifetime reduction is nevertheless difficult to quantify, as other factors affecting ageing have also to be considered.

From an economic point of view, the investment/asset is negatively affected in terms of efficiency and depreciation.

10.2 MECHANICAL IMPACTS

The increase of the thermal resistance of soil/backfill leads to increased temperature with resulting mechanical effects on the cable system, which may be observed in particular in the bends at the location of accessories.

For extruded insulation cables, tight bends in the route are the most vulnerable locations for potential mechanical damage due to high temperature. If the maximum operating temperature of the cable is exceeded by a significant margin then damage to the insulation is possible at bends due to axial forces developed within the conductor [61].

The literature survey, however, has not pointed out mechanical constraints combined with a change of soil/backfill performance to have been an important cause of failures. Nevertheless, the answers to the questionnaire and general experience show that problems/failures encountered at the location of accessories and attributed to the change of performance of the soil/backfill are highly significant.

10.3 CONSEQUENCES IF DETERIORATION CONTINUES WITHOUT MITIGATION

Moisture migration may take months or even years to fully manifest itself and, generally speaking, as load flows rise and become more stochastic, cables are likely to be subject to the most onerous conditions well into their service life. These matters, not strictly related to any degrading or changes in backfills or native soils, are nevertheless related to long-term performance and are therefore worth mentioning in this context.

Consequences of the increase of the thermal resistance of soil/backfill are mainly the increase of thermal and mechanical stresses on the cable system and premature ageing. If no proper action is taken (see Chapter 11), outages may occur earlier than the expected end of life of the cable system, and faults are likely to become more frequent. Although the encountered problem of soil/backfill performance may be localised, it can lead to the premature replacement of a larger length of cable with associated costs. As an example, this could occur where a long cable section length has been installed with the HDD technique and thus is obviously inaccessible for a localised repair, necessitating the replacement of the whole length.

11. MITIGATION OF LONG-TERM AGING OF SOIL AND BACKFILLS

11.1 TYPES OF DETERIORATION

Moisture Content (of soil or backfill): At a given location, this parameter can change with time (season) and as a function of the cable load (heat output – W/m), the associated heat flux (W/m²) and duration (hours). The impact of the change (depletion) of moisture results in an increase of thermal resistivity of the soil/backfill. Therefore for a given load, the interface temperature will increase. For short duration and cyclic load, the moisture usually returns to the normal level. However for extended high load (emergency load situation), the moisture may be driven away from the interface zone and create a thermally unstable environment; especially if the initial moisture level was at or below the 'critical' moisture level, see Chapter 5. A change (drop) in the natural water table can also reduce the moisture content of granular type soil/backfill material that has low moisture retention capacity. Moisture depletion is a common occurrence in the vicinity of large trees.

Density: The in-situ dry density usually does not change except when and if the material is not installed properly (poor compaction). Very seldom, if ever, the density will decrease except when affected by changes in the water level or by fast flow of water that may washout the fines. This can happen primarily with granular type materials of high porosity and also with poorly cement-stabilized backfills (cement bound sand, low strength flow-fill or sand-cement slurry). The effect becomes evident resulting in the creation of voids, uneven settlement of the pavement and development of cracks at the surface.

Physical and Mechanical: This applies mainly for cables installed in Concrete Duct-bank, HDD bores, PVC, HDPE and Concrete Pipes used in jack & bore and micro-tunnels. During soil boring activity it is quite common for the bore diameter to be significantly larger than the carrier pipe. For HDD installation it is assumed that the drilling mud occupying the annulus after the last back-reaming run will prevent any future settlement. However, if the native soil is soft and 'sensitive', it can collapse and the load on the carrier pipe can deform or break it. In installations where the annular space between the bore wall and the casing/pipe is grouted (with an appropriate slurry), this risk is minimized. Settlement and deformation in sub-surface soil is a long term process and its effects on the system components may not become evident in a short period. In order to improve the cable rating (Ampacity) thermal grout is often pumped in the annulus of carrier pipes to fill-in the air space between the cable conduits and the pipe. All the pipe joints whether glued, epoxied or fusion welded must stay leak-proof for the life of the system. Failures as a result of poor workmanship and long-term changes have been documented. Where bentonite grouts are used to fill voids or ducts, then if they are not adequately sealed partial or full drying of the grout may occur leading to shrinkage and voids, causing increased thermal resistance.

Design and Material Tolerances: Factors such as variability of the quality of the originally installed backfill, resulting in higher TR than expected, together with tolerances, constructional variations and perhaps failure to construct the installation according to the design dimensions are also factors to be considered.

11.2 MAINTENANCE STRATEGIES

Appropriate maintenance strategies are very much situation-dependent, but in general, the following measures should be considered:

- routine feeder route patrol
- rigorous field inspection during any third party work in the cable vicinity
- periodic serving tests
- periodic thermal resistivity measurements
- installation of a DTS system

In CIGRE TB 279 [62] there is a mention of routine route inspection, mainly to detect third party works and to limit damage from such activities. A case study, carried out in France, is mentioned where the cost of such regular inspections is compared with their relative effectiveness at limiting damage. In that case the most economical solution is for monthly patrols. Routine patrols and field inspections should, of course, also take note of less obvious threats to the system performance such as significant growth of vegetation or an increase of depth of a circuit due to third party civil works.

In the case of serving test failure, the opportunity will probably arise during the repair process, to make a visual assessment of the condition of the backfill as a minimum and perhaps to make resistivity measurements.

Periodic thermal resistivity measurements are less common, generally being reserved for cases where the original rating may be in doubt or uprating of a circuit is under consideration; see CIGRE TB 606 [63]. Clearly measurements with thermal probes in the vicinity of a cable must be made with suitable precautions to ensure safety and avoid any damage to the circuit under evaluation.

While it will generally not be practical to install a new DTS system (fibre cable and instrumentation), in some cases DTS cables have been installed without the necessary instrumentation. In this situation the addition of instrumentation at a later stage may be appropriate.

11.3 AVAILABLE MITIGATION TECHNIQUES AND STRATEGIES

Installation of external heat sources (steam and oil lines, distribution class power cables, underground subway stations etc.) may change heat transfer balance and create "hot spot" areas. Special considerations and measures are necessary, such as replacing poor backfill, thermally isolating external heat sources (if this can be done without compromising heat dissipation away from the cables of concern) and external cooling systems, e.g., closed-loop gravitational cooling systems [64]. Note that most active cooling systems add complexity and secondary systems with their own reliability issues. [64] provides a solution which, where suitable, offers the benefit of being virtually maintenance free.

Nearby tree root systems may dry surrounding areas, changing originally designed thermal backfill parameters. This, and many other subjects related to unfavourable thermal environments are covered in [65].

Utilities that are facing these problems have to take decisions. Several options are offered:

- derating of the cable system. It means recalculate the current rating of the cable system with the new thermal resistivity values.
- taking into account in-situ measurements of thermal resistivities. It can be sufficient, especially when the first rating has been established with standardised resistivity values of the soil.
- when relevant, improve locally the thermal resistance of the soil/backfill in order to match the initial rating.
- reinforcement of the cable system (new link) , this option is discussed extensively in [63]
- increase conductor size locally. When the area of concern is well localized, a section of larger cable diameter can be installed. Suitable transition joints must be installed at each extreme. For cables installed in ducts Civil Works might be limited to the jointing areas as cables can be removed and pulled into the ducts from the extremes.

If materials have been installed poorly, then their density can perhaps be improved by compaction at the surface or by excavation of the top layers, closer to the cable installation. Injection grouting is also an option that can be investigated, specifically for granular type backfills with higher porosities. Injection grouting may help where cavities or voids have been created by flooding, tidal changes, changes in the water table or water mains leakage. Situations warranting such measures are not very likely, however, and every circumstance should be evaluated on a case by case basis.

National guidelines may also exist which are relevant to the environmental conditions likely to be encountered in given regions. The UK reference [66], for example, understandably focuses on the environment from the tree's perspective, but also gives some very general guidelines on avoiding damage to apparatus by trees, noting that utilities should be consulted when there is any likelihood of trees being planted near existing underground apparatus. There is no specific mention of the risk of moisture depletion

from cable environments via tree roots, so it is up to electricity utilities to inform the relevant authorities of this. Care when removing tree stumps is mentioned, and this is of particular relevance to underground cables, which in some cases may be pulled up with the stumps. [66] makes reference to another set of guidelines covering the avoidance of danger from underground services [67]. This guide mostly focuses on the potential danger of live underground cables to the public, aptly illustrating the risk with a photo of a cable in flames, apparently due to arcing following penetration with a sharp object. Advice is duly given to take care and to carefully use hand-tools around power cables, but little is said about the impact of the environment on the cables (other than sharp objects penetrating cable sheaths).

With regard to backfill, concrete is discouraged in [67], due to the difficulty of safe excavation, and the main comment about backfill is that sharp edges, such as large pieces of rock and hard core, should be avoided. The reports stress that electricity utilities should be contacted and liaised with during any work that may impinge on their surroundings, but do not directly refer to the thermal requirements of cable backfills, and it might be conjectured (and has been observed by WG members) that on the basis of such general guides, civil contractors might reinstall entirely unsuitable backfill when cable trenches have been exposed.

Indeed, a variety of meanings are likely to be ascribed to the term backfill. Backfill might, for example, be chosen for good drainage properties (rather than moisture retention), good thermal insulation for district heating pipes (rather than aiding heat dissipation), and it is well to remember that sand backfills were originally chosen to provide a firm but smooth bedding for underground cables.

11.4 PROPERTIES ACCESSIBLE FOR MITIGATION

For a given system, the two parameters that have the highest impact on cable rating are the **temperature** (at burial depth and at surface) and the **thermal resistivity** of the external thermal environment (soil and backfill).

Temperature: Natural seasonal ambient temperatures cannot be controlled or modified; they can only be monitored and taken into account for the cable rating. However, for a given load, the temperature rise above the ambient at the burial depth can be controlled by modifying the thermal environment. Lower thermal resistivity will reduce the cable/backfill interface as well as the conductor temperature.

Thermal Resistivity: For the native soil we have limited choices – (1) in-situ thermal resistivity and (2) a different thermal resistivity (usually slightly higher) if it is replaced as the backfill. For an imported backfill we have numerous choices including; the type of material, its thermal characteristic and the size of the envelope around cables. However, once installed at its **optimum design for performance**, the choice is limited to controlling the change in its **moisture content**.

Moisture Content and Density: Moisture content at the burial depth (typically ~1.2m) may change marginally as a function of the time of the year but may change significantly as a function of the cable load and its duration; especially in the immediate vicinity of the cables. Since moisture content has the highest influence on the thermal resistivity for a given density, the question is - how to maintain a high level of moisture in the immediate surrounding? Options such as embedding perforated water pipes in cable trench to increase the moisture content in the backfill and running parallel water pipes to dissipate the heat have been implemented with little success.

Heat Pipes: mitigation of hot-spots by the installation of self-contained heat-pipes has proved to be a very efficient and cost effective technique. Heat pipes made of copper, aluminium, steel, etc., filled with a variety of 'working fluids' are available for various applications.

Modification or Replacement of Backfill Material: This can be an expensive proposition unless it is for a very short section such as a hot-spot or directly buried cable joint (failure repair). Modification may be possible to improve the in-situ density by addition of water and additional compaction effort. However, this should be considered only after taking samples of the existing material to determine its density and moisture content in order to assess the degree of improvement. Replacing existing soil or backfill with a thermally superior material such as 'fluidized backfill' is a good option.

11.5 ECONOMIC VIABILITY

Ultimately, any decision for major work on a cable connection should be based on economics and reliability, and this involves quantifying reliability, a challenging topic in itself.

For example, if a cable connection is nearing its end of life, and is already proving unreliable, a complete replacement strategy with larger conductors and state-of-the art backfill technology might be advisable. In other cases, localised hot-spot mitigation as discussed in earlier sections might be preferable.

The optimum course of action involves the usual assessment of fixed and variable costs over several decades, a process that requires the estimation of such variables as interest rates, load growth, and the price trajectories of all the material and components involved. Calculating the present worth of investment costs, running costs (including the cost of I^2R losses, see part 3.2 of [1]), maintenance costs and reliability, the latter of which must be weighted to maximum demand, energy, and customer type. All these cost components impact each other, and an electricity transmission or distribution company will also have to put these considerations into a system-wide context, exploring such options as network reconfiguration to reduce load and increase the service life of assets such as aging cables.

A further aspect to be considered nowadays is the environmental impact of a connection. Carrying out such assessments, which include some financial considerations, are also beginning to be introduced as has been reported recently in CIGRE TB 689 [68].

These latter considerations are obviously out of the scope of this brochure, but it is prudent to remind the reader that we are dealing with critical components in a wider system context, and the context should not be forgotten.

12. USERS GUIDE

12.1 INTRODUCTION

This chapter is designed as a user's guide to provide a straightforward method to assess the backfill performance for a particular cable circuit.

The technical details of backfill installation, potential problems, mitigation methods and performance assessment of backfills have all been described in detail in the preceding chapters. In this Guide flowcharts and outlines will provide a systematic method of assessing a backfill with references to relevant chapters in the document. Following the flowcharts is a more detailed description of the proposed methodology for the assessment and implementation for remedial action, where required.

There are 5 key components of a backfill performance assessment:

Rating Assessment (Figure 12.1):

1. A rating review of a circuit could indicate the need for a detailed backfill assessment due to:
 - a cable failure which may be due to overheating
 - known or suspected problems with the quality and/or installation of the backfill
 - plans to increase the loading of a circuit within the original designed rating
 - a need to increase the rating of the circuit
 - suspected overheating or potential of overheating of a cable circuit; caused by e.g. an increase in burial depth due to other construction activities, change of surface condition (asphalt instead of grass), new heat sources (steam mains), long term degradation of the backfill such as moisture depletion due to vegetation, change in ground water level, loss of fines, changes in the degree of compaction, change in seasonal load pattern.

Backfill Assessment (Figure 12.2):

A backfill assessment can be initiated by a rating assessment due to the reasons listed above. It can also be started when work in the cable vicinity indicates soil or backfill degradation or there is a reasonable suspicion that the backfill performance has undergone a long-term change and no longer meets the design assumptions.

2. Desktop Analysis
 - The backfill performance assessment will start with a review of the existing design and environmental parameters, system operations data, and identification of new or previously known hotspots or bottlenecks.
3. Route Survey
 - A route survey will identify the actual cable location, offer information for a possible failure assessment and provide information on sampling locations.
4. Site Based Analysis
 - On-site testing will support visual inspection as well as collection of soil and backfill samples and parameters.
 - It will permit the installation of thermocouples and thermal probes for post analysis monitoring.
5. Laboratory Based Analysis
 - Samples taken on-site will be analyzed in a lab
 - Data collected in the previous steps and data generated on lab samples, and an expert assessment will lead to the selection of a correct critical moisture content and specific thermal resistivity (ρ) for the circuit under assessment.
 - The appropriate values of ρ will be applied in the final rating calculations

12.2 FLOWCHART FOR RATING ASSESSMENT

The flowchart shown in Figure 12.1 guides the user in making an assessment of the rating of a circuit. In the event that the rating is more than adequate to meet the predicted load then no further action may be necessary. However if it is demonstrated that the calculated rating is marginal or less than the predicted

loading then one of the aspects that may need assessment is the thermal performance of the installed backfill. If such an assessment is required then the following section gives recommendations in this respect.

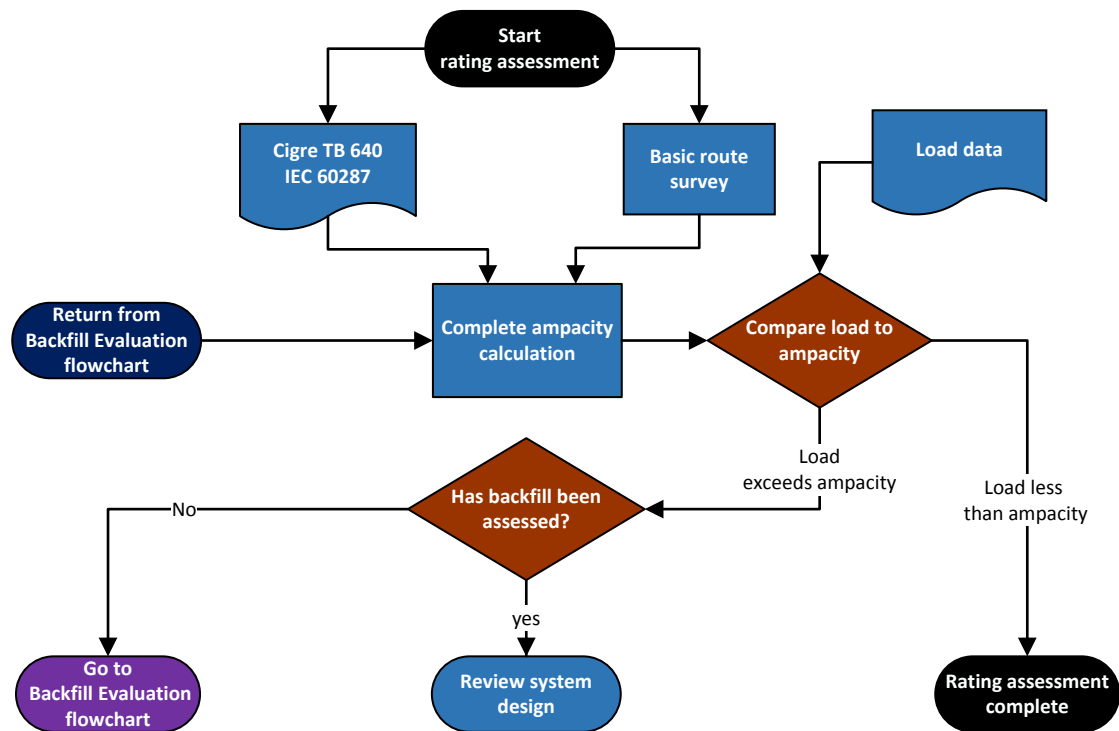


Figure 12.1 Flowchart showing assessment of circuit rating

12.3 FLOWCHART FOR BACKFILL ASSESSMENT

The flowchart shown in Figure 12.2 guides the user in assessing the thermal performance of the backfill in an existing cable installation.

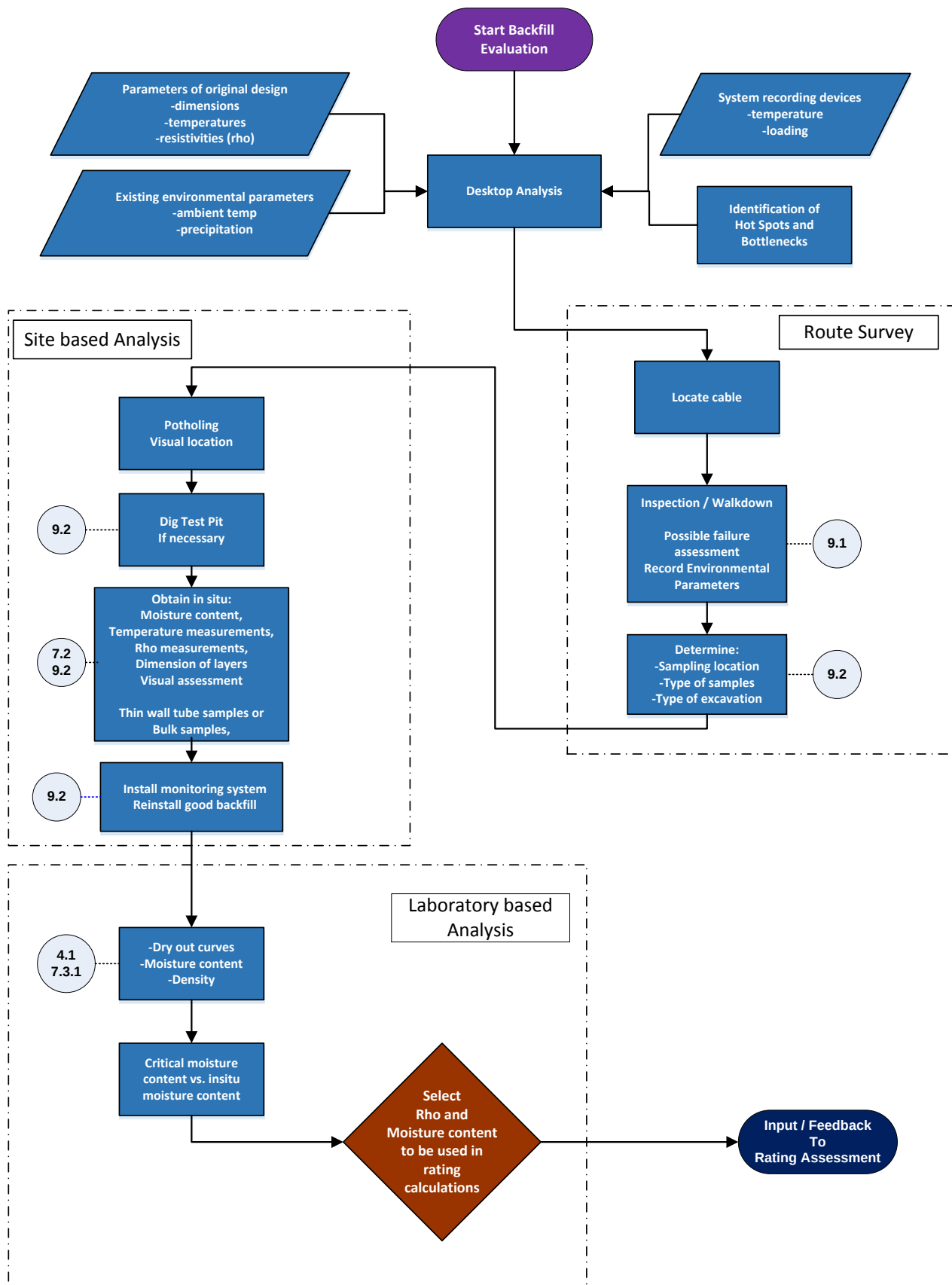


Figure 12.2 Flowchart for assessing backfill performance

A list for a typical procedure for assessing thermal conditions of a cable installation with reference to Figure 12.2 is given below:

1. In order to evaluate any changes, the parameters used in the original system design before commissioning must be known and taken into consideration for comparison with the new findings. What were the design values of the ground surface temperature, earth ambient temperature at the cable depth, and the thermal resistivity of the soil and backfill? For the dimensions and elevations of the cable trench, cable spacing, duct-bank, etc. refer to the installation drawings for the design dimensions of the installation.
2. Take into account available environmental data including trends: air and earth ambient temperature records, precipitation and frost from environmental agencies, airport authorities, research institutes, etc..
3. Check if the system was installed with any type of temperature measurement devices. In some cases thermocouples were installed on the cable(s) at selected locations. Temperature records in conjunction with the load and load pattern can be used to extrapolate future temperature rises.
4. It is highly unlikely that the parameters along the entire cable route would change (deteriorate) by the same magnitude. Therefore it is important to identify hot-spots for measurement and mitigation. Obvious hot-spots are: deeper burial depth such as at road and railroad crossings, proximity to other power cables and steam mains, areas with native soil of higher thermal resistivity, below asphalt surfaces, in proximity to large trees, at a change in elevation as a result of construction, etc..
5. If no detailed data are available or when a high level of data accuracy is necessary, a route thermal survey should be conducted. This will involve excavation of test pits, soil/backfill sampling, laboratory analysis of soil and backfill samples and in-situ thermal testing and installation of instrumentation on the circuit, see Section 9.2.
6. Depending on the cable route length and other parameters such as changes in soil conditions or hot spots, distances between test pits are typically between 150 m and 1 km. In urban areas, the spacing can be even closer because of the presence of other utilities.
7. Once the cables have been precisely located a test pit can be dug which may be as small as 1 m wide by 1.5 m long. The purpose is to expose the cables (or the duct-bank) and the surrounding backfill and native soil. An air-knife and vacuum system is the safest method for excavation and for precise location of cables. Water jetting (water-knife) should not be used as it will change the moisture content of the soil and backfill. Once the cables are exposed, excavation of the remaining area must be done with utmost care.
8. Measurement of temperatures should be conducted as soon as possible after excavation to evaluate the thermal gradients. This can be done with thermal probes installed in the end walls of the test pit at the cable surface and at several distances above it. For example, 25mm, 75mm, 150mm, 300mm, etc.. Once the temperatures are recorded, thermal resistivity in-situ measurements should be made as described in Section 9.2. This will give a profile of temperature and thermal resistivity that can be used in ampacity calculations.
9. Make accurate measurements of all dimensions of the system components within the test pit (i.e. X-Y coordinates for the ampacity program) for verification of the design layout. Make a visual assessment and take photographs.
10. Collect soil/backfill samples at each thermal probe position for the measurement of moisture content. These samples can be small, i.e. about 250 g each. In addition, collect undisturbed tube samples (76 mm diameter by 150 mm long) for the measurement of density (see 7.2.2) and a larger sample (2 kg) for determination of the thermal dryout curves (see Section 7.3.1). If the backfill and the native soil is non-cohesive (granular), it may not be practical to collect undisturbed samples. In this case, about 2 kg bulk samples should be collected which should then be reconstituted at the lab with the in-situ moisture content and at the density that will give a thermal resistivity close to the in-situ value.
11. Although the moisture and density values are not used in the cable rating calculations, their influence on thermal resistivity and critical moisture content are significant. In-situ thermal resistivity and moisture content values should be identified on the thermal dryout curve, together with the critical moisture content. Compare these values with the ones used for the cable rating and/or conductor sizing, if available. If the in-situ moisture content is close to or lower than the critical moisture content, this can be an indication of onset of thermal drying, see Section 9.2

12. Before backfilling the trenches, it is suggested that temperature sensors are installed for future measurements and verification of thermal conditions of soil and backfill for rating calculations. The findings will help not only the assessment the current status of the system but also forecasting and calculation of future performance. Appropriate mitigation methods and materials can be evaluated for specific locations along the route or for future projects, see Chapter 11
13. Input all the relevant data into the ampacity program for comparison and verification of original and current ampacities. If the new ampacity is higher, there may be spare capacity, but if it is lower, the circuit will need to be de-rated unless mitigation measures can be employed, see Chapters 10 and 11.

Use of the above methodology will give a good indication of the current and potential rating capability of a circuit and will help to provide information as to potential problem areas.

13. CONCLUSIONS

This TB provides a guide to determine the condition and long term performance of soil and backfill systems for underground cable. The scope covers AC and DC, extruded and laminar transmission cables, landings of submarine cables as well as medium voltage cables of high importance.

Existing and up-rated cable systems are loaded increasingly higher. In all cases the higher loads result in higher operating temperatures for the backfill even if the rated operating temperatures remain the same. Many of the existing circuits have been in service for 40 or more years when engineered backfills were in their infancy. Limited knowledge is available of past backfill design parameters and their change over time. Recent work showed that properties have changed, such as degree of compaction and stratification of backfill components. Of particular interest is how high load conditions, change in physical properties and environmental changes will impact aged backfill and soil conditions.

There are no common standards or guidelines on how and when to check backfill properties and what tests to be performed. Each utility has its own procedures but long term performance of backfill is not addressed. Experts have observed dry regions around direct buried cables or pipes indicating backfill drying but in many cases without resulting failures.

Historically little was known about the backfill behaviour (backfill design, installation, or moisture migration). This changed in the early 1960s when research efforts started to focus on this topic. Because of lack of knowledge, earlier cable installations can be at risk as being based on too optimistic design parameters. Even nearly 60 years later, there is still ongoing research into new backfill materials; e.g. some of the techniques used in the civil / construction industry are being implemented, especially in 'renewable energy' projects.

At first the Working Group B1.41 conducted a worldwide survey of utility experiences. A total of 46 responses from 17 countries was received and analyzed. Of these responses, 22 from 11 countries indicated problems and/or deterioration of installed backfill systems. The main conclusion of the survey was that there are not an extensive number of failures attributed to backfill conditions. This can be due to conservative designs by utilities, low loads and benefits due to system contingencies. However higher loads in the near future can reduce these safety margins. A summary of the survey is provided in Chapter 3 while details of this analysis are given in Appendix C.

Experiences in different countries around the world are listed in Chapter 3 and Appendix B, while extensive literature references are given in Chapter 2 and Appendix B.

This brochure provides means for investigation and assessment of marginal backfills. For novices in this field an easy to apply Users Guide in Chapter 12 provides flowcharts and sequential instructions. The more experienced user can find a great number of references and expert contributions with regards to soil and backfill characterization parameters, moisture migration, installation topics, test methods, changes of soil and backfill parameters over time, degradation and consequences of resulting inaction as well as mitigation methods when unacceptable conditions are encountered.

One of the most important parameters impacting the thermal properties of soil and backfills is moisture content. Thus Chapter 5 is solely focused on moisture and moisture migration in great detail.

Correct installation of backfill has been identified as another key parameter. This topic can be found in Chapter 6 including what needs to be considered for a proper compaction. Often the quality control of backfill installation has been found lacking. As this greatly impacts the long term performance of the cable system, QC audits should be conducted when installation quality is in doubt to avoid potential of problems in the future. In Chapter 7 readers can review relevant international standards and test methods.

Special conditions such as landing of submarine cables or cables installed for renewable wind or solar generation systems are dealt with in relevant chapters. Renewable generation have different load patterns compared to utility systems and are also governed by different design considerations.

What can be anticipated when backfill conditions and parameters change over a long time? A general assessment is provided in Chapter 8 followed by topics such a migration of fines, chemical changes, long-

term changes in environmental and climatic conditions. Fines are an important backfill component. If fines separate or stratify out of the particle matrix due to water movement and / or improper compaction, the thermal conductivity is negatively impacted. Brazilian research shown in Appendix B.5.3 illustrates this effect.

Chapter 9 together with a User Guide in Chapter 12 are powerful tools to determine a possible degradation of soil and backfill conditions. The outlined audit is advisable when marginal backfill conditions are encountered or are suspected. The analysis shown in these procedures will provide reliable results for rating calculations such critical moisture content and actual thermal resistivity.

Consequences of performance changes on cable systems are analysed in Chapter 10. The most stringent impact of soil or backfill degradation on cable system is thermal, then at a much lower level mechanical. No significant electrical impact has been reported.

When marginal backfill conditions are confirmed, systems might have to be de-rated unless mitigation methods, shown in Chapter 11, can be deployed. While limited in nature, mitigation options such as improving the thermal backfill around hot spots or applying a dynamic rating system can increase or at least maintain the rated ampacity of circuits.

The main factors impacting the thermal, mechanical and chemical performance of soil and backfill systems have been identified as being moisture content, thermal properties of the backfill constituents, gradation of particles, movement of fine particles and quality of compaction. The Working Group found limited problems due to chemical degradation arising from, for example, the presence sulfites. This TB provides necessary tools to assess existing conditions and to develop options to correct unsatisfactory backfills.

14. REFERENCES

- [1] IEC 60287 Electric cables - Calculation of the current rating.
This standard currently has 9 parts as follows:
Part 1-1: Current rating equations (100 % load factor) and calculation of losses - General
Part 1-2: Current rating equations (100 % load factor) and calculations of losses - Section 2: Sheath eddy current loss factors for two circuits in flat formation
Part 1-3: Current rating equations (100 % load factor) and calculation of losses - Current sharing between parallel single-core cables and calculation of circulating current losses
Part 2-1: Thermal resistance – Calculation of thermal resistance
Part 2-2: Thermal resistance - Section 2: A method for calculating reduction factors for groups of cables in free air, protected from solar radiation
Part 2-3: Thermal resistance - Cables installed in ventilated tunnels
Part 3-1: Operating conditions - Site reference conditions
Part 3-2: Sections on operating conditions - Economic optimization of power cable size
Part 3-3: Sections on operating conditions - Cables crossing external heat sources
- [2] IEC 60853 Calculation of the cyclic and emergency current rating of cables.
- [3] IEC TR 62095 Electric cables - Calculations for current ratings - Finite element method.
- [4] CIGRE TB 640 F. De Wild et.al., "A Guide for Rating Calculations of Insulated Cables".
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APPENDIX A. WG B1.41 TOR

The terms of reference (ToR) for WG B1.41 were approved by Study Committee B1 and are as follows:

Existing and up-rated cable systems are loaded increasingly higher. In all cases the higher loads result in higher operating temperatures for the backfill even if the rated operating temperatures remain the same. Many of the existing circuits have been in service for 40 or more years when engineered backfills were in their infancy. Limited knowledge is available of past backfill design and how it will change over time. Recent work showed that properties have changed, such as degree of compaction and stratification of backfill components. Of particular interest is how high load conditions, change in physical properties and environmental changes will impact aged backfill and soil conditions.

The Working Group shall cover the following topics

To review the literature (experience, history) on the subject

To establish the appropriate terminology and characterization parameters.

To review methods to measure the thermal, mechanical and chemical soil / backfill properties and stability.

To review methods to measure the aging and long-term stability of soil and backfill properties over system life

To review technical methods how to mitigate deterioration of soil and backfill conditions including moisture depletion by vegetation or other utilities

To evaluate the consequences, if no action is taken, such as loss of ampacity, including cost and overheating of the cable system.

To integrate the information in a practical users guide.

To apply to extruded, paper, and paper-laminate cable systems

To apply to HV AC and DC land cable systems including direct buried, direct buried ducts or pipe, duct bank / manhole systems, and Horizontal Directional Drill (HDD) installations

To apply to HV AC and DC submarine cable systems including ploughing, jetting, trenching and HDD installations

To apply to MV AC cable systems of high importance

At the conclusion of the work, the WG shall prepare a Technical Brochure, a summary in Electra and a Tutorial.

The work is scheduled to complete in August 2015.

APPENDIX B. BIBLIOGRAPHY, REFERENCES AND ADDITIONAL EXPERIENCE IN DIFFERENT COUNTRIES

Appendix B contains the results of the literature survey introduced in Section 2., followed by additional experience collected from different countries.

B.1. CIGRE PAPERS

Table B.1 lists the papers with some relevance to the WG topic published by CIGRÉ from 1960 to 2012. Some of the papers come close, but none directly address the core subject of this working group.

Table B.1 Cigré Papers

Type of document	Date	Title	Remarks
Session doc	21-3-1970	Thermal design parameters used for high capacity EHV cable circuits in Great Britain	SC 21 Skipper, Holdup, Endacott, et al.
Session doc	21-6-1970	Contribution to the study of the Thermal environment of buried cables	SC 21 Arrighi, Ridon, Benard et al.
Session doc	21-1-1974	The external thermal resistance of cables in soils as non-linear environment	SC 21 Slaninka
Session doc	21-09-1974	High power cables with internal or external water cooling	SC 21 Schuppe, Birnbreier, Brakelmann et al.
Session doc	21-05-1980	The optimum use of controlled backfills for underground electric cables	SC 21 Germa, Steffens, Van Melsen et al.
Electra n. 84	October 1982	The conditions controlling the drying out of soil around power cables	Vermeer, van de Wiel, Snijders
Electra n. 98	January 1985	The calculation of effective external thermal resistance of cables laid in materials having different thermal resistivities	SC 21 WG 02
Electra n. 104	January 1986	Current ratings of cables buried in partially dried out soil	SC 21 WG 02
Electra n. 145	December 1992	Methods for calculating cycling rating for buried cables with partial drying out of the surrounding soil	SC 21
Electra n. 145	December 1992	Determination of a value of critical temperature rise for a cable backfill material	SC 21
Session doc	2000	Analysis of temperatures and daily field data on the soil thermal parameters of operating circuits	SC 21 – 204 Lyall Barnewall
Session doc	2002	Managing the ratings of underground cable circuits	SC 21 – 101 Bucea, Zhao et Al.

B.2. JICABLE PAPERS

Table B.2 Jicable Papers

Date	Reference	Title of the Paper	Authors	Issue
1991	A.4.4	Thermo-mechanical behaviour of XLPE cables under normal and short-circuit conditions	Head J.G., Crockett A.E., Wilson A., Williams D.E.	Describes an investigation of the thermo-mechanical performance on bends, of XLPE supertension cable.
1995	A2.2	Fluidized thermal backfill for increased ampacity of underground cables	D. Parmar, H.S. Radhakrishna, J.E. Steinmanis	Thermal characterisation and performances of backfills
1995	D.2.11	Thermal performance of underground power cables with constant and cyclic currents in presence of moisture migration in the surrounding soil	D. Freitas, A. Prata, A. De Lima	Impact of moisture migration on rating, software CYMCAP
1999	A7.1	Transmission capacity design of underground power cables	Okada M., Karasaki, Furuhashi H., and Al	Increase of transmission capacity
1999	C5.8	Estimation of thermal loss of life of MV/LV cables taking into account moisture migration	Caramia P., Losi A., Russo A., Verde P.	Influence of the moisture migration on the cable useful life
2003	C.8.1.7	On-site follow up of the characteristics of controlled backfill, using the TDR moisture measurement method	Gille A., Geerts G., Xanthoulis D., Fonder N., Mella J-P.	Example of in-situ moisture content measurement
2007	B4.5	Re-evaluation of 150 kV cable capacity	Van Houwelingen D., Van Rossum J.	Thermal characterisation of soils, resistivity measurement /example of practical characterisation of the surrounding environment of an underground link
2007	C5-2-2	Application results of real-time ampacity estimation system and intelligent power cable system	S.H. Nam and al	Change of location of hot spots over time
2007	C.5.2.3	Adaptive monitoring program for dynamic thermal rating of power cables	H. Brakelmann, H. Hirsch, A. Röhrich, H-P. Scheiffarth, J. Stammen	Adaptive rating calculation
2007	C5-2-5	Development on the mortar material for cable systems in a directional drilling	Y. Nishiyama, H. Mashima, M. Uesawa, and al	Development of a new infill backfill with low thermal resistance. Directional drilling
2011	C.2.3	Improving distribution system reliability	Donald Bainbridge	Effect of combination of hot summers and higher loads. A paragraph is dedicated to "climate change".
2011	C.7.1	Improvements of underground cable distribution systems in the	Robles E., Sierra	Change of thermal environment

		area of Mexico City	V.	due to expansion of city
2011	C.7.6	New characterization and design of power cables systems	J. Orella, J. Besteiro, V. Macias	Thermal characterisation of soils, resistivity measurement /example of practical characterisation of the surrounding environment of an underground link
2011	C.8.6	Research of upgrading ampacity for high voltage XLPE power cable	Fan Y. and al	Increasing ampacity
2011	C.9.2	Dynamic rating of transmission cables	E. Jacobsen and al	This paper shows possible rapid changes of the thermal characteristics of cable surroundings
2011	C.9.6	Impact of moisture migration on the current rating of high operating temperature cables	J. Pilgrim, D. Swaffield, P. Lewin	Modelling of moisture migration (allowing overloading of the cable system)
2011	C.10.2	A new backfill material with an extremely high thermal conductivity	H. Brakelmann, J. Stammen, J. Dietrich, R. Böing, H-P. May	Development of new backfill with improved thermal performance
2015	A2.6	Study on thermal backfill materials for directly buried cables	Burceanu, M, Van der Borght, Pieter	Covers the fundamentals, with a survey of six different backfill materials. Stresses the importance of good installation practice, in particular the laying and compaction of the backfill material
2015	D9.6	Thermal ratings of submarine HV cables informed by environmental considerations	Hughes, T, Henstock, T, Pilgrim, J.A., Dix, J, Gernon, T., Thompson, C.	Attention is paid to convective heat transfer through the sediment present above submarine HV cables. The simulations show that sediment permeability has a significant influence on the nature of the heat transfer, and hence the thermal ratings of submarine HV cables. Migration of sedimentary bedforms can result in variations in the depth of the seabed of up to 5m per year through mechanisms that are not operative on land.
2015	E2.1	A guide for rating calculations of insulated cables	De Wild et al	Covers the topic of cable rating very broadly, based on the extensive and comprehensive work of WG B1.35. A survey reveals a lot about installation practice, which in turn gives some information about the things WG B1.41 should look out for.
2015	C8.1	Ampacity and other design considerations for medium voltage cables used in	E. C. Bascom III, P. C. Schenectady, R.	A timely publication on the additional challenges imposed by renewable energy cable connections making the point that

		renewable energy applications	W. Allen Jr.	some installation problems may well appear after warranty periods when ownership has been transferred
2015	C8.2	Enhanced medium voltage cable ratings by improving cable trench design and thermal conditions	Meijer, S, De Wild, F, Al Aghbari, A, Al Neaimi, M, Ashaar, M, Alabbadi, M.	An interesting case study from Dubai, showing the benefit of careful selection and use of local materials to produce good results in a challenging environment
2015	F9.2	Computationally light two-zone moisture migration modelling for underground cables - critical temperature vs. Critical heat flux	Millar, R.J., Degefa, M., Lehtonen, M., Ekström, J., Brakelmann, H.	The main development of this paper modelling moisture migration is to delineate dry from wet regions with a critical heat flux rather than the traditional critical isotherm. The results, based on measurements in a graded sand backfill surrounding a cable-scale heating tube indicate that, at least for the experimental setup, the critical heat flux produces slightly better results.
2015	E2.2	Gravitational cooling of cable installations	Brakelmann, H., Wasch, V.	Describes a system for cooling hot spots by installing water pipes near the cable and near to the ground surface. These pipes are connected to each other such that the water circulates by gravitational flow, providing cooling without any active elements (e.g. pumps).
2015	E3.6	A novel cooling solution for an intersection of a 2x2 duct bank with hv cables crossed by a steam pipe	Brakelmann, H., Anders, G., Cherukapalli.	Presents the results of ampacity studies and proposes remedial actions for the situation where a steam pipe and several distribution cable circuits cross a duct bank with HV transmission cables.

Table B.2 contains a list of the Jicable papers found. The most relevant are summarised below:

- In paper 2007 B4.5, original design values of thermal resistivity of soils (calculated about 40 years before the study) are compared with the present and actual values at different critical locations. The surroundings have changed over time resulting in a lower transport capacity. Nevertheless, the use of the cyclic rating described in IEC 60853 permits a prolonged use of the 150 kV cables with moderate reduction in transport capacity.

- In paper 2007 C7.1, climate changes during the dry season in the area of Mexico City or overload conditions, combined with the extension of the city (construction of roads, sidewalks) is responsible for overheating of 23 kV cables. The choice of modern cables with a reduced insulation thickness and a higher operating temperature allowed the installation of cables with a larger cross section of conductor while not requiring any additional civil work.

- In paper 2011 C2.3, the effect of climate change on 22 kV cable systems in the state of Victoria (Australia) is clearly described:

"In Victoria, hotter summers have increased air-conditioner installation rate and usage resulting in increased hot weather loading on the electrical system. Soil temperature maximum was 20°C when the cable system

was designed in the 1970s. In recent years soil temperature at cable depth has been measured at over 25 °C in southern Victoria.

Rainfall average has decreased and water evaporation from the soil has increased since the 1990s. The clay soil has dried and contracted causing cracks in the soil thus increasing the thermal resistance of the soil.

The cables are running hotter due to:

- increased current loading of air-conditioners
- increased soil temperature
- increased soil thermal resistance of dry, cracked soil, which traps in heat from cables.

The contraction in the dry clay is visible as cracks in the surface and as undulations in the roadways.

The undulation in the roadways, cracking and mal-alignment of concrete footpaths driveways and gutters also suggests that there is soil movement around the cables and joints "inducing mechanical forces into joints."

- In paper 2015 C8.1 the challenges of installations, especially wind plant connections, are discussed in some detail, noting that usually assumed soil characteristics may be overly optimistic for renewable projects.

B.3. PAPERS AVAILABLE FROM IEEE EXPLORE

The papers in Table B.3 are mostly from AIEE/IEEE and IEE/IET journals, but some conference publications are also included.

Table B.3 Papers available from IEEE Explore

Year	Journal	General topic / Title	Authors	Content summary and relevance to WG B1.41	Digital Object Identifier
1949	AIEE Transactions, vol. 68, issue 1	The Temperature Rise of Buried Cables and Pipes	J.H. Neher	An early classic paper on cable rating with some relevance to the WG, in that it includes a plot of ambient temperature vs. depth.	10.1109/T-AIEE.1949.5059897
1956	Proceedings of the IEE - Part A: Power Engineering, vol. 103, no. 11	An evaluation of two rapid methods of assessing the thermal resistivity of soil	M. W. Makowski, K. Mochlinski	Introduces and compares methods for thermal resistivity measurements, including the now ubiquitous thermal probe. A nomogram is also presented by which the TR of sand-clay mixtures can be estimated	10.1049/pi-a.1956.0123
1957	Power Apparatus and Systems, Part III. AIEE Transactions, vol. 76, no. 3	Thermal and Mechanical Problems on 138-Kv Pipe Cable in New Jersey	A.S. Brookes, T.E. Starrs	A real-world installation with intensive site testing revealing substantial variance in TR along a typical cable route, and generally highlighting some of the potential concerns of WG B1.41	10.1109/AIEEPA S.1957.4499655
1958	Power Apparatus and Systems, Part III. AIEE Transactions, vol. 77, no. 3	Control of the Thermal Environment of Buried Cable Systems-Part II	L.H. Fink, L.L. Smerke	Covers some interesting issues, including backfilled trench dimensions and an appendix that covers the ambient temperature increasing effect of using more thermally conductive backfills. However, this latter effect, while interesting, was considered insignificant in terms of its effect on cable ratings	10.1109/AIEEPA S.1958.4499882
1964	IEE Proceedings, vol. 111, no. 5,	Characteristics of Soil Affecting Cable Ratings	A.G. Milne, K. Mochlinski	Contains a clear description of moisture movement in porous media and good overall data from the south of England, plus results from 16 field studies spanning seven years	10.1049/piee.1964.0160
1964	IEE Proceedings, vol. 111, no. 5, paper	Influence of soil-moisture migration on power rating of cables in h.v.	A.N. Arman, D.M. Cherry, L. Gosland, P.M. Hollingsworth	System operation and the influence on cable loading are considered. In particular the case of the failure of two 132kV 0.6in ² gas-filled cables	10.1049/piee.1964.0159

	4465P	transmission systems	h	due to thermal instability of the backfill is described. A method of calculating ratings taking into account migration of moisture is described. A revision of the suitable critical isotherm for cables under long-term high loading is suggested, which may be the origin of the 50 °C isotherm.	
1965	Power Apparatus and Systems, IEEE Transactions, vol.84, no.9	Evaluation of Soil Thermal Characteristics	L.H. Fink	2-stage sampling scheme to ascertain the upper limit for a cable route's thermal resistivity	10.1109/TPAS.1965.4766260
1965	IEE Proceedings, vol.112, no.12,	Thermal analysis of power cables in soils of temperature-responsive thermal resistivity	H.N. Cox, R. Coates	A mathematical paper showing how the effect of drying out can be taken account of in rating calculations. Early treatment of 2-zone moisture migration, with consideration of summer and winter conditions. It talks about the "current trend" of measuring cable temperatures in service and a need for the same if naturally cooled cables are to be fully exploited.	10.1049/piee.1966.0202
1968	Power Apparatus and Systems, IEEE Transactions, vol. PAS-87, no. 4	The Thermal Behaviour of Cable Backfill Materials	J.I. Adams, A.F. Baljet	Includes an analysis of various backfill materials, moisture migration and some long-term monitoring	10.1109/TPAS.1968.292094
1973	IEE Proceedings, vol. 120, no. 1	Moisture Control of Cable Environment with Particular Reference to Surface Troughs	E.J. Brooks, C.H. Gossling, W. Holdup	Cable irrigation to restore and improve TR, cement bound sand. Focus is on trough installations near surface, also with consideration of seasonal changes in ambient moisture, rainfall cycles, etc.	10.1049/piee.1973.0012
1975	IEE Proceedings, vol. 122, no.11, paper no. 7517P	Developments in UK cable-installation techniques to take account of environmental	H.N. Cox, H.W. Holdup, D.J. Skipper	Gives an account of the evolution of backfills in the UK as of 1975. Special backfills, drying out etc. taken into account in design. In the discussion the method of	10.1049/piee.1975.0307

		thermal resistivities		testing sands for moisture migration tendency (or lack of it) is introduced with the proposal of using a TR of 0.9 K·m/W for sands passing the proposed test.	
1978	6th International Heat Transfer Conference, Proceedings of.,	Simultaneous diffusion of heat and moisture around a normally buried EHV cable system	R.J. Preece, J.A. Hitchcock	This would seem interesting, and is referenced by Groenvald et al, and more recently by Pilgrim et al, but has not yet been found	
1979	IEE Proceedings, vol. 126, no. 6, Proc. IEE paper no. 8341P	Soil thermal and hydrological characteristics in designing underground cables	F. Donazzi, E. Occhini, A. Seppi	Soil characteristics in relation to moisture migration stability. Characteristics of soil described and test for moisture migration introduced in detail. Proposed that TR of a soil can be determined solely from its porosity and saturation degree and that the hydrological stability is a function of the saturation degree only. Hydrological equilibrium is then also dependent on the thermal gradient. The method enables ratings to be optimised using natural soil backfills with good hydrological characteristics. Very clear and useful methodology, although some question about oversimplifying in terms of assumption that backfill has only one constituent bulk material with known TR.	10.1049/piee.1979.0119
1979	IEEE/PES Transmission and Distribution Conference and Exposition	Predicting Thermal Stability And Transient Response Of Soils Adjacent To Underground Power Cables	J.G. Hartley, W.Z. Black	Thorough, and backed up with tests. This is good source material on critical moisture content and time to dryout	10.1109/TDC.1979.712683
1980	Power Apparatus and Systems, IEEE Transactions, vol. PAS-99, no. 3	Thermal Instability and its Prediction in Cable Backfill Soils	H.S. Radhakrishna, F.Y. Chu, S.A. Boggs	This provides interesting data, techniques, and a description of moisture equilibrium and thermally induced moisture migration, and includes comments by other experts in a very interesting discussion.	10.1109/TPAS.1980.319714

1981	Power Apparatus and Systems, IEEE Transactions, vol. PAS-100, no. 9	Practical Aspects of Applying Soil Thermal Stability Measurements to the Rating of Underground Power Cables	M.A. Martin, R.A. Bush, W.Z. Black, J.G. Hartley	Focusing on heat flux, the time to dry-out, etc.	10.1109/TPAS.1981.316975
1982	IEE Proceedings A, Physical Science, Measurement and Instrumentation, Management and Education - Reviews, vol.129, no.3	Natural and forced-cooling of HV underground cables: UK practice	D.E. Williams	A very comprehensive review paper, with motivation for forced cooling circuits from a 1962 cable failure. The various types of forced and natural cooling systems are discussed, with consideration of the reliability and failure alarms of secondary equipment.	10.1049/ip-a-1.1982.0023
1984	IEE Proceedings C, Generation, Transmission and Distribution, vol.131, no.2	Improved method to calculate the critical conditions for drying out sandy soils around power cables	G.J. Groeneveld, A.L. Snijders, G. Koopmans, J. Vermeer	A clear paper adding hysteresis to the Philip and de Vries modelling of moisture migration	10.1049/ip-c.1984.0007
1987	IEE Proceedings C, Generation, Transmission and Distribution, vol.136, no.6	Soil physical routed survey and cable thermal design procedure	G. Koopmans, G.M.L.M van de Wiel, L.J.M. van Loon, C.L. Palland	This paper utilises the moisture migration methodology published by Groeneveld et al. in 1984 to develop and demonstrate a thermal rating procedure. See Section 2.5.9	10.1049/ip-c.1989.0047
1991	Circuits and Systems, 1991, Proceedings of the 34th Midwest Symposium on	Effect of Trench Geometry, Cable Size and Top Layer Thickness on the Heat Dissipation in Buried Cables	M.A. Hanna, A.Y. Chikhani, M.M.A. Salama	The change to surface conditions over old cable installations may relate to the analysis in this paper	10.1109/MWSCAS.1991.252005

1992	IEEE Transactions on Power Delivery, vol. 7, no. 2	Field test program and results to verify HPFF cable rating	J.A. Williams, E.C. Bascom, T. Aabo, B.M. Horgan	Forced cooling for hotspots - it looks at an existing cable, and existing cables are in large part what we are on about, but nevertheless, it doesn't address our area	10.1109/61.127114
1994	IEEE Transactions on Power Delivery, vol. 9, no. 1	Controlled Backfill Optimization to Achieve High Ampacities on Transmission Cables	J.A. Williams, D. Parmar, M.W. Conroy	Practical paper about the application of Neher-McGrath for rating, installation techniques, backfill properties and thermal resistivity testing, with a great discussion section	10.1109/61.277727
1997	CIRE 97, 2-5 June 1997, Conf. Pub. 438	Long Term Investigation of Thermal Behaviour of 110 kV Underground Transmission Lines in the Belgrade Area	M.R. Sredojević, R.M. Naumov, D.P. Popović, M.D. Simić	An empirical report on the long-term drying out parts of a underground transmission connection, with observations on the other things that can change ampacities - district heating pipes, etc.	10.1049/cp:19970548
1998	IEEE Transactions on Power Delivery	Thermal Analysis of Power Cable Systems in a Trench in Multi-Layered Soil	M.A. Hanna, A.Y. Chikhani, M.M.A. Salama	The effect of cable spacing, diameters, trench width, etc. May aid re-rating of installations that have other heat sources (cable or other) installed after original installation	10.1109/61.660894
1998	Power Engineering Journal, June 1998	Auckland Central Business District Power Failure	Bryan Leyland	A cautionary tale, where original rating for winter conditions was no longer relevant for peak summer loading – indicating that seasonal moisture content and ambient temperature should be modelled	10.1049/pe:19980303
1998	Power Engineering Journal, vol. 12, no. 6	Auckland Central Business District supply failure: the ministerial inquiry	Bryan Leyland	More technical detail than the earlier article	10.1049/pe:19980605
1999	IEEE Transmission and Distribution Conference, vol. 1	Rating of Cables in a non-uniform Thermal Environment	G. Anders, J.M. Braun, M. Vainberg, S. Rizzetto, H. Brakelmann	Delivers derating factors for street crossings - outlines a dynamic feeder rating system	10.1109/TDC.1999.755321
2000	PowerCon 2000	Probabilistic Cable Rating Based on Cable	H.C. Zhao, J.S. Lyall, G. Nourbakhsh	Uses thermal spheres for long term TR measurements and correlation with rainfall	10.1109/ICPST.2000.897169

		Thermal Environment Studying			
2001	CIREC 2001, 18-21 June, Conf. Pub. 482	Increasing the Capacity of Cable Systems using Cable Asset Management Based on Thermal and Mechanical Properties	B.J. Grotenhuis, J.E. Jaspers, A. Kerstens, A.H. van der Wey, F.H. de Wild	This is more information on KEMA's real-time glass-fibre based monitoring system, and the statement in the paper related to subsidence in soft and weak ground suggests that this type of degradation is of some concern in the Netherlands.	10.1049/cp:20010804
2003	2003 IEEE PES Transmission and Distribution Conference and Exposition, vol. 3	Ampacity Evaluation of High-Pressure Gas-Filled (HPGF) Pipe-Type Cables Under Bridgeport Harbor	E.C. Bascom, M.A.Pasha, W. Zenger	A good report on a complicated ampacity problem, and an evaluated set of proposals to mitigate the problem	10.1109/TDC.2003.1335045
2006	IEEE PES GM	Major Factors Affecting Cable Ampacity	Francisco de León	Clear overview of ampacity-affecting factors, including installation conditions. Uses Cymcap, a commercial standard-based rating program	10.1109/PES.2006.1708875
2007	CCECE 2007	Advanced Thermal Field Sensitivity Analysis of Power Cables	M.S. Al-Saud, M.A. El-Kady, R.D. Findlay	Another paper concerning non-homogeneous thermal environments, and so of some interest	10.1109/CCECE.2007.40
2008	IEEE Transactions on Power Delivery, vol. 23, no. 2	Effects of Backfilling on Cable Ampacity Analyzed with the Finite Element Method	F. de León, G. J. Anders	This paper reverse engineers T4 from FEM simulations, but then covers a wide range of backfill/native soil ratios and non-homogeneous installations to show the limits of the standards	10.1109/TPWRD.2008.917648
2013	IET Generation, Transmission & Distribution	Quantifying possible transmission network benefits from higher cable conductor temperatures	J. Pilgrim, P. Lewin, A. Gorwadia, F. Waite, D. Payne	This work develops a <i>Dynamic Backfill Model</i> , a development of Philip and de Vries, implemented in FEM software. High temperature cable research puts new emphasis on backfill modelling	10.1049/iet-gtd.2012.0004
2015	IET Generation, Transmission &	Thermal modelling and analysis for offshore submarine high-	Z. Huang, J. A. Pilgrim, P. Lewin, S. Swingler and G.	This paper applies FEA for submarine cable rating, including sophisticated modelling of two-phase	10.1049/iet-gtd.2015.0551

	Distribution	voltage direct current cable crossings	Tzemis	porous media	
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The following paragraphs summarize just a few of the most relevant publications, noting that none of the publications fit our topic perfectly, but that they address the key environmental factors that we should look out for, in terms of possible degradation that affects the rating (ampacity) of important cable connections.

The first paper in Table B.3, is an early paper by J. H. Neher, "The temperature rise of buried cables and pipes," 1949, which is mentioned in this context because of an early plot of ambient temperature variation as a function of burial depth, reproduced here as Figure B.1. There is also some interesting advice on in-situ thermal resistivity and diffusivity testing using spherical and short cylindrical radiators.

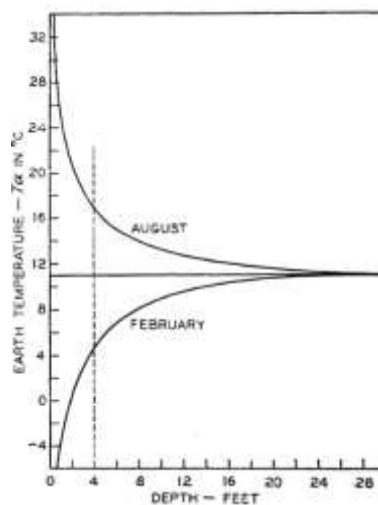


Figure B.1 Typical vertical temperature gradients in the earth at extreme seasons of the year.

The 1957 paper by Brookes and Starrs, "Thermal and Mechanical Problems on 138-Kv Pipe Cable in New Jersey," highlights the variation in TR measurements along a cable length and, applying the findings and theory earlier developed by Fink and Neher, tests various backfills, noting that a well graded, well compacted high quartz content backfill yielded good results. Steep gradients are treated, but more for the mechanical concerns of pipe cables than for thermal environment reasons. The paper is of some historic interest, as it involved extensive on-site testing.

"Influence of soil-moisture migration on power rating of cables in h.v. transmission systems" is inspired in part by cable failures in 1962 in Belvedere, London, that were likely caused by drying out of the cable backfill at temperatures not previously thought to have been a problem in this regard. The observation is made that moisture migration is likely to occur in summer if cable surface temperatures are in the vicinity of 70 °C. This was reinforced by results from a drought in 1959, and then, dramatically, from a failure of two quite new (at that time) transmission circuits in Belvedere, London in 1962, caused by severe drying out of the sand surrounding the cables. The paper looks at cable loading from a system point of view, and investigates the aforementioned cable failures. Tree roots were found close to one of the fault locations. The paper also discusses general rating procedures and temperature mitigating procedures such as water cooling. Section 3.3 of the paper may be one source of the famous 50 degree isotherm and some European cable rating practices based on conductor temperatures of 65 °C. Quoting from Section 3.3:

"General experience had indicated that, where heating is intermittent, as with cyclically loaded cables, migration of moisture becomes serious only at temperatures well in excess of 50 °C. Moreover, the figures from countless route surveys had shown that, in normal moist soils, thermal resistivities were often much less than the 1.2 K·m/W on which current ratings were based, and so cable temperatures remained well below the nominal full-load value even when loading was continuous."

There is a lot more practical and theoretical material in this paper, including a brief consideration of 'weak concrete mixture', with some concern expressed about the possible mechanical stresses this might impose

on the cables. Another paper, also replete with field experience, is "Characteristics of soil affecting cable ratings" by Milne and Mochlinski. This gives a good general description of moisture movement in porous media, and gives results from a wide scale test covering 16 locations over a period of 9 years, investigating the effect of ground water level and rainfall on thermal resistivity. From WG B1.41's point of view, however, the only long-term trend noted is that minimum winter thermal resistivities tend to decrease with time due to the slow improvement of compaction of the bedding soil/backfill around the cables. What may be of interest is possible change in what the paper refers to as moisture deficit, which is associated with the relationship between rainfall, evaporation and ground water level. Further, WG B1.41 suggests that if this balance can be shown to be changing due to climate change or more obvious thermal intrusions due to above ground or subterranean construction, then long-term changes in thermal resistivity (and/or external thermal resistance) might be expected.

The long-term (one year) monitoring reported in "Thermal Behaviour of Backfill Materials," by Adams and Baliet, points to one consideration worthy of our attention, which is that if uniform sands are installed in wet conditions, or well-graded backfills are installed without adequate compaction, there may be subsequent severe degradation in the thermal performance after long-term moisture migration has occurred, which might not be picked up in initial tests during installation in initially wet conditions. Otherwise, this paper also covers moisture migration, with some moisture content related measurements of vapour flow and suction pressure, comparing uniform sand with well-graded stone screenings. It is interesting to see the understanding of moisture migration evolving over time.

By 1979 the methodology for analysing moisture migration was starting to be more clearly elucidated by the adoption of the equations by Philip and de Vries, for example, an Italian paper, "Soil thermal and hydrological characteristics in designing underground cables" by Donazzi, Occhini, Sen, and Seppi. They formulated the net thermal resistivity of a porous media in terms of saturation degree, porosity, dry density, the thermal resistivity of the constituent bulk material and the thermal resistivity of still water. WG B1.41 notes that the formulation presented in this paper presupposes such knowledge of the bulk material, and that there is only one constituent material in the backfill. This paper analyses the hydrological behaviour of porous media in a simple and clear way, also discussing the likelihood of drying out at cable burial depths due to evaporation from the ground surface between rainfalls, which was not, at least in this paper, considered a major issue in temperate climates. The other paper in 1979 was by Hartley and Black, "Predicting thermal stability and transient response of soils adjacent to underground power cables," which introduces the concept of time to dryout.

The following year produced a paper by Radhakrishna, Chu and Boggs from Ontario Hydro, "Thermal Instability and its Prediction in Cable Backfill Soils." Black, Bush and Martin took exception to the statement in the paper, "...the sole factors determining cable backfill thermal stability are moisture content and compaction," and in the discussion give evidence of why, illustrating that waiting a bit longer would have shown moisture movement. In other respects, the two-power method that Radhakrishna et al propose seems pragmatic, provided one waits long enough to check that moisture migration will not occur after a 'long enough' time interval.

Martin, Bush, Black and Hartley got their turn again in 1981, producing a paper, "Practical Aspects of Applying Soil Thermal Resistivity Measurements to the Rating of Underground Power Cables," which observed that, for a single circuit, thermal stability is a function of heat dissipated from the cable per unit length, soil moisture content and cable diameter, with, all other things being equal, the time to dryout being proportional to the square of the diameter of the cylindrical heat sources. They also discuss multiple cable systems.

The 1982 paper by Williams, "Natural and forced-cooling of HV underground cables: UK practice," is a very comprehensive review paper, with at least some of the motivation for forced cooling circuits coming from the famous 1962 London cable failures. Various types of forced and natural cooling systems are discussed, with consideration of the reliability and failure alarms of secondary equipment.

"Controlled Backfill Optimization to Achieve High Ampacities on Transmission Cables" by Williams, Parmar and Conroy is a combination of Neher-McGrath theoretical modelling and real-world application, followed by an expert discussion. Some very high thermal resistivities were encountered in the volcanic installation region (Hawaii).

The title of the paper “Long Term Investigation of Thermal Behaviour of 110 kV Underground Transmission Lines in the Belgrade Area,” by Sredojević, Naumov, Popović and Simić would seem to be promising for WG B1.41. The main aim of this paper, however, was to investigate the application of a special backfill to mitigate hot-spots in an existing connection. The test facility referred to in the paper seems to be very useful for long-term monitoring, but the paper uses it to show the improvement in hot-spot performance due to installing a well-graded sand with weak mix cement backfill. Otherwise, it seems that the temperature responses along the cable route did not change much in 8 years. The paper describes a promising test set-up on a real 110kV cable circuit and reaffirms that the subsequent installation of heat generating services in the thermal vicinity of an underground cable, long periods of dry weather, and the poor choice of backfill in the first place are things to watch out for.

Another high profile HV cable failure that had a number of contributory factors is related in “Auckland Central Business District Power Failure” by Leyland. Quite simply, the thermal environment of the cables was not what was assumed, due to poor backfill, a shift in peak loading from winter to summer, and perhaps some inadequacy in the jointing practice, not to mention the unavailability of spare parts and the virtual impossibility of getting permits to provide Auckland with a long overdue new HV connection (the cable failures expedited the granting of the necessary permits...). This provides a cautionary tale, because, although most cities are not built on and in the vicinity of old volcanoes, the assumptions often made about the thermal behaviour of sand backfills may not hold true – however, this relates to incorrect assumptions at installation and a subsequent failure to reassess the ampacity of the cables when the time of peak loading changed, not really long term degradation.

Another paper from the southern hemisphere, “Probabilistic Cable Rating Based on Cable Thermal Environment Studying” by Zhao, Lyall and Nourbakhsh uses Lyall’s thermal spheres for long term TR measurements. Good correlations with rainfall are revealed. Such installations may also prove useful in the future, for monitoring long-term degradation of the thermal environment.

“Increasing the capacity of cable systems using cable asset management based on thermal and mechanical properties” by Grotenhuis, Jaspers, Kerstens, van der Wey and de Wild provides information on KEMA’s real-time fibre optic monitoring system. Of interest to us is a paragraph near the end, directly quoted here:

“Another very frequent cause of underground cable failures in the Netherlands is the extreme subsidence of cables in the softer and weaker grounds. Ground subsidence will lead to an increased mechanical strain in the energy cable, which will result in a significantly shortened cable life-time...”

This statement would indicate that ground subsidence is a likely form of degradation to watch out for, at least in the Netherlands.

Another report on a complicated rating (ampacity) problem “Ampacity Evaluation of High-Pressure Gas-Filled (HPGF) Pipe-Type Cables Under Bridgeport Harbor” by Bascom, Pasha and Zenger evaluates a set of mitigation proposals.

B.4. PAPERS FROM EPRI, ERA AND MISCELLANEOUS SOURCES

Table B.4 EPRI Papers

General topic / Title	Journal	Year	Authors	Content summary and relevance to WG B1.41	Abstract
Backfill Materials for Underground Power Cables, Phase 1	EPRI EL 506 project 7841-1	1977	Univ. of California, Berkeley	Trenching, Additives, Soil TR, Soil Thermal stability, Higher operating	Improved installation methods & treatments to reduce TR and improve thermal stability. Special thermal needle method. Cement, waxes and asphalt: low TR even when dry (0.3 to 0.6 K·m/W). Appear durable. Testing not completed. Tests planned

				temps.	on moisture migration and field tests on novel backfills. Wax treatment seems favoured and field, durability and thermal stability tests were planned on this.
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Table B.5 ERA Papers

General topic / Title	Journal	Year	Authors	Content summary and relevance to WG B1.41	Abstract
Thermal properties of soil	ERA internal F/Aj/T2	1954	Panel Aj	An abstract of progress reports carried out from 1930 to 1935	Tests of moisture movement in 300mm long horizontal tubes. Problems with clay due to cracking; In sand depends on temp gradient & dry density.
Soil moisture migration away from cylindrical heaters: proving tests in natural soil	ERA internal F/A/T420	1957	Sub-committee A	Investigating moisture migration and drying in tests up to 100 hours	
Soil moisture migration away from cylindrical heaters: time required for the commencement of runaway in sandy soils	ERA internal 4B/5 (series T)	1961	Committee 4B	Theoretical and practical studies of onset of thermal runaway using cylindrical heater model	Acceptable agreement between theory and practice. 3 equations presented. Commencement of drying at 40°C independent of soil type.
The structure of the soil and a critical review of the mechanisms of soil moisture retention and migration	ERA internal F/T193	1961	M J Vanner	Examines published theories about moisture migration	Considers effects on cables taking into account the structure of the soil. This is studied in detail. Forces holding water in the soil are considered and mechanisms of movement.
The ERA transient needle probe	ERA internal 69-25	1969	R G Oliver		Describes the construction and use of a needle probe for the measurement of thermal resistivity of cable backfills in-situ.
Assessment of soil ambient temperatures	ERA internal No 5289	1969	K Mochlinski	A study of existing records of soil temperatures in the UK	Instructions given to determine limits of expected soil temperature at any point in the UK at any time, down to 6m depth.
Field investigation of thermal properties of	ERA internal 69/72	1969	ERA	A 10 year program of measurements at 17 cable	An investigation was carried out over 10 years at 17 cable installations. Measurements were made of cable and soil temperatures. Load, and rainfall and

soil and their effects on cable ratings				test sites. Relevant to B1.41	water table details were recorded when possible. Specific conditions relating to each site were also considered. Different soil types were considered. Excavations were carried out at 3 sites at the end of the 10 year period. A difference in density between the original soil and the bedding/trench fill (lower density) was indicated by results from all 3 sites. The biggest difference was at a site backfilled under adverse weather conditions. Moisture contents were higher in the backfill than the original material in all case, correlating with a decrease in dry density "it reflects perhaps the common opinion that cable trenches act as drains". Dry density of virgin soil increased with depth, however there was not a variation of moisture content with depth (based on one set of measurements). At one site, where the water table was always high in winter, the TR showed a continuous fall from 0.7 -0.8 K·m/W in 1956 to 0.6 - 0.65 in 1963. "This is conceivably due to slow consolidation of the bedding".
A mechanism of liquid phase moisture migration due to thermal gradients in sand	ERA internal report no. 84-0025	1984	Highgate	Method of producing samples for moisture migration tests	Measurements of moisture migration are described and a new theory of liquid phase migration introduced which is proposed to explain long term effects observed.
A method of assessing cable external thermal resistance based on considerations of moisture migration	ERA internal report no. 84-0026	1984	Highgate	Review report of work in 1960s	Mechanism of liquid phase moisture migration propose, based on variation of suction characteristics with temp. Implications on cable ratings reviewed.
Rapid moisture migration in soil	ERA internal report no. 84-0144	1984	Highgate	Theoretical and experimental study to confirm theory of liquid and vapour phase flow contributing to moisture migration	The theory that liquid phase flow occurs was found to explain qualitatively all phenomena observed in the experimental work. The liquid phase transfer is a rapid process that only occurs under specific conditions. Under these conditions the wet/dry front can move at 10mm per hour when a certain temperature difference exists in the soil. The application of the theory to several cable installations are discussed. The following conclusions follow when the

					theory is applied: Backfill materials should have high suction characteristics. The backfill should have a higher suction characteristic than the site soil. Improved compaction reduces the likelihood of moisture migration. In poor site conditions (e.g. dry) beneficial materials are: sand with a large clay content ("heavy sands") or chalk or crushed limestone. Such materials require very careful compaction to avoid large voids.
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We conclude this Appendix with a German report written in English, Heinrich Brakelmann, "Physical principles and calculation methods of moisture and heat transfer in cable trenches," *etz-Report 10*, VDE-VERLAG GmbH – Berlin und Offenbach 1984

This work showed, some thirty years ago, much the same conclusions that more recent work has shown using FEM (Pilgrim et al, 2011) and 2-zone analytical modelling backed up by measurements (Millar, 2006).

B.5. EXPERIENCES FROM DIFFERENT COUNTRIES

This section contains a range of related information from various countries, in some cases describing a country or utility's general site inspection, testing and installation practices, and in other cases, dealing with topics more directly connected with the long-term behaviour of the installed environment.

B.5.1 Australia - experience of cable backfill aging in TransGrid Australia

B.5.1.1 Introduction and general problems

TransGrid is the owner, operator and manager of the NSW high voltage network connecting generators, distributors and major end users in NSW and the ACT.

This report describes backfill performance observed on Cable 41 (commissioned in 1979) and Cable 42 (commissioned in 2004). Both circuits are generally direct buried. A mix of 14:1 sand/cement was used as cable bedding and a 20:1 sand cement mix was used as backfill (above protective slabs). Subsequent circuits, which are not discussed in this document, were installed in concrete duct banks.

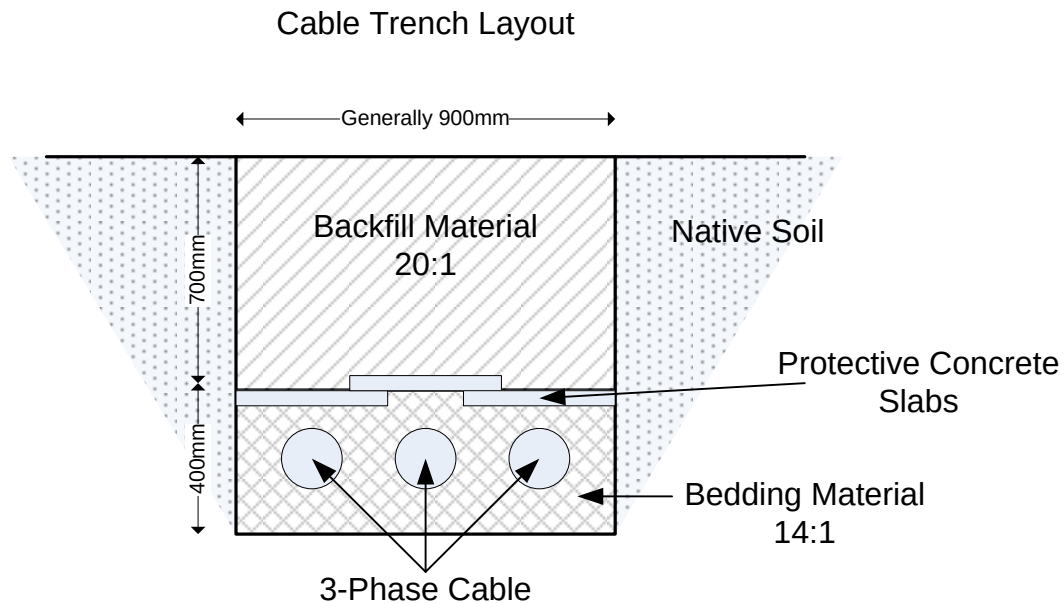


Figure B.2 Cable trench layout

Deterioration of 14:1 exposed to air

About 320 metres of the Cable 41 is laid in tunnel. The tunnel is approximately 10 metres above sea level, naturally ventilated and not lined. Cables are in flat formation in a bedding mix of 15:1 sand/cement (according to the contract). During the last inspection the following was found:

The floor consisted of cemented sand with loose sand on top. The loose sand layer was up to 50mm thick in some parts.

The material underneath the loose layers was still quite solid

It appears that the top exposed layer has lost some mechanical performance. This could be due to the following:

The layer exposed to air dries out with naturally and loses strength, like old mortar.

Water aging – water entering the tunnel at other locations, which had a muddy layer, flowing down towards the tunnel drain, reacting with the cement and/or removing small grain sizes.

Aging caused by water movement

Case - Water leak near Cable 42 - 2012

A leak from a water main near the trench eroded the bedding exposing the cable. It appears that the water flowed down the space left when the shoring was removed, an issue discussed in the WG. The flow of the water eroded the bedding around the cable. This was relatively new (8 years old at the time) 14:1 bedding.

It is appears that sand/cement bedding and backfills may not provide adequate long term protection. Slow leaks can be left for decades until being repaired, removing the cement and fine particles from the cable installation. While the water keeps the cable cool, mechanically the system is compromised. If the water main is repaired the air voids would cause the cable to run hotter and possible even fail.

It is possible that ground water could have a similar effect. Seasonal changes in the water table and flows of rainwater may effectively age the backfill as the water runs through the trench like a gutter. Chemical reactions should not be discounted as groundwater does not have a neutral pH. If the water is acidic then reactions with the cement may occur.



Figure B.3 Inspection of backfill near leaking water pipe

Performance Issue – Bedding loss with adjacent dewatering works – 2005 (26 years old backfill)

During installation of a sewer connection, sediment was removed as part of the dewatering process. The excavation was 3 m from the edge of the cable trench and the sewer pipe was 7.5 m deep. Excessive amounts of sediment were removed during the dewatering process.

The road surface had some very large cracks, one extending approximately 1.6 m deep indicated that a large area over the cables had been affected by the subsidence. The cable was excavated and sag of 110 mm over 2 m was observed. The trench collapse also caused movement in two cable joints which required to be replaced.



Figure B.4 Subsidence due to sewer excavation 3m to one side

The backfill was also found to have lost all cement. Documentation found states that the sand cement ratio was found to be 65:1.

Aging factors at this location

- High risk of acid sulfates
- Close to the water table
- Low point on the cable route more likely to be subject to water aging effects.
- Previous water main leaks resulting in free flow of water through the trench.

The thermal resistivity of the bedding and backfill in this location was found to be poor, between 1.75 and 3.3 K·m/W (fully dried out).

Service crossing C41 – no cement

During the installation of a service crossing the backfill was found to be very soft (very little cement). Acid sulfate soils at this location are unlikely (approximately 40 metres AHD, not noted on risk map). The trench appears to be acting like a channel for water, possibly from a water main leak uphill from location.

Aging factors at this location

- Water main leaks or groundwater flowing through the trench, removing cement and fine particles.
- Slow chemical actions of the water (slight acid) with the cement.



Figure B.5 Cable trench layout

B.5.1.2 General mechanical findings

It appears that sand cement backfill mechanically ages at a lower rate when not exposed to water. During a 132 kV cable relocation, completed by the distributor, it was reported that when the native ground surrounding the cable trench was dry sandstone, backfill material was rock hard and had to be broken away with hard wood wedges; it was a very slow and tedious task.

A similar observation was found when taking a backfill sample at one of the highest points on the route of TransGrid's Cable 41. The backfill and native ground was very dry with rock underneath. At this location there was not much opportunity for moisture to get into the trench. The material was taken from above slab so should not have been unduly affected by underground water flows.

When sensors were installed on Cable 41 in 2008 the inspector stated that "the material was not hard but varied in difficulty with most just scraped out". The substrata at most of these locations were fairly porous (generally clay). TR test results taken during this installation are shown in Section 5.

Discussion with the distributor found the condition of their sand/cement backfill material varied along any cable route. This adds further weight to the theory that substrata material may determine the rate of aging. The distributor has also reported poor TR values on some of their circuits. Some of these installations were purely sand to begin with, as sand/cement was only used since the mid 1970s. This poor backfill has resulted in derating of some circuits, bringing forward capital works.

During discussions with an Australian cement company, their opinion is that 14:1 sand cement is only stabilised sand and has a very short life span. This implies that its use for cable backfill may be unsuitable.

B.5.1.3 Installation issues

It is possible that poor installation of the backfill may have contributed issues observed with Cable 41. Tests conducted in the late 2002 as part of the Cable 42 installation found that it was impossible to achieve the required compaction more than 2 hours after time of batching.

Compaction is important for achieving acceptable TR. It was found that with an in-situ density of less than 1.9, it was uncertain whether a sample would achieve a fully dried out TR target of 1.05 K·m/W.

B.5.1.4 Thermal aging

In early 2016 TransGrid completed a comprehensive backfill and native soil sampling program on Cable 41. The properties of the 20:1 sand cement mixture above the slabs were significantly different than design assumptions (1.2 K m/W). Only two backfill sample out of the 29 locations had a TR of less than 1.2 K m/W.

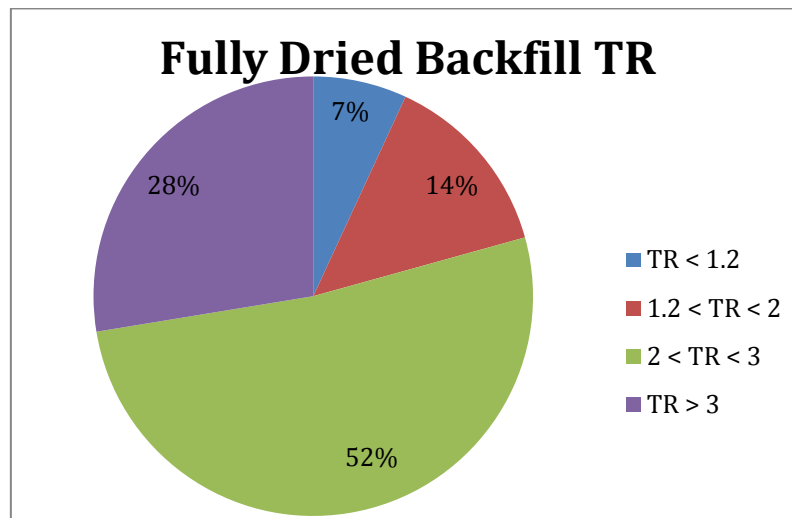


Figure B.6 Results from TR sampling on Cable 41

Surprisingly, there did not appear to be any correlation between TR and height above sea level. It was expected that areas at the top of the cable profile, whilst more likely to dry out, would have been subjected to less fines washout, resulting in a more favourable fully-dried TR.

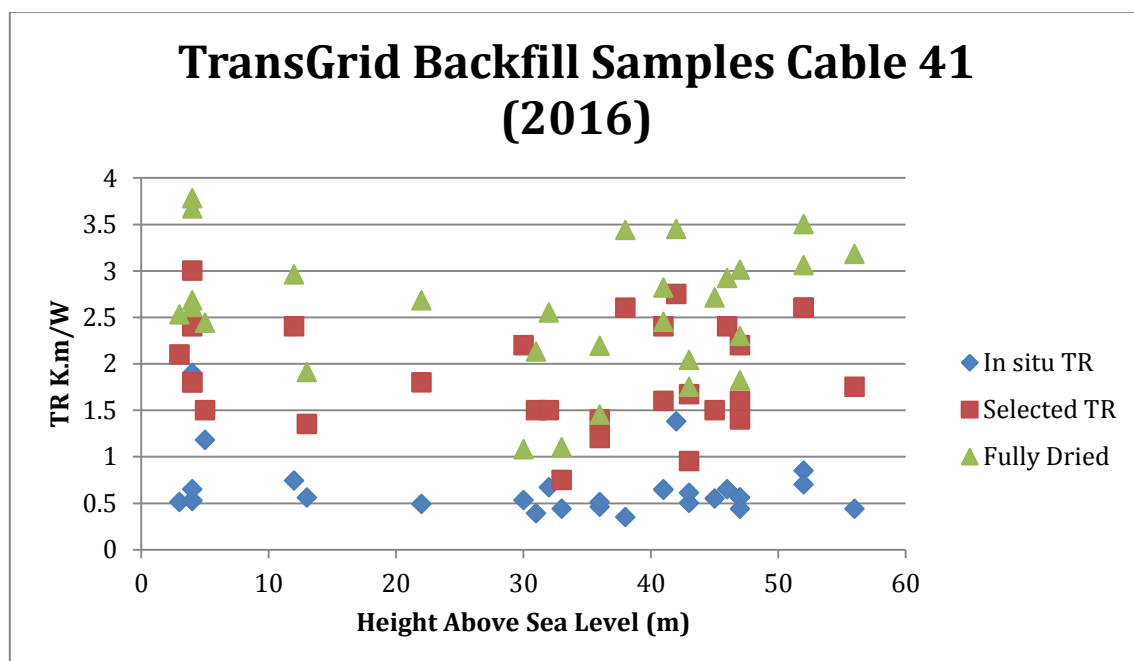


Figure B.7 TR values for samples taken from Cable 41

For on-going ampacity purposes the TR ("Selected TR" in graph above) at the following moisture contents were assumed:

- 2% for sandy backfills
- 5% for clay soils
- 3-4% for a mixture of sand and clay or stone and clay.
- Fully dried TR value of 2.88 K m/W below the cover slabs (not sampled in 2016) was used.

Following on from this study, a continuous section, approximately 60 metres in length, near the constraining location was excavated in the process of obtaining a cable sample. The bedding material around the cables was found to be very inconsistent. In some places, it was found to be extremely hard, like concrete, in others soft like sand. This inconsistency means that it remains uncertain whether or not the sample of cable was representative of the worst condition along the cable route, given the inconsistency over a very small length.

Thermal Resistivity Testing Methods

In 2002, when evaluating new sand/cement backfill material testing methods the following was determined:

- Thermal Resistivity increases by 0.07 K·m/W per hour until time of compaction.
- 14 days curing in water will achieve a 10% lower TR over one day curing in air.
- 28 day curing in water will achieve a 20% lower TR over one day curing in air.
- Drying out the test samples at 90°C as opposed to 105°C achieved lower fully dried out results. The maximum temperatures of the cable jackets could be considered if it is worth diverging from the IEEE test methods.

B.5.1.5 Acid sulfates

Refer to Section 8.3 for a background of acid sulfate soils.

It is possible that acid sulfate soils could contribute to the loss strength by chemical reactions with the cement.

In 1993, a cable bridge on Cable 41 subsided due to tunnelling work underneath. When the cables were exposed during repairs they were not parallel. At certain points the outer cable was touching the trench wall. The backfill and bedding were reduced to mainly sand. Reports at the time stated that they never had to use wedges to remove most of the backfill. There may have been short lengths where the sand/cement was still bonded to the cable jacket, but overall they just shovelled the sand out. This area is identified as a high acid sulfate risk on soil maps.

The bedding loss during dewatering described in Section 3.2 may have also been degraded by acid sulfates as it is also in a high risk area.

B.5.1.6 Mitigation of acid sulfates at time of installation

Acid sulfates were considered during the planning stage of Cable 42, which was commissioned in 2004. In locations where potential acid sulfates were identified, the bedding/backfill material was wrapped in plastic. There were only two sites where this was required: near the riverbeds of the Georges River and Cooks River.

B.5.1.7 Further works

TransGrid considered a project to replace the backfill above the concrete slabs with TSB. The cable has already been derated from 663 MVA to 575 MVA (cyclic ratings) due to higher ground temperatures and backfill issues noted in this document. This derated value made assumptions on some ground moisture improving TR. The latest comprehensive samples detailed above identified the requirement for further derating (to 426 MVA) to allow for long periods without rain. The aim of the TSB was to ensure a firm 575 MVA cyclic capacity.

Due to an emerging issue with the joints (not thermal related) and the construction challenges of backfill remediation within major roads it is intended to replace the lost capacity with new cables over a different route.

B.5.1.8 Conclusions

Installation measures are paramount to ensure correct backfill performance. When conducting quality checks, sand/cement backfill will have a lower TR when fully cured.

Exposure to air and water can weaken sand/cement mix. Water leaks from nearby services and ground water movement appear to mechanically age the backfill.

Aged backfill with good mechanical properties may not necessarily have good thermal properties.

Acid sulfate soils may accelerate ageing of sand/cement backfills, as the sulfate can react with the cement.

- There is only a very small part of the TransGrid cable routes that are within acid sulfate risk zones.
- With regards to WG B1.41, it is worth noting that submarine cable landing areas would often be subject to acid sulfate soils.

Some other conditions that may influence the aging process are:

- Compaction density: loosely compacted material allows increased free water flow through the trench
- Quality and composition of backfill material
- Substrata material, when the base is very dry, such as when going through rock, can reduce external aging influences, such as water flowing through the trench.

Further note: impacts of acid sulfate soils:

It has been hypothesized that the presence of acid sulfate soils may degrade sand/cement backfills.

Sulfates are able to react with the free lime released during the hydration of Portland cement and with the calcium aluminates present in the cement. The reaction results in the formation of a range of sulfate compounds. The key mechanism is the replacement of calcium in calcium silicate hydrates that form much of the cement matrix. This leads to a loss of the binding properties^[1].

In concrete, the addition of fly ash is used to combat sulfate attack. As there is no fly ash in sand/cement backfill it could be assumed susceptible to acid sulfates.

In concrete, the failure mode is expansion and associated cracking because these compounds occupy a greater volume than the original concrete.

The following are risk factors for the potential of acid sulfate soil ^[2]:

- Sediments of recent geological age (Holocene, i.e. last 10,000 years)
- Land and soil at elevations less than 10 m AHD (Australian Height Datum, 0 is mean sea level). Note some sources quote 10 m AHD.
- Sediments and tidal lakes of marine origin
- Coastal wetlands and swamps, waterlogged or scalded areas, inter-dune swales or coastal sand dunes (if deep excavation or drainage is proposed), coastal sand sheets
- Areas where the dominant vegetation is mangroves, reeds, rushes
- Other swamp tolerant, salt tolerant or marine vegetation
- Older estuarine sediments of Pleistocene age (only an issue if these have been preserved in an anaerobic state since they were laid down).

[1] Sulfate attack on concrete – What it is and how to stop it. – *Fly Ash technical Notes No.1 1995* – Ash Development Association W. Barry Butler

[2] Acid Sulfate Soils, Assessment Guidelines – Acid Sulfate Soils Management Advisory Committee, NSW Agriculture

B.5.2 Belgium

In Belgium a controlled backfill is laid around all the HV cables installed in trenches and bentonite is used to fill cable ducts in the case of high thermal resistivity of the soil around the cable ducts. Medium voltage network operators do not perform a soil analysis before cable laying.

The following documents, are used for thermal assessment:

- Electrical Research Association – “Methods of determining thermal resistivity of soil in situ for the calculation of cable ratings” – Technical report F/T 181-1955.
- CIGRE B1 Session 1970 – “Contribution to the study of the thermal environment of buried cables”
- ASTM D5334-08 Standard test method for determination of Thermal conductivity of soil and soft rock by thermal needle Probe Procedure
- Electra N°145 1992 “Determination of a critical temperature rise for a cable backfill material”

Thermal resistivity is measured by the transient heating of a thermal probe fixed in the centre of a cylindrical mould containing the compacted material. This procedure enables thermal resistivity, water content, saturation degree, porosity and specific gravity to be measured on the same sample. After measuring samples in the wet state, the samples are dried in the oven (72 hours, 105 °C) and are then re-measured.

The assessment of the thermal characteristics are different for backfills and for natural soils.

In order to assess the thermal characterisation of a backfill, the grain size distribution and composition are analysed.

The laboratory analysis allows a first selection of the backfill materials, but the final selection is performed after an in situ research of the backfill's thermal characteristics.

The Normal Proctor curve is determined and then the thermal resistivity is measured for the water content obtained at the Optimum Proctor point. For backfills, thermal resistivity measurements are performed for densities around 93-98% of the Optimum Proctor density for different water contents (dryout curve determination).

Measurements of compaction level are performed on site in order to determine which is the minimum number of passages of the vibrating plate machine necessary to obtain a compaction level around 93-98% of the Optimum Proctor of the backfill. The in-situ compaction level is measured. Laboratory measurements are then performed after the in-situ analysis on samples with thermal characteristics measured on site (dry density and water content). The thermal resistivity should be lower than 0.9 K·m/W.

In case of a native soil, thermal resistivity is measured on bulk samples of the material that is prevalent on site. The thermal characteristics are measured (dry density, porosity, saturation degree, water content and thermal resistivity in dry and wet state).

In order to decide whether the soil/backfill is in danger of dryout, an interpretation of the results is performed, taking into account the dry out curves for different types of soils described in Electra N°145 1992 “Determination of a critical temperature rise for a cable backfill material”.

Belgium is laying a controlled backfill around all HV cables installed in trenches and is using bentonite to fill the cable ducts where there exists high thermal resistivity of soils around the cable ducts.

B.5.3 Brazil

This section contains a truncated translation (from the original Portuguese) of report RE – 357/243, the evaluation of backfill grain characteristics due to the displacement of water table.

Introduction

Although satisfactory laboratory results, relating to mechanical and thermal stability, had been achieved in previous tests made with the backfill (composed of granitic rocks), they were not considered conclusive, because these tests did not fully represent the real conditions that this material is subjected to in directly buried power cable installations. Hence, the work covered in this report adds to the knowledge regarding the more specific and challenging conditions that may occur for this backfill.

Objective

The main objective of this work was to evaluate the variation in the grain size distributions of backfill composed of granitic rocks (used in the Norte - Miguel Reale underground transmission line , 345 kV , 450 MVA/circuit , two circuits , 15.0 km long) after it had been subjected to an accelerated displacement of water (simulating water table movement).

Methodology

A sample of backfill (0.48 m^3 , compacted to a dry density of 2 g/cm^3) was subjected to an accelerated displacement of water according to the following daily cycle, which was performed over a period of 2 months:

- 6 hours without water
- 6 hours with the water level increasing
- 6 hours immersed
- 6 hours with the water level decreasing

This sample was contained in a fibreglass box surrounded by sand and gravel, according to Figures B.8 to B.12.

The water flow variation was admitted slowly (over 6 hours) and controlled by two flow meters (Figure B.12).

The water was introduced from the top of the box, dripping the water into a blanket or cotton textiles to avoid causing a depression in the sand.

Tests to verify the grains size distribution of the backfill were performed at 15 specific points (according to Figures B.8 and B.9), before and after the sample has been submitted to water level variation.

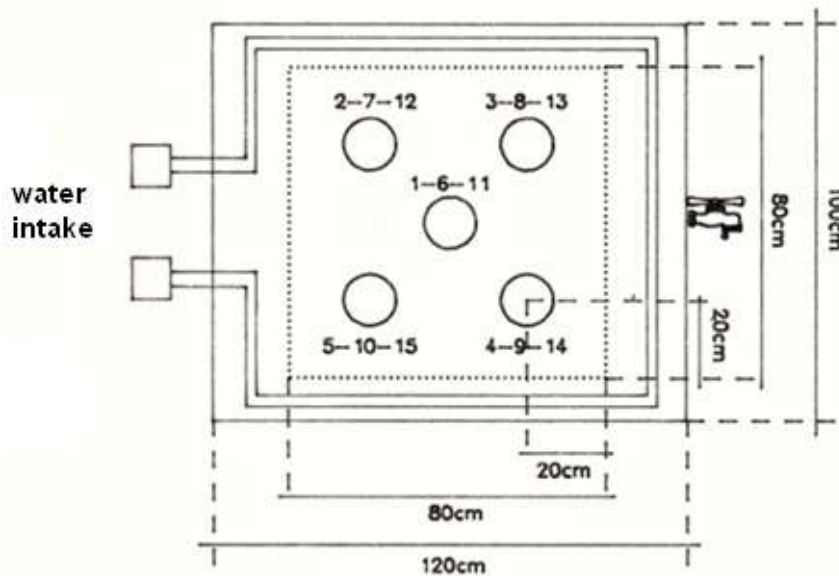


Figure B.8 Plan view of the box containing backfill and the points where grain size distribution tests were performed

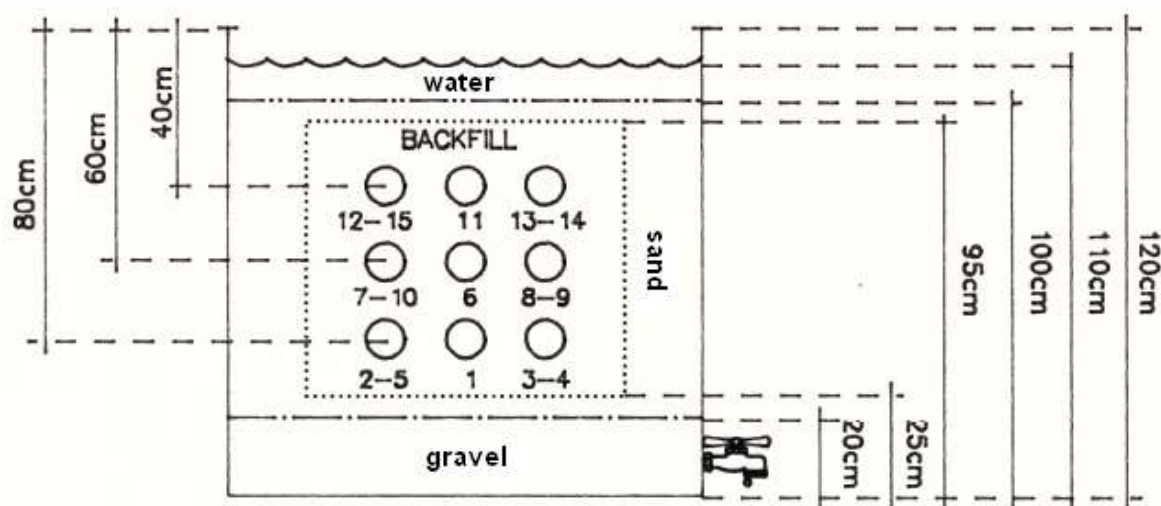


Figure B.9 Elevation view of the box containing backfill and the points where grain size distribution tests were performed



Figure B.10 General view of the box

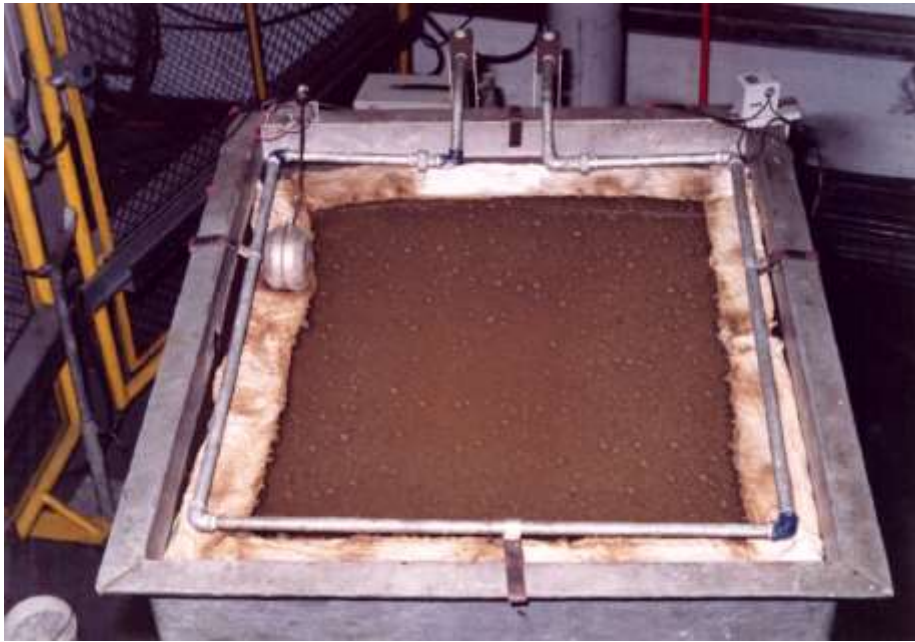


Figure B.11 View of the water system



Figure B.12 Measurement of the water flow

Sample composition:

The backfill mixture comprised:

- 60% of course material (with grain diameters between 10 mm and 1 mm)
- 40% of fine-grained material (with grain diameters below 1 mm)

This material, compacted with a dry density of 2g/cm^3 , has a thermal resistivity of $1.0\text{ K}\cdot\text{m/W}$ (totally dry).

Description of tests and results:

The results of tests to verify the grain size distribution are shown in Figures B.13 to B.18. The smooth lines represent the theoretical upper and lower limits of grain size distribution of the backfill.

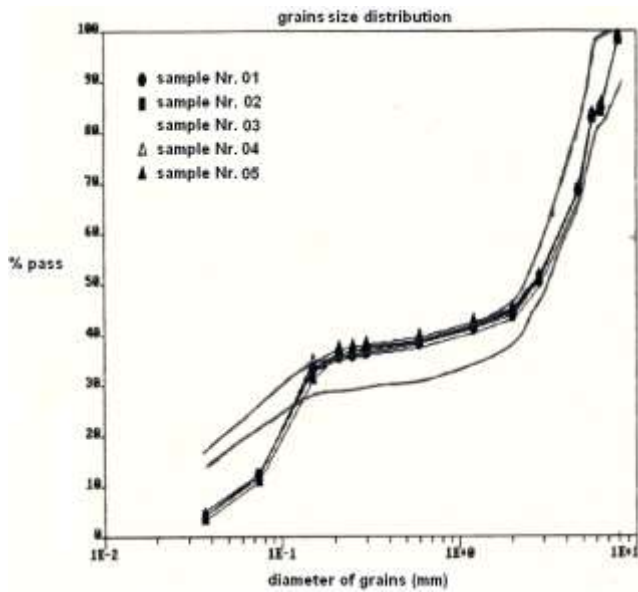


Figure B.13 Grain size distribution at points 1 to 5 before test

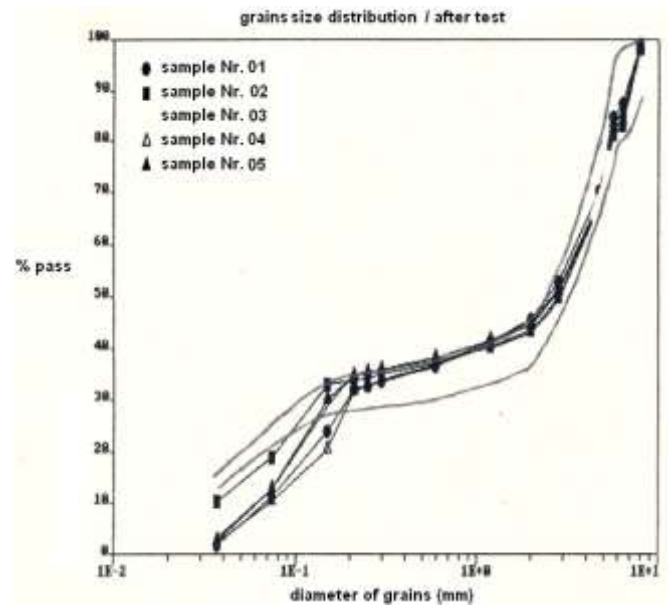


Figure B.14 Grain size distribution at points 1 to 5 after test

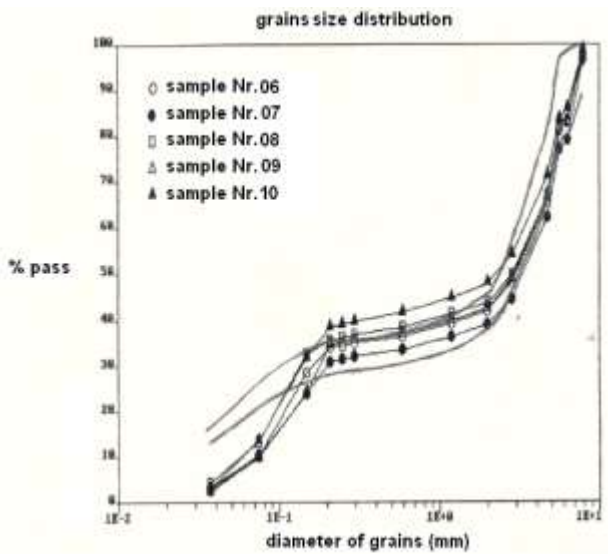


Figure B.15 Grain size distributions at points 6 to 10 before test

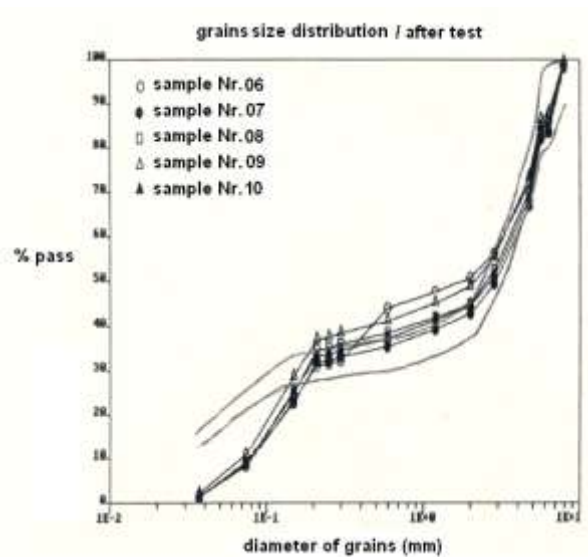


Figure B.16 Grain size distributions at points 6 to 10 after test

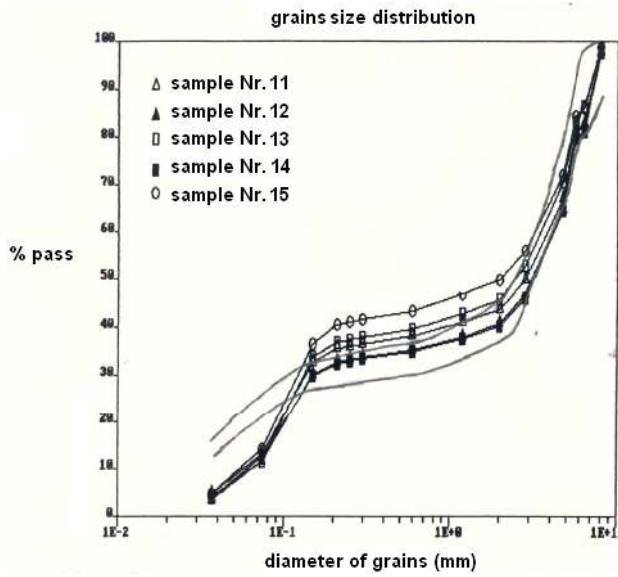


Figure B.17 Grain size distributions at points 11 to 15 before test

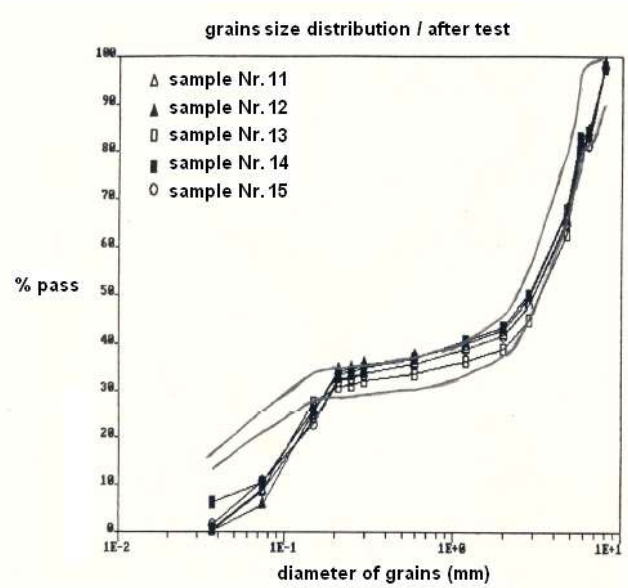


Figure B.18 Grain size distributions at points 11 to 15 after test

Figure B.19 shows the grain size distribution of the sand used to simulate the soil surrounding.

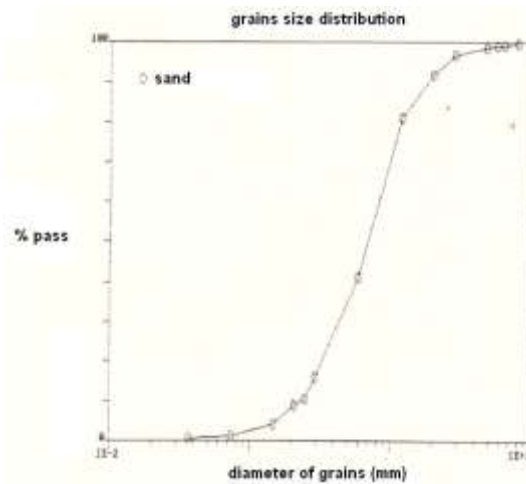


Figure B.19 Grain size distribution of the sand used to simulate the soil surrounding

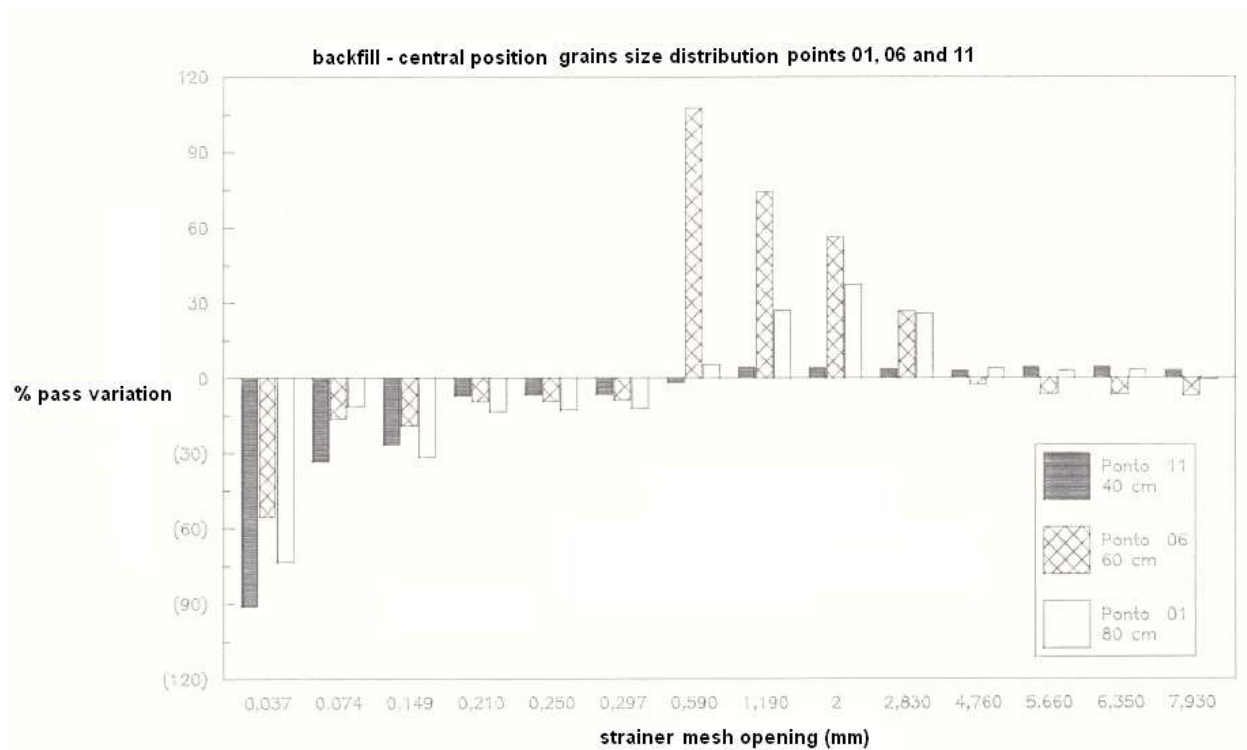


Figure B.20 % pass variation – central position - points 01, 06 and 11

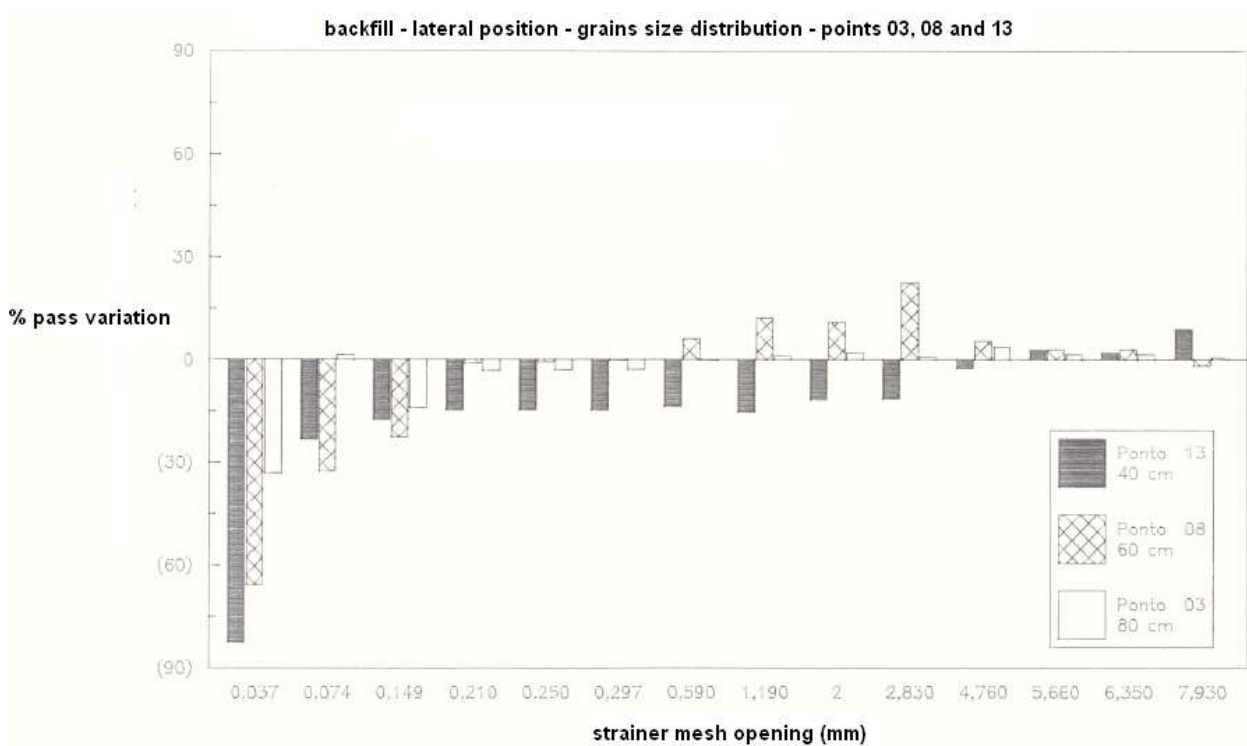


Figure B.21 % pass variation – lateral position - points 03, 08 and 13

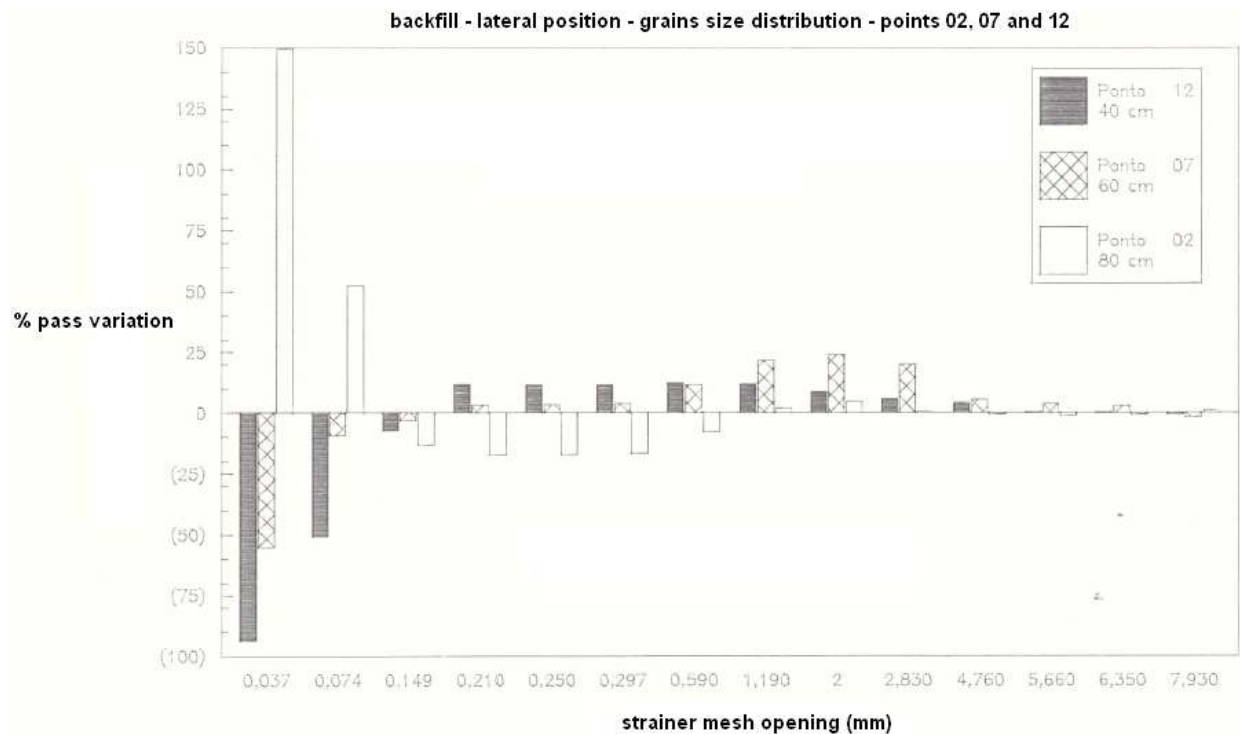


Figure B.22 % pass variation – lateral position - points 02, 07 and 12

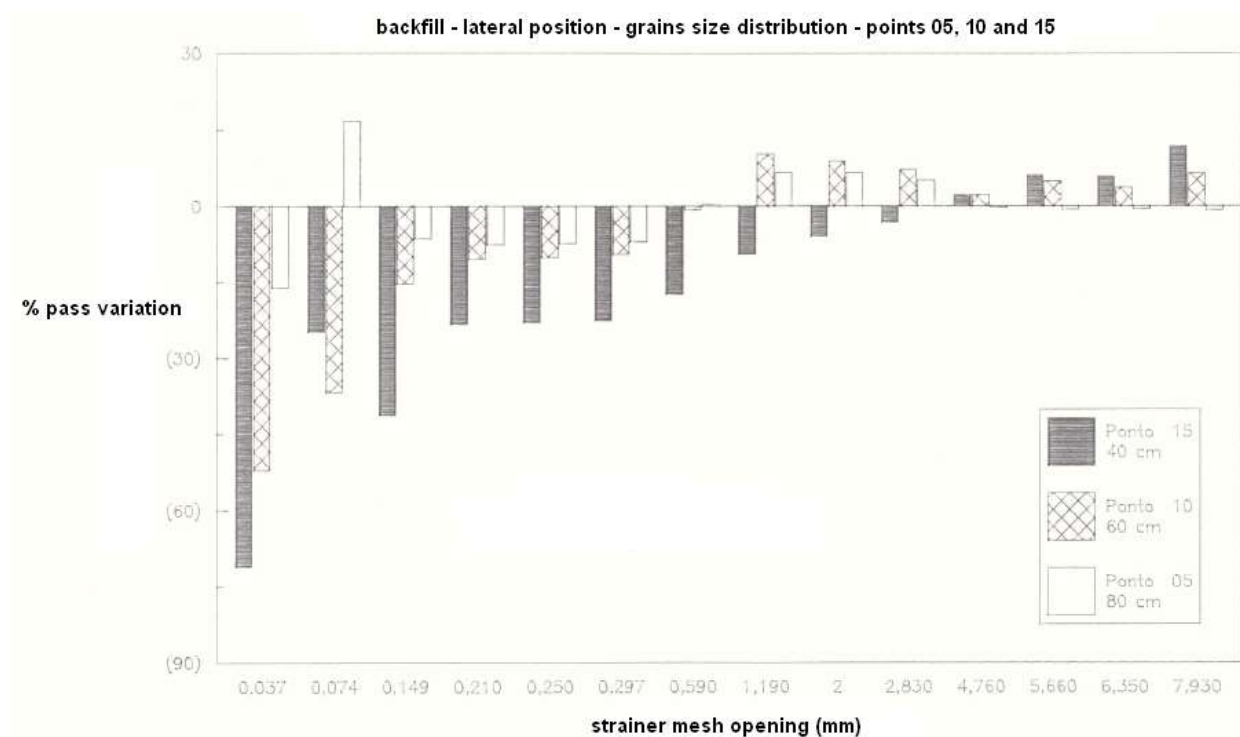


Figure B.23 % pass variation – lateral position - points 05, 10 and 15

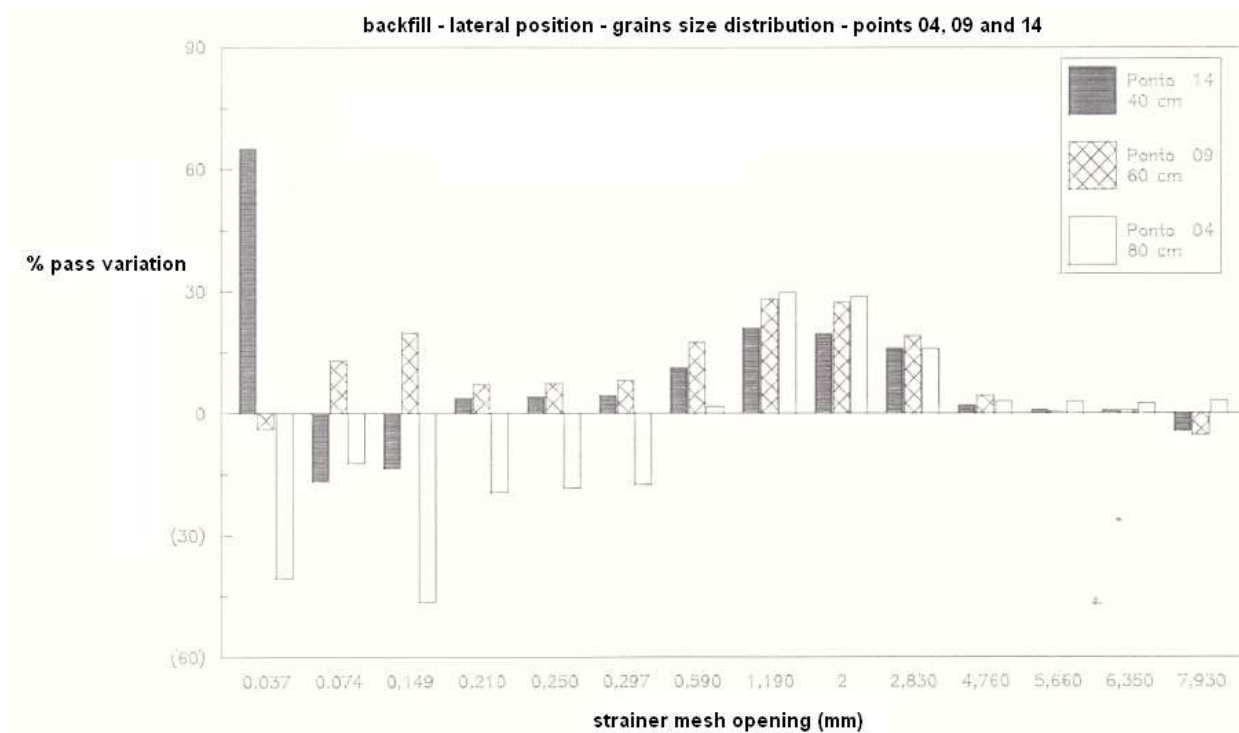


Figure B.24 % pass variation – lateral position - points 04, 09 and 14

Comments

How can we interpret the results? For example, comparing Figures B.13 and B.14 we can conclude that for:

Sample No. 1 (central position at the bottom):

grains with diameter below 0.5 mm were displaced (the sample lost these grains)

grains with diameter above 0.5 mm were displaced from the other points to this point, i.e., the sample at point 1 received grains from the other points

Evaluating the grain size 0.037 mm in Figures B.13 and B.14, the % passes of material were 32 % and 22 %, respectively. Figure B.18 represents this variation ($1 - 22/32 = 31.25\%$). The bar below the zero line means the grains lost from the sample.

Sample No. 2 (lateral position at the bottom):

Evaluating the grain size 0.037 mm in Figures B.13 and B.14, the % passes of material were 5 % and 10 % respectively. Figure B.20 represents this variation ($1 + 5/10 = 150\%$). The bar above the zero line indicates the grains received by the sample at that point. As points 7 and 12 lost 0.037 mm grains, these grains were probably displaced to point 2 during the water flow.

Comparing the tests of grain size distribution performed in each point before and after the water application, referring to Figures B.20 to B.24, we can note that the displacement of grains of diameters 0.037 mm, 0.074 mm and 0.149 mm was greater than the displacement of larger grain sizes.

Figures B.13 to B.18 show that the grains with diameter below 1.2mm remained within the established proportion of 40% of material. There was, however, significant variation in the number of grains with diameters below 0.15 mm.

Cigre WG B1.41 comments on the above work:

It should be noted that the backfill used in this extensive study was designed in a laboratory to have a dry density and dry thermal resistivity of 2 g/cm³ and 1 K·m/W, respectively, but the corresponding mechanical and thermal properties of the material were not evaluated after the test simulating the water table movement.

This work, however, makes very interesting observations about the displacement of small particles, which may well be a problem in real cable installations where, either due to poor installation or subsequent disturbance of backfilled areas, poor compaction combined with water movement may lead to severe compromising of backfill thermal performance.

B.5.4 Canada

Thermal Properties Database of Canadian Soils

Field and laboratory geotechnical and thermal resistivity testing were carried out on 100 soil samples taken from all across Canada. The soils were grouped according to the Unified Soil Classification System (modified) and thermal dryout curve (thermal resistivity versus moisture content) enveloped defined. Although the thermal dryout curve for a given soil may be placed within a specified envelope, its exact shape and position are determined by other parameters (i.e. dry density, organic content, mineral composition, particle shape, fines content, etc.) and their complex inter-relationship.

The report also considered the elements of a comprehensive soil thermal survey. The theory and application of the transient thermal probe technique were also discussed and standardized testing using an automatic instrument was suggested (see an example in section 7.2.1). Soil geotechnical properties affecting thermal resistivity were discussed as well as some semi-empirical estimation techniques that attempt to account for these. This database should allow a more realistic definition of thermal properties of Canadian soils both for scientific and practical engineering applications.

The objectives of this research study were as follows:

- Review, collate and list existing literature reporting soil thermal properties.
- Develop a databank of existing measurements in the literature with appropriate information, such as: methods, conditions of measurement, accuracy of measurement, soil type, and geotechnical properties.
- Describe and assess existing physical models relating thermal properties to other soil properties.
- In conjunction with the scientific authority select up to 50 sites across Canada, including northern Canada, which adequately represent the soil type (gravels, sands, silts, clays, peats, permafrost, etc.) and climatic and geological history of the Canadian landmass.
- Conduct in-situ measurements of thermal properties and collect core samples at 1-metre depth at each of the selected sites for laboratory analysis. Thermal property measurements in the field and laboratory should be standardized using an automatic instrument.
- Measure thermal properties (resistivity, diffusivity, heat capacity, thermal dryout curves) in the laboratory covering such conditions as might be encountered naturally or during development.
- In addition each sample should be analyzed for standard geotechnical properties (i.e. density, water content, gradation, composition, Atterberg limits, specific gravity, etc.).
- Examine the thermal and other physical data for trends and statistical correlations between properties.
- Test existing models with new data to examine their validity in predicting thermal properties from other physical properties.
- Develop, where needed, appropriate new models to use in thermal property prediction, including variation of temperature.
- Make recommendations on the use and validity of thermal property estimation using models.
- Recommend models to be used and the best correlated geotechnical properties.

B.5.5 China

Thermal resistivity vs. depth

For the work reported in clause 3.3.5. soil samples were taken at various depths, including 5m, 4m, 3m, 2m and 1m, in the Wuhan test centre of CEPRI. The thermal resistivity values were determined adopting laboratory testing methods before the test and the temperatures were tested in the field. The soil samples had naturally settled over a period of nearly 40 years without outside disturbance. Figure B.25 shows a 5m deep cable trench.



Figure B.25 The 5m depth cable trench

The thermal resistivity measurements showed rather low values (see Appendix B.5.5), presumably corresponding to quite moist conditions.

Soil samples from different depths were taken from a cable trench, and were then sealed in a plastic bag.

Table B.6 Measurement results of soil from different depths

depth m	thermal resistivity values K·m/W	temperature °C
1.2	0.80	about 22.9
2.0	lost	19.7
3.0	lost	16.2
4.0	lost	15.3
5.0	0.57	14.2

The date was Aug. 25, 2008. The thermal resistivity was measured for the various materials used in the installation. The clay soil excavated from the trench was used as backfill.

Three types of medium were measured:

1) Backfill soil at 0.7m depth: the composition of the backfill soil is mainly clay. Compared with the native soil, the backfill soil is dense, but the moisture and the compaction degree may be different.

- 2) Fine sand in cable trench
- 3) Native soil in test site

The tests were performed using a QTM-500 quick thermal conductivity meter.

Table B.7 Measurement results of thermal resistivity, K·m/W

medium	Result 1	Result 2	Result 3	average
"backfill soil"	0.84	0.81	0.82	0.82
Fine saturated sand	0.48	0.50	0.49	0.49
native soil	0.54	0.55	0.54	0.55

The thermal conductivity meter is shown in Figure B.26a), and the test field show in Figures B.26b)-d)

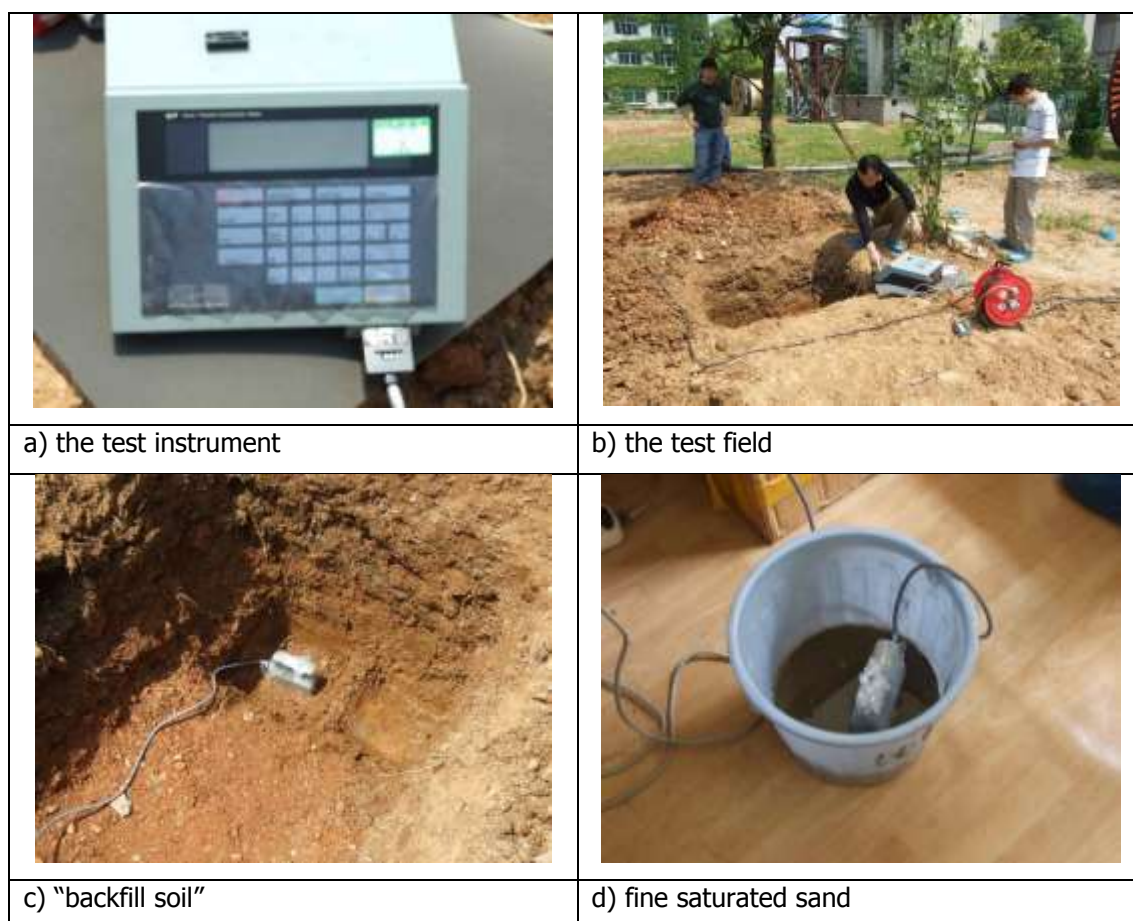


Figure B.26 Thermal conductivity meter and the test field

Continuous monitoring of soil temperature at a depth of 1m

A continuous in-situ monitoring of the soil temperature was accomplished at the Wuhan test site over a period of 1 year (December 2007–December 2008). Temperature values at a depth of 1m in four different test points were recorded and the average value is plotted in Figure 2.1.

According to the experimental results, the soil temperature in Wuhan test site varied significantly in response to changing climate, with a highest value of 25.8 °C (Aug. 01, 2008) and a lowest value of 7.2 °C (Jan. 31, 2008).

Status analysis

The thermal resistivity of soil is a key parameter in ampacity calculation for underground power cable lines, which is dependent on the degree of compaction (dry density) and the moisture content.

The cable environment varies considerably depending on the geographic location in this large country. For example, the western region is subject to droughts and the southern region in China can experience high rainfall, with the moisture content in the south showing particularly strong seasonal change. This, and the seasonal temperature rating, points to the need for seasonal cable rating.

Flowable thermal backfill

Flowable thermal backfill material has been applied to improve the current rating of power cables in China. In order to evaluate the effect of flowable material on the ampacity of power cables in ducts, a full-scale experimental setup including a 3×3 duct bank and a cable trench was built. It was found that when the duct was filled with backfill, the ampacity of the cable in the 3×3 duct bank improved by 17.2% (and the ampacity of a single cable in a duct improved by about 14.5%) compared to the situation without the flowable backfill. The material has now been applied to an actual installation.

In summary, CEPRI have clearly made some investigation into the local thermal conditions for cable installations, both in terms of seasonally dependent ambient temperature and moisture content, although this does not directly indicate the thermal conditions of existing cable environments.

B.5.6 Finland

HV cables, meaning 110 kV sub-transmission cables, normally XLPE cables with stranded aluminium conductors (3 x AHXLMK-W 1x800), are normally laid in trefoil on a bedding of backfill, see Figure B.27. The concrete trough (about 50x40 cm) is then slightly overfilled with crushed rock or graded sand backfill. A concrete cover is then placed on top, and the trench is filled in. The burial depth to the bottom conductors in trefoil is typically 1.1 m. Some newer important connections have had larger copper conductors and have been installed in flat configuration with cross bonding of the sheaths. Some older connections still have oil-paper cables.



Figure B.27 Typical concrete trough for installation of 110 kV cables (lid not shown)

MV cables (20 or 10 kV) are installed at depth of about 70 cm, nowadays in unfilled composite plastic conduits, earlier under inverted concrete U-sections. The environment of MV cables may or may not be controlled backfill, as other services have often subsequently been placed in the same region.

Large sections of the fault-prone rural MV network is now being replaced with underground cable, usually using some kind of ploughing technique. It should be noted that the cable environment in Finland is mostly (~70%) a combination of granite bedrock and large rocks.

Occasionally for, e.g., road crossings, HV cables are also drawn into composite plastic tubes. These tubes are not usually filled (with bentonite or anything else). A typical PE tube is shown in Figure B.28.



Figure B.28 Typical composite plastic tube (usually used for MV cables, but occasionally for HV road crossings, etc.)

The thermal resistivity under normal moist conditions of the backfills tested in the Helsinki region is about $0.7 \text{ K}\cdot\text{m}/\text{W}$, but when dry rises to about $2.3 \text{ K}\cdot\text{m}/\text{W}$. Recently, weak mix cement backfills have been used to provide greater thermal stability.

The following is a typical specification for weak mix:

"The sand which will be used shall consist of different grain sizes e.g.

grain size 1.0-2.0 mm 40 %, grain size 0.5-1.0 mm 26 %, grain size 0.2-0.5 mm 18 % and grain size 0.1-0.2 mm 16 %

The cement (6 percentage of weight) and water (10 percentage of weight) shall be blended to above mixture of sand grains (weak mix to be blended before filling into trench). It is also essential to compact (tamp) the weak mix in the trench before the cement starts curing.

Thermal resistivity of weak mixes should always be tested in laboratories in advance. Same material and blend proportions should then be used throughout the project."

There have been no major HV cable failures in the Helsinki region in the past decades, except where there was operational error, or dig-in faults big digging machines. On the whole, the underground transmission cables are lightly loaded, according to N-1 security criteria. The transmission network (>200 kV) in Finland is almost all overhead. There is, however, some concern that with extensive subterranean building in urban areas (car parks, etc.) and the removal of snow from mostly sealed surfaces (a large proportion of the annual precipitation is snow), that there may be long-term dryout of some cable environments (due to lack of rewetting), which may have a detrimental impact on ampacity.

B.5.7 France

Rating calculations are conducted according to the definitions and thermal analyses given in IEC 60287.

"By default", the standard **thermal resistivities** of native soils are:

$1.20 \text{ K}\cdot\text{m}/\text{W}$ during the summer,

$0.85 \text{ K}\cdot\text{m}/\text{W}$ during the winter.

As a result of a study based on the analysis of meteorological data and soil temperature measurements, France can be divided in 3 zones as shown on the following map, Figure B.29:



Figure B.29 The division of France into 3 temperature zones

Temperatures to be considered in rating calculations in relation to the climatic zones and seasons are given in the following table:

Table B.8 pH Classification of Soils

Climatic zone	Summer season (May – Oct)	Winter season (Oct – May)
Cold	20°C	13°C
Intermediate	22°C	15°C
Hot	25°C	17°C

B.5.8 Netherlands

Overview of studies on the thermal properties of soil in the Netherlands

The following is a selection of soil thermal property studies relevant to the rating of cables and is not intended to provide a complete overview of all studies that have taken place in the Netherlands in this field.

In 1952 D.A. de Vries received his PhD from the University of Leiden in the Netherlands with a thesis [26] entitled "Het warmtegeleidingsvermogen van grond" (the thermal conductivity of soil). This contains a theoretical and experimental study on the dependence of the thermal conductivity of soils on their composition, water content and temperature. Subjects in this thesis are explanations and comparisons of different theories, heat transfer related to vapour diffusion, measuring the thermal conductivity of soil in the field and in the laboratory, calculation of the thermal conductivity of soil and a comparison of the theoretical and experimental values.

In 1957 J.R. Philip and D.A. de Vries wrote the paper "Moisture movement in porous materials under temperature gradients" [27]. In this paper a theory about moisture movement in porous material is developed which describes the transport of heat and moisture driven by a temperature and moisture gradient.

The book "Physics of plant environment" [27], edited by professor of Physics at the Agricultural University Wageningen, W.R. van Wijk, contains a chapter "Thermal properties of soils" written by D.A. de Vries. In this chapter the heat capacity and thermal conductivity of soils are described, along with the theory of the thermal conductivity of granular materials and the influence of moisture movement. The thermal conductivity of four types of soil is calculated as an example.

In 1969 P.E. Rijtema did research on the prediction of the moisture content of soil. His results are presented in a paper "Soil moisture forecasting" [29]. The accuracy of the prediction of the moisture extraction depends not only on the accuracy of the soil properties, but also on the climatological conditions and the depth of the groundwater table. For a series of standard soils the moisture characteristics are measured and these can be used for forecasting the soil moisture conditions and capillary rise above the groundwater table.

In the book "Heat and Mass Transfer in the Biosphere. I: Transfer Processes in Plant Environment" from 1975, written by D.A. de Vries [30], a chapter contains the subject "Heat transfer in soils". Besides the transport of heat by conduction, the transport of heat by water vapour is also taken into account. The present theories with their limitation are discussed and some suggestions are made for future studies.

At end of the '70s till the beginning of '90s in the twentieth century at lot of research has been performed by KEMA and Heidemij (a company engaged in the development of agricultural land, construction timber, etc.), such as:

- Article "Het warmte-afvoerend vermogen van grond rondom hoogspanningskabels" (The heat removal capacity of soil around power cables), 1978, by J.Th.J. Beeren et al [31]. This paper describes a procedure to determine the thermal resistivity of soil in an existing or new cable route. It gives the factors that will affect the transmission of heat through the soil and some guidelines for the composition of backfill soil with low thermal resistivity.
- Article "Veldexperimenten ter bestudering van de uitdroging van grond rondom hoogspanningskabels" (Field experiments for analysing the drying-out of soil around power cables), 1979, by G.M.L.M. van de Wiel et al [32]. An experiment with cables and dummy cables buried in two different kinds of sand is described. The experiment, which lasted more than four years, shows the influence of rainfall and soil temperature on the heat conducting properties of the two types of sand. It demonstrates that not only the temperature is important for drying-out of the sand but also the heat flux density. It also shows that irrigation is a good method to prevent drying-out of sand.
- Report "Moisture migration and drying-out in sand around heat dissipating cables and ducts. This theoretical and experimental study" dated 1981 [44] was the result of collaboration between KEMA and Heidemij, and describes the physical mechanism of the drying-out phenomenon of sandy soil around power cables. This is done by extensive experiments, both in the field and in the laboratory. It shows the relation between the water retention capacity of sandy soils and the critical conditions where drying-out starts. The theory of Philip and de Vries on moisture and heat transfer is developed into a mathematical model which describes the distribution of temperature and moisture in sandy soil with the drying-out phenomenon.
- Paper "The conditions controlling the drying-out of soil around power cables" from J. Vermeer et al in 1982 [33]. The dissipated heat of power cables may cause a dry zone around the power cable. To prevent drying-out of soil, the conditions which cause drying out have to be known, such as: specific soil properties, degree of compaction of the soil, moisture content of the soil, heat flux density from the power cables, and the temperature of the jacket of the power cable, which are all described in this paper.
- Paper "Improved method to calculate the critical conditions for drying-out sandy soils around power cables" by G.J. Groeneveld et al, 1984 [34]. A model was developed that calculates the critical thermal conditions at which drying out in sandy soils starts. The model is based on the theory of Philip and de Vries. The results of this model are compared with the results of the experiments of power cables surrounded by different kinds of sandy soil. The knowledge of this study enables quantification of the risk due to thermal runaway.
- Paper "Transport of heat and moisture in soils with hysteretic moisture potential" by G. Koopmans et al [35], 1985. Two types of moving boundaries were encountered, one between fully dried-out soil and moist soil and one between decreasing and increasing moisture content.

- Report "Critical temperature for drying-out of soils as a function of circumstances", 1987, by G. Koopmans et al [36]. Based on the previous published model of the drying-out of sandy soils, a critical temperature expression is obtained. A single and three equally loaded cables in flat formations are considered in different conditions. The paper provided a practical method to get the highest (worst case) values along the cable route.
- Paper "Soil physical route survey and cable thermal design procedure" by G. Koopmans et al [37], 1989. The previous model for the prediction of drying out is used for a thermal rating procedure. The model needs the soil dry density, suction tension curve and saturated hydraulic conductivity. The accuracy of this procedure was investigated over nine years in a field experiment.
- Report "The critical temperature for drying-out of soils around power cables as a function of local conditions" dated 1990 by G. Koopmans et al [38]. A practical method was developed to obtain the worst case parameters from a soil survey along the cable route.

Soil samples and collecting additional information

The following briefly describes the procedure of performing a soil survey in the Netherlands. The survey consists of the following parts, bottleneck analyses, field survey, laboratory tests and thermal resistivity.

Bottleneck analysis

By executing a soil survey the first part is a desk study to investigate the potential bottlenecks in the cable route. Aspects that need the attention in this bottleneck analyses are for instance elevation changes in the landscape, type of vegetation's, waterways and other heat sources etc. Also places where it is expected that the soil will be replaced, like a HV power station, could be a bottleneck. In this bottleneck analyses the locations are determined for the field survey which will be executed by a soil expert.

Field survey

The activities carried out by a soil expert in the field is preparatory to the civil engineering of the cable system. In the bottleneck analysis the locations are globally determined. The exact position of the sites should be made on a drawing which contained also the plan of the cable route.

To make a good estimation of the existing soils, soil layers, at actual and lowest average groundwater level on the locations a hand drilling of at least 2 m below the surface will be carried out.

To determine the dry density and moisture content of the different soil layers two undisturbed soil samples (insert ring samples) will be taken of each layer (Figure B.30).



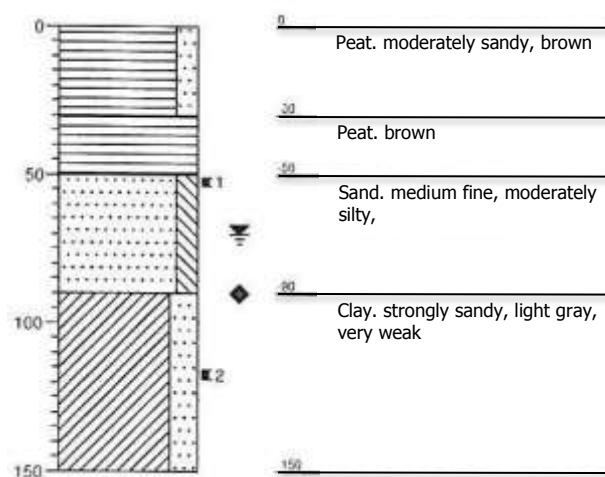
Figure B.30 Taking undisturbed soil samples

The following procedure is proposed (by a soil expert) at the locations for new cable connections:

The soil expert will make photos of the environment of the site.

Perform hand drilling to approximately 2 m below the surface. Based on this drilling a borelog will be drawn which contains the different soil types and thickness of soil layers. The actual groundwater level and an indication of the average lowest groundwater level (if necessary drilling deeper than 2 m) will also be determined

While performing hand drilling, 2 undisturbed ring samples are taken at approximately 0.60 m above cable depth, 2 undisturbed ring samples are taken at cable depth and a sample of approximately 7 kg of sand is taken from around the cable depth. Figure B.31 shows a typical cross section of a cable trench.

**Figure B.31 Typical cross section of trench**

For existing cable connections a trench is made in which the undisturbed and disturbed samples are taken, at the depths described above.

Laboratory tests

In the laboratory the dry density in kg/m^3 and the moisture content as a percentage of the dry density is determined for the 4 undisturbed soil samples. From the sand around the cable a Proctor test and a Sieve test will also be performed.

The Proctor test is a laboratory test which is carried out a few times using sand with each time a different moisture content. This sand will be compacted by a standardized procedure with a specific energy of approximately $0,6 \text{ MJ/m}^3$. After the compaction the dry density is determined. The highest dry density that is determined during this test is the Proctor density. This value is used to determine the dry density that can be achieved after compaction. Typically is 92 – 95% of the Proctor density.

Soil consists of composite particles of different size and material. The Sieve test will separate the soil into several particle classifications with sieves of different aperture sizes. Usually the wet Sieve analysis is performed so that the agglomerated fine particles separate from each other. The results of the Sieve test and observation in the field will be used to determine the type of soil and select the corresponding pf curve of the Rijtema Catalogue. This catalogue contains the most common soil types in the Netherlands.

Thermal resistivity

The above mentioned activities will result in the representative value of thermal resistivity of the soil for the average lowest ground water level. This will be calculated based on the internal procedures of DNV GL.

To calculate the equivalent thermal resistivity of the soil above the cable the Conformal Mapping Method will be used which is described in Electra no. 98.

Conclusions

This procedure will determine the thermal resistivity of the soil at the worst conditions, namely by the average lowest groundwater level.

B.5.9 Sweden

The climate in Sweden and the fact that most cables are normally used to supply households means that the load follows a daily pattern with a seasonal variation. The highest peak loads are in the coldest days during the winter. The cables for household supply are, based on security of supply, normally designed with 100% redundancy at estimated peak load. Thus, the cables are normally only loaded to a maximum of 50% of peak load. The thermal design of the cables is normally based on a 1 m laying depth, ambient soil temperature of 15 °C and a thermal resistivity of 1.0 K·m/W . Considering the normal load pattern and seasonal variation these thermal design criteria are conservative.

Some twenty-five to thirty years ago a lot of cabling was done in, for example, Stockholm. When the thermal design and the route and installation planning for this cabling were completed, studies were made on the thermal protective backfill normally used at that time. The conclusion was that the thermal properties were equal to or better than assumed and that there was no risk for drying out. The thermal backfill has very good properties, such as, for example, retention of moisture. Thus, the thermal backfill in Sweden is generally considered to be very stable, with thermal properties equal to or better than assumed for the thermal design of the cables which are used for household loads with the load pattern and seasonal variation applicable in Sweden. There are a few cases where cables connecting generating plants have been loaded very heavily during relatively long periods of time during warm and dry periods of the year and where drying out of the soil has been reported. In some cases sprinkling above the cables has been necessary to restore the moisture content and the thermal conductivity to acceptable values.

The recent trends with increased cabling on high and extra high voltages for both AC and DC transmission have put more focus on the thermal properties and stability of the thermal backfill. The load pattern on these transmission cables is expected to be different than the cables for household supply near city centres. Especially the very long and large DC transmission systems being built in Sweden at the moment are expected to have different load patterns, not correlated to the load pattern and seasonal variation in Sweden. In the South West Link project (2x720 MW, 300 kV DC, 190 km land cable) extra focus was put on the thermal backfill. A detailed specification with requirements for the thermal backfill was developed after studies and testing of the thermal backfill from several sand and rock pits. The specification is used also for NordBalt (720 MW, 300 kV DC, 40 km land cable) and for 400 kV AC cables where high and continuous transmission and cable temperatures are expected.

The specification includes two types of thermal sand for backfill, one normal backfill and one backfill which could be used during route and installation planning where the thermal conditions need special attention. For the two types of thermal backfill the specification includes the following requirements:

- Requirements on the thermal backfill
 - quartz content
 - grain distribution
 - dry density
- Requirements for packing of the thermal backfill in the cable trench
- Control and testing of the thermal backfill
 - Control and testing before supply of backfill from sand and rock pit

- Control and testing during the delivery time
 - Control and testing when laying and packing
- Svenska Kraftnät's control

Quite a lot of effort has been put into the thermal design of the cable system for the South West Link. Approximately 40 installation section cases have been developed for the route and installation planners to use in the detailed route and installation planning. As given above, two qualities of thermal backfill were also given for the route and installation planners to use in the detailed field planning. In addition to this, two cable cross sections were also given: one "normal cable" and one cable with larger cross section for thermally difficult conditions.

Since the South West Link is a new type of cable installation, and is also one where we do not know the expected load pattern and seasonal variation, we have introduced a R&D project. The aim of this project is to evaluate the thermal design of the cable system, including the thermal backfill. A DTS system is installed on the main part of the 190 km length, and this will be used to evaluate the thermal design and properties of the backfill and soil.

APPENDIX C. SURVEY

C.1. SURVEY AS CIRCULATED

CIGRE WG B1.41



Long term performance of soil and backfill of cable systems

Survey

Experience with Long-term Performance of Underground Cable Soil and Backfill Systems

Introduction

The thermal properties of native soil and trench backfill can have significant impact on underground cable ratings. In the past, soil parameters were roughly estimated along the cable routes. The task of this CIGRE Working Group is to gather available field information on utility experience with underground cable system soil and backfill performance over a long (10+ years) period of time.

Scope

The most important factors affecting cable rating are the thermal resistivity of the soil, which may change over time due to moisture and soil particle migrations.

The thermal resistance of the cable environment is mainly related to the thermal resistivity of the trench backfill, surrounding native soil and the burial depth of the cables. A cable's buried depth and/or arrangement may change over time due to construction activities in the vicinity of the cable, "sinking ground" effect, cable movement due to expansion and contractions etc.

The scope of this survey is to gather available information on the long term soil/backfill thermal performance of underground cable installations. If you have experienced visual or measured soil/backfill parameter changes, flooding, soil migration, cable failures due to local overheating, or suspect water/sewer breaks in the vicinity of the cable trenches, please share your experience with us.

The results of this survey will help us analyse the long term performance of cable system soil and backfill, develop useful recommendations for the frequency and locations of soil testing and suggest improvement measures for overall cable system rating.

Utility Survey: Experience with Long-term Performance of Underground Cable Soil and Backfill Systems

PLEASE ANSWER THE FIRST QUESTION. IF YOU HAD EXPERIENCED ANY CHANGES IMPACTING CABLE RATINGS PLEASE CONTINUE. IF THERE ARE NO NOTICABLE CHANGES YOU CAN SKIP QUESTIONS 2 TO 8.

Utility Name: _____ Country: _____

(All information given is considered confidential and will not be published)

	Question	Answer
1	Have you experienced any visual or measured changes in the soil/backfill thermal and mechanical parameters of cable systems after a long period of operation? - if NO – skip questions 2 to 8 - If YES – please describe the observed changes and continue to answer questions 2 to 8.	
	All subsequent questions relate to changes caused by long term operation only!	
2	With what type of backfill did you experience problems?	
3	When did you determine soil/backfill thermal resistivity? If not, would you consider doing so to verify the design parameters for the existing loads (Ampacity audit)?	
4	Has any cable failure occurred due to local overheating? If so, did you investigate the cause?	
5	Please describe the type of installation you experienced change. 1. Directly buried 2. Duct bank 3. Steel or plastic pipes 4. Trenchless technologies	
6	Have you experienced flooding or water/sewer breaks in the vicinity of a cable system?	
7	Has the cable thermal environment changed as a result of construction activity or natural subsidence?	
8	Do you have any type of cable temperature monitoring systems installed?	

C.2. SUMMARY OF RESULTS OF SURVEY

The working group circulated the simple two page questionnaire, given in this appendix, to utilities worldwide, via SC B1 regular members. 46 responses were received from 17 countries. Of those replies 22 gave a positive answer to question 1 (i.e. they had experience of a problem with thermal backfill, and thus continued to complete the rest of the questions).

The answers are analysed in the following table:

	Question		Question answers	Remarks
2	With what type of backfill did you experience problems?		Most of the answers refers to sand backfill (11). 6 answers are relevant to native soil without backfill, one to both the above. In one case problems are reported in a sand cement mix and one in concrete (most probably weak mix).	Sand backfill is the most critical one, followed by natural soil.
3	When did you determine soil/backfill thermal resistivity? If not, would you consider doing so to verify the design parameters for the existing loads (Ampacity audit)?		Is not possible to organize the answers, some of them seems not relevant to the questions. In some cases (4) the measurements have been taken after a failure or in case of problems. In 2 cases measurements are taken during the design stage , and in 1 case during the installation works.	From the answers it seems that the question has not fully understood: in 7 cases the answer was referred to the specific failure or problem in a cable circuit, in others the answer covers a more general behaviour of the Utility. Some of them are not doing any TR measurement at all, others at the design stage and one both at the design and installation stage. It is difficult to find out a correlation
4	Has any cable failure occurred due to local overheating? If so, did you investigate the cause?		14 positive answers, 5 recognized as caused by soil drying out. In three cases the failures are relevant to accessories. For four cases there is no explanation of the causes.	The soil drying out is one the major reason for faults due to soil overheating.
5	Please describe the type of installation you experienced change.		Multiple answers possible	It is questionable if the high number reported for the directly buried cables reflects more the diffusion of such methods or the fact that this type of installation is the one more subject to changes in soil properties
	1. Directly buried		18 answers	
	2. Duct bank		2 answer	
	3. Steel or plastic pipes		6 answers	
	4. Trenchless		1 answer	

	technologies			
6	Have you experienced flooding or water/sewer breaks in the vicinity of a cable system?		12 positive answers out of 22, but only two reported serious issues for the cables backfilling	Despite the relatively high number of positive answers, only in two cases the cables were affected. This issue seems in general not critical.
7	Has the cable thermal environment changed as a result of construction activity or natural subsidence?		13 positive answers but not linked to thermal failures or issues. 6 out of 20 do not have info or are not sure of the situation.	The awareness of changes of the environment around the cables is clear in many cases, but this, except one case, has not triggered any action to correct the situation. On the other side it must be underlined that several answers show that the actual situation around the cables is not clear or unknown.
8	Do you have any type of cable temperature monitoring systems installed?		17 out of 22 have DTS systems, but in 3 cases these DTS are not used anymore and in 1 case it is for system testing purposes. Most of the answers are relevant to few in-service systems.	The use of DTS systems seems quite widespread although only for selected lines, may be the most critical one/s for the operator. In five questionnaires where the answer to question 1 is no, this point has been addressed, with 3 negative and two positive answers.